

Inter Pipeline Propylene Ltd.
Propane Dehydrogenation Project



DYNAMIC ANALYSIS OF 10-CG-10001 TAIL GAS COMPRESSOR PILE AND
FOUNDATION in 2NH
PDH-02-ST-CAL-0009



This document has been revised as indicated below. Please destroy all previous revisions.

Rev	Date	Revision Description	PREP	CHKD	APP'D	MANAGEMENT	
						Fluor	IPPL
0	18-Jan-2018	Issued for Information – Project 1202	LL	SC	-	-	-

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PROJECT: Williams Alberta PDH Project
Location: Fort Saskatchewan, AB
Title: 10-CG-10001 TAIL GAS COMPRESSOR FOUNDATION
Calculation No.: PDH-02-ST-CAL-0009
By: Lebing Liu

Contract No.: A6KV
Date: August 10, 2016
Check Level: 2B
Chk'd:

Drawing Numbers - Titles: DYNAMIC ANALYSIS OF 10-CG-10001 TAIL GAS COMPRESSOR FOUNDATION

Design Specifications:

Structural Design Criteria: See specification PDH-00-ST-SPC-0001 Structural Engineering Design Criteria

Approvals for First Issue

	Name (Print)	Signature	Date (yr/mo/day)
Originator	Lebing Liu		Aug. 10 2016
Checker	Skyler Chen		2016.09.20 15:12:19 -06'00'
Engineer of Record	Skyler Chen		

Record of Revisions

Rev. No.	Reason for Revision (Only for changes after first issue)	Date (yr/mo/day)	Engineer	Checked By

0		Issued for Design		
Rev.	Date	Revision Description	Lead	Design Supervisor



PROJECT:	Williams Alberta PDH Project	Contract No.:	A6KV
Location:	Fort Saskatchewan, AB	Date:	August 10, 2016
Title:	10-CG-10001 TAIL GAS COMPRESSOR FOUNDATION		
Calculation No.:	PDH-02-ST-CAL-0009	Check Level:	2B
By:	Lebing Liu	Chk'd:	

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3. CALCULATION SUMMARY



PROJECT:	Williams Alberta PDH Project	Contract No.:	A6KV
Location:	Fort Saskatchewan, AB	Date:	August 10, 2016
Title:	10-CG-10001 TAIL GAS COMPRESSOR FOUNDATION		
Calculation No.:	PDH-02-ST-CAL-0009	Check Level:	2B
By:	Lebing Liu	Chk'd:	

3. CALCULATION SUMMARY

This report presents detailed design calculations for the dynamic analysis of pile foundation supporting Williams PDH project area tail gas compressor 10-CG-10001, and associated Motor. The vendor drawings are present in Appendix C.

Compressor 10-CG-10001 is an 1100 HP compressor with operating speed of 1200 RPM. A common pile cap is used to support the compressor. The pile cap is supported by a group of 15 – 610 mm diameter x 12.7 mm wall thickness – 20 m long steel pipe piles. For foundation IFC drawings, please refer to Appendix A.

Soils condition for the foundation was based on the soils report by Thurber Engineering. Related geotechnical information is attached in the Appendix B of this report.

Dynamic analyses of the foundation were carried out by using Dyna-5 program. The foundation is considered as a rigid block supported by a group of driven steel piles when analyzing the combined amplitudes of foundations adding effects of 3 compressors together. There are relative thin links between each pedestal, the results of analyzing as a rigid block may not be conservative. Thereby, an additional check considering only the mass and piles contributed to one pump are performed. In other words, the foundation is analyzed as 3 independent virtual rigid blocks to check if the amplitude is within the allowable limits. Due to the similarities of the 3 blocks, only results of the virtual block of pump A are presented in this calculation.

The vibration responses were checked against the standard industry practice charts for the human perception and machine vibration limits to check the acceptability of the foundation system. Normal standard industry practice is to avoid resonance within +/- 20% of the operating frequency. Therefore, the analysis was performed for operating frequencies between 960 rpm and 1440 rpm.



PROJECT: Williams Alberta PDH Project
Location: Fort Saskatchewan, AB
Title: 10-CG-08001 COMPRESSOR FOUNDATION
Calculation No.: PDH-02-ST-CAL-0002
By: Donna Liu

Contract No.: A6KV
Date: 12-8-2015
Check Level: 2B
Chk'd:

4. A UNCERTAINTY AND RISK (SEE JDB 6.99 ATTACHMENT 04)

Uncertainty	Mitigation (Design Requirement)	Remaining Risk	Comments	Resolution of Uncertainty

5. COMPUTER FILE INDEX

The purpose of this form is to keep track of “calculation computer files” and to assign filenames using a standard format. This will assist in organizing data at a later date. All calculation files shall be stored under

P:\215 STRUCTURAL\215E ENGINEERING DESIGN DATA\215E 014 R Proj Calculations Sketches\PDH-02-ST-CAL-0009_02NH-CC-FDN 10-CG-10001ABC Compressor

and the respective subdirectory for the calculation folder in accordance to the Job Data Book.

File/ Folder Name	Description	Directory Location
PDH-02-ST-CAL-0009 Tail Gas Compressor.pdf	Current report.	\
10-CG-10001_Tail Gas.xls	Manual CG and Load Calculations.	\10-CG10001ABC_Tail Gas Compressor
DYNA 5.4 Steel Pile Lower Bound Runs	DYNA 5.4 Input for pump and motor for foundation system supported on steel pipe piles and lower bound geotechnical properties.	\2NH-CC10001ABC DYNA
DYNA 5.4 Steel Pile Upper Bound Runs	DYNA 5.4 Input for pump and motor for foundation system supported on steel pipe piles and upper bound geotechnical properties.	\2NH-CC10001ABC DYNA
Emails and Correspondence	PDF copy of all emails sent and received.	\Emails and Correspondance
Vendor Drawing PDFs	Vendor Drawings	\Vendor
PDH-02-ST-CAL-0009_02NH-CC- 10-CG-10001ABC Tail Gas.docx	Native File for present report	\



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6. EQUIPMENT LIST (SEE APPENDIX C FOR VENDOR DRAWINGS)

Equip Tag	Drawing/ Doc No	Revision	Description
10-CG-10001	A6KV-4-0306-01-00014	5A	General Arrangement DWG
10-CG-10001	A6KV-4-0306-01-00010	4	Foundation Plan
10-CG-10001	A6KV-4-0306-01-00010	1	Motor Pump Base Plate

7. REFERENCE DATA

Design carried out using the following NWR Specifications, reports and FLUOR Standards.

Specification or Reference Number	Description
PDH-00-ST-SPC-0001	Structural Engineering Design Criteria
PDH-00-ST-SPC-0003	Mechanical Equipment – Structural Data and Design Requirements
PDH-00-ST-SPC-0006	Concrete Materials
PDH-00-ST-SPC-0007	Concrete Construction
PDH-00-ST-SPC-0008	Concrete Reinforcing
PDH-00-ST-SPC-0011	Grouting
PDH-00-ST-SPC-0012	Driven Steel Piles
000.215.1235	Vibrating Machinery Foundations on Piles
19-3500-7	Thurber Engineering Williams PDH Project Geotechnical Report
19-3500-8	Williams PDH Project Supplemental Geotechnical Investigation

8 INTRODUCTION

This calculation is for the dynamic analysis of the piled foundations, which support the tail gas compressor 10-CG-10001 for the Unit 10 of Williams PDH Project. The foundation system consists of pedestals, pile cap and piles supporting the complete pump unit.

8.1 Design Basis

The Dynamic Analysis ensures that all the vibration limit & frequency requirements, as specified by the Vendor and the Williams specification are satisfied, thus ensuring the satisfactory operation of the equipment.

The combined foundation of 3 compressors are analyzed as a rigid block. Additional checks have been made considering the relatively small thickness of pile caps between pedestals; A separated foundation considering only the concrete and piles under one compressor are analyzed as isolated smaller blocks. The actual vibration amplitudes are expected between the "one rigid block model" and "isolated blocks" model. Only one block is analyzed for the "isolated blocks" model due to similarities between pumps.

8.2 Foundation Layout Sketches

These are shown in Appendix A

9 DESCRIPTION OF PUMP and MOTOR

Tag number are: 10-CG-10001A/B/C

Info from vendor drawing code A:

Motor 820KW(1100HP), speed 1200RPM;

Compressor weight 3866kg, motor weight 5125kg, base plate weight 2100kg, other weight 1500kg;

Base plate size: 2.032m x 5.61m

10 DYNAMIC DESIGN CRITERIA

Dynamic analysis is performed to ensure the satisfactory operation of the pumps. The associated foundation has to satisfy both the 'foundation acceptance criteria' and the 'frequency criteria'.

10.1 Foundation Acceptance Criteria

As per vendor requirement, the acceptable vibration speed is 2.8mm/s at 1200 RPM

10.2 Frequency Criteria

As per Fluor guideline 000.215.1235, foundations frequency should be outside the resonant range $\pm 20\%$ beyond the machine's operating range.

11 DYNAMIC ANALYSIS METHODOLOGY

The dynamic analysis is performed using DYNA 5 (version 5.4), which is extensively used for 'Dynamic Analysis of Foundation for the Effects of Harmonic and Impact Loadings'. DYNA 5 has the flexibility and convenience of inputting multiple points of excitation at a common operating speed; thus it is preferred to DYNAN where all excitations have to be converted into a single excitation for input.

12 GEOTECHNICAL INFORMATION FOR DYNAMIC ANALYSIS

The geotechnical information used for the present analysis is based on the report provided by Thurber Engineering LTD, titled "Fort Saskatchewan Propane Dehydrogenation Facility Supplementary Geotechnical Investigation", File No: 19-3500-8, Dec 4 2014; Hereafter refers to as the geotechnical report.

13 DYNA Input Calculations

13.1 Pile Information

Pile type:	Driven Steel Pipe Pile		
Pile diameter:	610 mm (new)	604 mm (corroded)	1
Pile Thickness:	12.70 mm		
Pile embedment in pile cap:	100 mm		
Total Pile length	20.35 m		
Pile length (under pile cap)	20 m	Used 20 m in analysis.	
Steel unit weight	77000 N/m ³		
Poisson's ratio:	0.3		
Material damping:	0.02		
Coefficient of rigidity:	1.11		
Modulus of Elasticity:	2 x 10 ¹¹ N/m ²		
Cross-sectional area, A:	0.024 (new)	0.021 m ² (corroded)	
Moment of inertia about Y, I _y :	0.001063 m ⁴ (new)	0.000931 m ⁴ (corroded)	
Moment of Inertia about X, I _x :	0.001063 m ⁴ (new)	0.000931 m ⁴ (corroded)	
I _z = I _y + I _x :	0.002127 m ⁴ (new)	0.001861 m ⁴ (corroded)	

1 Choice of pile size is preliminary and related to static load requirements.

13.2 Geotechnical Information

##

TABLE 4.1
REVISED RECOMMENDED DYNAMIC SOIL PARAMETERS

PROPERTY	MATERIAL	
	SAND/CLAY (0 – 15 m)	CLAY TILL (15 m – 25 m)
Shear Wave Velocity (m/s)	100 – 200	200 – 400
Dynamic Young's Modulus(MPa)	55 – 220	230 – 925
Dynamic Shear Modulus (MPa)	20 – 80	85 – 340
Poisson's Ratio	0.35	0.35
Bulk Unit Weight (kN/m ³)	20	21
Internal Damping Factor	0.02 – 0.04	0.02 – 0.04

14. Calculation of Speed at motor and compressor location based on result amplitudes

14.1 Speed for "Isolated Blocks Model", only considering one compressor at a time

Max considered frequency	1440 rpm	
Max amplitude	0.006 mm	(x direction lower bound)
Max speed	0.905 mm/s	< 2.8 mm/s

14.2 Speed for "one rigid model", considering accumulation of speed of 3 compressors running at the same time

Max considered frequency	1440 rpm	
Max sum of amplitudes of 3 compressors	0.007 mm	
Max speed	1.056 mm/s	< 2.8 mm/s

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13.4 DYNA Input Load Calculations

A. Vendor Data

Reference	Component	Operating Frequency (rpm)	Operating Frequency (Hz)	Rotor Weight (kg)	Rotor Weight (N)	Rotor Weight (As per vendor drawing A6KV-4-0306- 01-00010)
10-CG-10001A/B/C	Compressor	1200	20.00	260	2551	
	Motor	1200	20.0	983.0	9643	

B. DYNA Input Calculations

Reference	Component	Rotor Weight (kg)	Rotor Weight (N)	BQ _g (mm/s)	Frequency (rpm)	Frequency (rad/s)	SF	Force G 1.0 (N)	F* (DYNA) (kg m/rad ²)
10-CG-10001A/B/C	Motor	983.0	9643	6.25	1200	125.7	2	1544	0.098
	Compressor	260.0	2551	6.25	1200	125.7	2	408	0.026

Note:

1. $F^* (DYNA) = \text{Unbalance Force in } N / (\text{Frequency in rad/s})^2$

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C. Vibration Limits for Dyna outputs

Unit 30: 10-CG-10001A/B/C	
DYNA Input	

Reference	Operating Frequency (rpm)	Operating Frequency Hz(cps)	0.8 x Operating Frequency (rpm)	1.2 x Operating Frequency (rpm)	Vibration Limits by Vendor		Vibration Limits in Dyna	
					Displ. (um)	velocity (mm/s)	N	um
10-CG-10001A/B/C Dynamic Torques	Motor	-	960	1440		2.80		22.3
	Compressor	-	960	1440		3.50		27.9
	1st Order	20.00	960	1440				

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DYNA Input	

13.3 Coordinates and C.G. Calculations For a Virtual Block Separated from The Main Block

Coordinates relative to south-west corner of pile cap.

X-Coordinate Adjustment	820	mm
Y-Coordinate Adjustment	0	mm
Z-Coordinate Adjustment	2950	mm
Pile Cap Thickness	900	mm

(Top of pile cap is at EL 628.300)

A. Motor/ Compressor/ Pier

Reference	Description	Local Coordinates (Base on pump datum)			Global Coordinates (Base on SW corner of pump FDN)			Remarks
		X (mm)	Y (mm)	Z (mm)	X' (m)	Y' (m)	Z' (m)	
10-CG-10001A	Compressor -A	999	4495	975	1.819	4.495	3.925	
	Motor-A	999	2000	1145	1.819	2.000	4.095	
	Piping-A	820	1620	2055	1.640	1.620	5.005	
	Base plate-A	1250	3000	300	2.070	3.000	3.250	

Note: 1. The origin for the global coordinates, i.e., (X' = 0, Y' = 0 and Z' = 0), is at the bottom south-west corner of the pile cap. X' is +ve going east, Y' is +ve going north and Z' is +ve going upwards.

B. Concrete Blocks		=		2400		kg/m³	
Density of concrete							
Block Number	Description	Lx (mm)	Ly (mm)	Lz (mm)	C.G. of Blocks		
					X' (m)	Y' (m)	Z' (m)
1	Pile cap	3900	6000	900	1.950	3.000	0.450
2	Pier 1 (2.5W&6.0L)	2500	6000	2075	2.050	3.000	1.938
					Total:		125244
C. Machine Masses					kg		
Reference	Description	X' (m)	Y' (m)	Z' (m)	Mass (kg)		
10-CG-10001A	Compressor -A Motor-A Piping-A Base plate-A	1.819	4.495	3.925	3966		
		1.819	2.000	4.095	5125		
		1.640	1.620	5.005	1510		
		2.070	3.000	3.250	2100		
		Total:			12701.0		
					kg		

D.1 For centrifugal machines, foundation weight shall not be less than 3 times the weight of the machine:

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Reference	Description	X' (m)	Y' (m)	Z' (m)	Mass, M (kg)	M*X (kgm)	M*Y (kgm)	M*Z (kgm)
10-CG-10001A	Compressor -A	1.819	4.495	3.925	3966	7214	17827	15567
	Motor-A	1.819	2.000	4.095	5125	9322	10250	20987
	Piping-A	1.640	1.620	5.005	1510	2476	2446	7558
	Base plate-A	2.070	3.000	3.250	2100	4347	6300	6825
	Pile cap	1.950	3.000	0.450	50544	98561	151632	22745
	Pier 1	2.050	3.000	1.938	74700	153135	224100	144731
Total:					137945	275056	412555	218412

System CG:	X'	=	1.99395	m	DYNA CG:	1.99395	m	0.000015%
	Y'	=	2.99072	m		2.99072	m	0.000062%
	Z'	=	1.58333	m		1.58333	m	-0.000013%

OS_x = Distance between CG and center of resistance of pile group divided by
pile cap length (percent) = 1.13 % < 5% OK

OS_y = Distance between CG and center of resistance of pile group divided by
pile cap length (percent) = -0.15 % < 5% OK

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DYNA Input	

E. Points to be Analyzed in DYNA

E.1

Points to be assessed for Foundation Vibration Limits as per Vendor Criteria: See Appendix A for location of points.

Reference	Description	Global Coordinates			Coordinates with reference to System CG			Remark
		X' (m)	Y' (m)	Z' (m)	X'' (m)	Y'' (m)	Z* (m)	
P1 (Pump A CG)	See sketch in Appendix A	1.819	4.495	3.925	-0.175	1.504	-2.342	Note 1
P2 (motor A CG)		1.819	2.000	4.095	-0.175	-0.991	-2.512	Note 1

Note:

- The global Z in DYNA is +ve downwards which is opposite direction to the of the global Z' direction for the coordinates, hence the Z* is modified to be consistent with the DYNA conventions.

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F. Pile Coordinates (For concrete piles)

	Pile Number	Global Coordinates		Coordinates with reference to System CG	
		X' (m)	Y (m)	X" (m)	Y" (m)
Start		0.75	0.75		
Spacing		2.5	2.25		
Spacing		2.5	2.25		
	1	0.75	0.75	-1.244	-2.241
	2	3.25	0.75	1.256	-2.241
	3	5.75	0.75	3.756	-2.241
	4	8.25	0.75	6.256	-2.241
	5	10.75	0.75	8.756	-2.241
	6	0.75	3	-1.244	0.009

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13.4 Coordinates and C.G. Calculations of Combined Foundation

Coordinates relative to south-west corner of pile cap.

X-Coordinate Adjustment	820	mm
Y-Coordinate Adjustment	0	mm
Z-Coordinate Adjustment	2950	mm
Pile Cap Thickness	900	mm

(Top of pile cap is at EL 628.300)

A. Motor/ Compressor/ Pier

Reference	Description	Local Coordinates (Base on pump datum)		Global Coordinates (Base on SW corner of pump FDN)			Remarks
		X (mm)	Y (mm)	X' (m)	Y' (m)	Z' (m)	
10-CG-10001A	Compressor -A	999	4495	1.819	4.495	3.925	
	Motor-A	999	2000	1.819	2.000	4.095	
	Piping-A	820	1620	1.640	1.620	5.005	
10-CG-10001B	Base plate-A	1250	3000	2.070	3.000	3.250	
	Compressor -B	4679	4495	5.499	4.495	3.925	
	Motor-B	4679	2000	5.499	2.000	4.095	
10-CG-10001C	Piping-B	4500	1620	5.320	1.620	5.005	
	Base plate-B	4930	3000	5.750	3.000	3.250	
	Compressor -C	8359	4495	9.179	4.495	3.925	
	Motor-C	8359	2000	9.179	2.000	4.095	
	Piping-C	8180	1620	9.000	1.620	5.005	
	Base plate-C	8610	3000	9.430	3.000	3.250	

Note: 1. The origin for the global coordinates, i.e., (X' = 0, Y' = 0 and Z' = 0), is at the bottom south-west corner of the pile cap. X' is +ve going east, Y' is +ve going north and Z' is +ve going upwards.

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Unit 10: 10-CG-10001A/B/C
DYNA Input

B. Concrete Blocks		=		2400		kg/m ³	
Block Number	Description	Lx (mm)	Ly (mm)	Block Dimensions		C.G. of Blocks	
				Lz (mm)	X' (m)	Y' (m)	Z' (m)
1	Pile cap	11500	6000	900	5.750	3.000	0.450
2	Pier 1 (2.5W&6.0L)	2500	6000	2075	2.050	3.000	1.938
3	Pier 2 (2.5W&6.0L)	2500	6000	2075	5.730	3.000	1.938
4	Pier 3	2500	6000	2075	9.410	3.000	1.938
Total:							373140
							kg

C. Machine Masses							
Reference	Description	X' (m)	Y' (m)	Z' (m)	Mass (kg)		
10-CG-10001A	Compressor -A	1.819	4.495	3.925	3966		
	Motor-A	1.819	2.000	4.095	5125		
	Piping-A	1.640	1.620	5.005	1510		
10-CG-10001B	Base plate-A	2.070	3.000	3.250	2100		
	Compressor -B	5.499	4.495	3.925	3966		
	Motor-B	5.499	2.000	4.095	5125		
10-CG-10001C	Piping-B	5.320	1.620	5.005	1510		
	Base plate-B	5.750	3.000	3.250	2100		
	Compressor -C	9.179	4.495	3.925	3966		
	Motor-C	9.179	2.000	4.095	5125		
	Piping-C	9.000	1.620	5.005	1510		
				Base plate-C	9.430	3.000	3.250
Total:					38103.0		kg

D. System CG Calculations

D.1

For centrifugal machines, foundation weight shall not be less than 3 times the weight of the machine:

Machine Weight, Mm	=	38103.0	kg
Foundation Weight, Mc	=	373140	kg
Total		411243	kg
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Unit 10: 10-CG-10001A/B/C	
DYNA Input	

Mass Ratio, Mc/Mm

=

9.8

> 3.0 OK

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Location of System CG									
Reference	Description	X' (m)	Y' (m)	Z' (m)	Mass, M (kg)	M*X (kgm)	M*Y (kgm)	M*Z (kgm)	
10-CG-10001A	Compressor -A	1.819	4.495	3.925	3966	7214	17827	15567	
	Motor-A	1.819	2.000	4.095	5125	9322	10250	20987	
	Piping-A	1.640	1.620	5.005	1510	2476	2446	7558	
	Base plate-A	2.070	3.000	3.250	2100	4347	6300	6825	
10-CG-10001B	Compressor -B	5.499	4.495	3.925	3966	21809	17827	15567	
	Motor-B	5.499	2.000	4.095	5125	28182	10250	20987	
	Piping-B	5.320	1.620	5.005	1510	8033	2446	7558	
	Base plate-B	5.750	3.000	3.250	2100	12075	6300	6825	
10-CG-10001C	Compressor -C	9.179	4.495	3.925	3966	36404	17827	15567	
	Motor-C	9.179	2.000	4.095	5125	47042	10250	20987	
	Piping-C	9.000	1.620	5.005	1510	13590	2446	7558	
	Base plate-C	9.430	3.000	3.250	2100	19803	6300	6825	
	Pile cap	5.750	3.000	0.450	149040	856980	447120	67068	
	Pier 1	2.050	3.000	1.938	74700	153135	224100	144731	
	Pier 2	5.730	3.000	1.938	74700	428031	224100	144731	
	Pier 3	9.410	3.000	1.938	74700	702927	224100	144731	
Total:					411243	2351372	1229890	654070	

System CG:	X'	=	5.71772	m	DYNA CG:	5.72862	m	-1.090021%
	Y'	=	2.99067	m		2.99067	m	0.000015%
	Z'	=	1.59047	m		1.59047	m	0.000005%

OS_x = Distance between CG and center of resistance of pile group divided by pile cap length (percent) = -0.28 % < 5% **OK**

OS_y = Distance between CG and center of resistance of pile group divided by pile cap length (percent) = -0.16 % < 5% **OK**

Date	09-Aug-2016
CONT. NO.	A6KV
BY	Lebing Liu
CHK'D	
SHEET NO.	
CALC.	PDH-02-ST-CAL-0009

E. Points to be Analyzed in DYNA

E.1

Points to be assessed for Foundation Vibration Limits as per Vendor Criteria: See Appendix A for location of points.

Reference	Description	Global Coordinates			Coordinates with reference to System CG			Remark
		X' (m)	Y' (m)	Z' (m)	X" (m)	Y" (m)	Z* (m)	
P1 (Pump A CG)	See sketch in Appendix A	1.819	4.495	3.925	-3.899	1.504	-2.335	Note 1
P2 (motor A CG)		1.819	2.000	4.095	-3.899	-0.991	-2.505	Note 1
P3 (Pump B CG)		5.499	4.495	3.925	-0.219	1.504	-2.335	Note 1
P4 (motor B CG)		5.499	2.000	4.095	-0.219	-0.991	-2.505	Note 1
P5 (Pump C CG)		9.179	4.495	3.925	3.461	1.504	-2.335	Note 1
P6 (motor C CG)		9.179	2.000	4.095	3.461	-0.991	-2.505	Note 1
P8 (Motor B center)					0.000	0.000	0.000	Note 1
					0.000	0.000	0.000	Note 1

Note:

- The global Z in DYNA is +ve downwards which is opposite direction to the of the global Z' direction for the coordinates, hence the Z* is modified to be consistent with the DYNA conventions.

Date	09-Aug-2016
CONT. NO.	A6KV
BY	Lebing Liu
CHK'D	
SHEET NO.	
CALC.	PDH-02-ST-CAL-0009
DYNA Input	

F. Pile Coordinates (For concrete piles)

		m		Global Coordinates		Coordinates with reference to System CG	
		m		X' (m)	Y (m)	X" (m)	Y" (m)
Start Spacing Spacing		0.75	0.75			-4.968	-2.241
		2.5	2.25	3.25	0.75	-2.468	-2.241
		2.5	2.25	5.75	0.75	0.032	-2.241
Pile Number		8.25	0.75	8.25	0.75	2.532	-2.241
		10.75	0.75	10.75	0.75	5.032	-2.241
		0.75	3	0.75	3	-4.968	0.009
		3.25	3	3.25	3	-2.468	0.009
		5.75	3	5.75	3	0.032	0.009
		8.25	3	8.25	3	2.532	0.009
		10.75	3	10.75	3	5.032	0.009
		0.75	5.25	0.75	5.25	-4.968	2.259
		3.25	5.25	3.25	5.25	-2.468	2.259
		5.75	5.25	5.75	5.25	0.032	2.259
		8.25	5.25	8.25	5.25	2.532	2.259
		10.75	5.25	10.75	5.25	5.032	2.259

Total number of piles

Sum of machine and fdn weight in kg

Sum of machine and fdn weight in kN

Compression Pile capacity of 610 dia

steel pile 20 m long

50% of pile capacity

Average load per pile

1.25 * Load average percentage

15

411243 kg

4034 kN_u

1271 kN_f

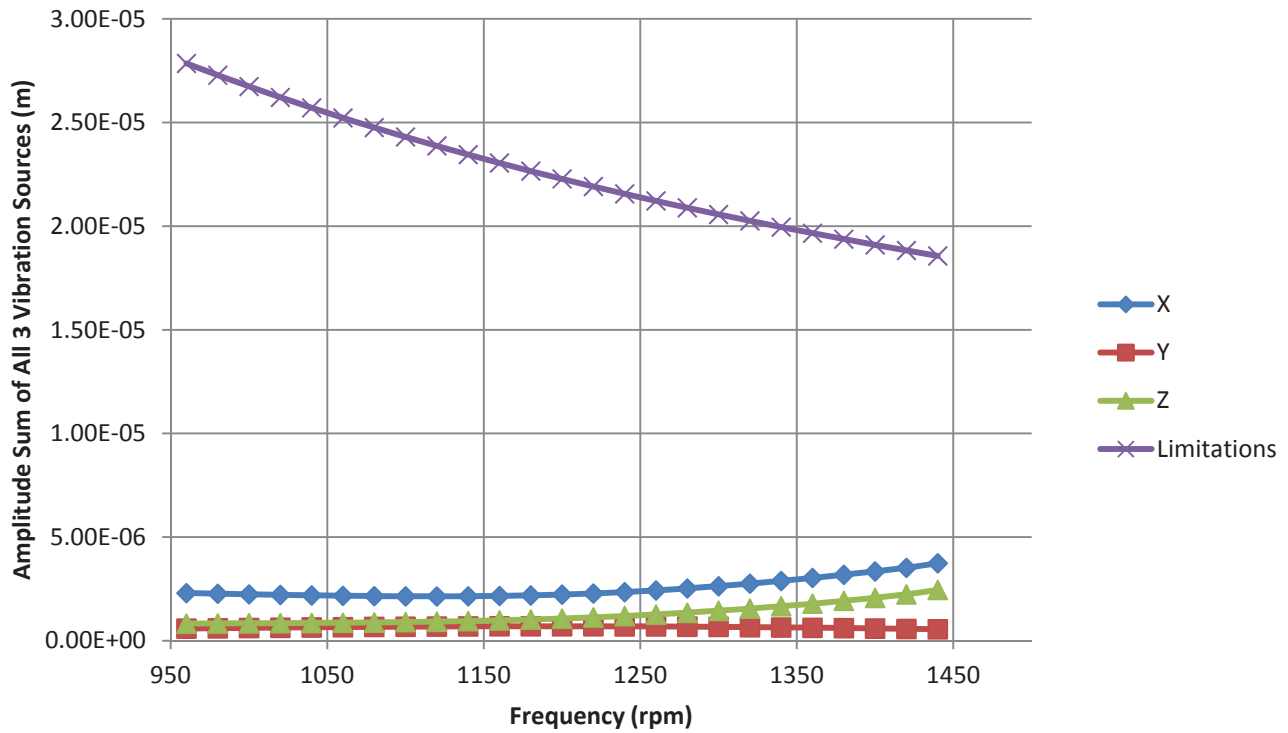
635.5 kN_f

269 kN_u

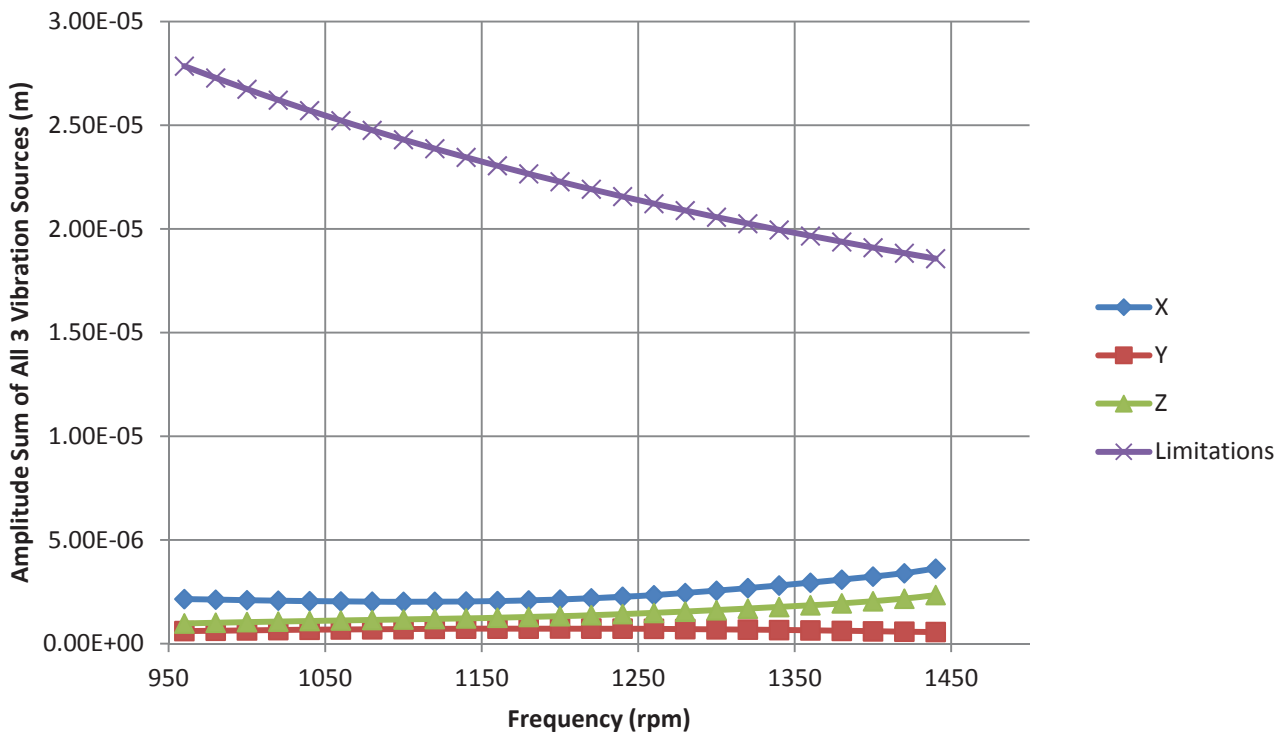
53% < 100 % Okay. This assumes the system CG coincides with the pile centre of resistance and check is done against 50% available capacity.

13.5 DYNA 5 Lower Bound Input & Output for Combined Foundation

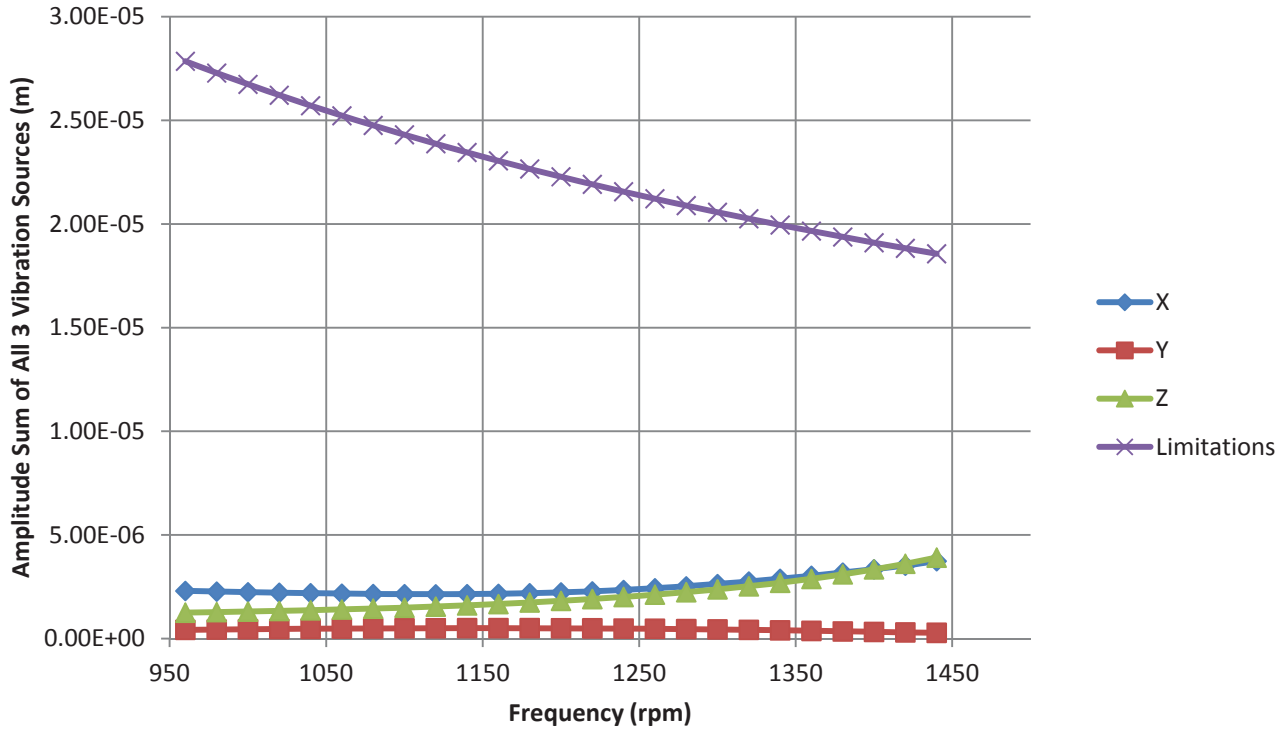
Translations at CG10001 Compressor A (LB)



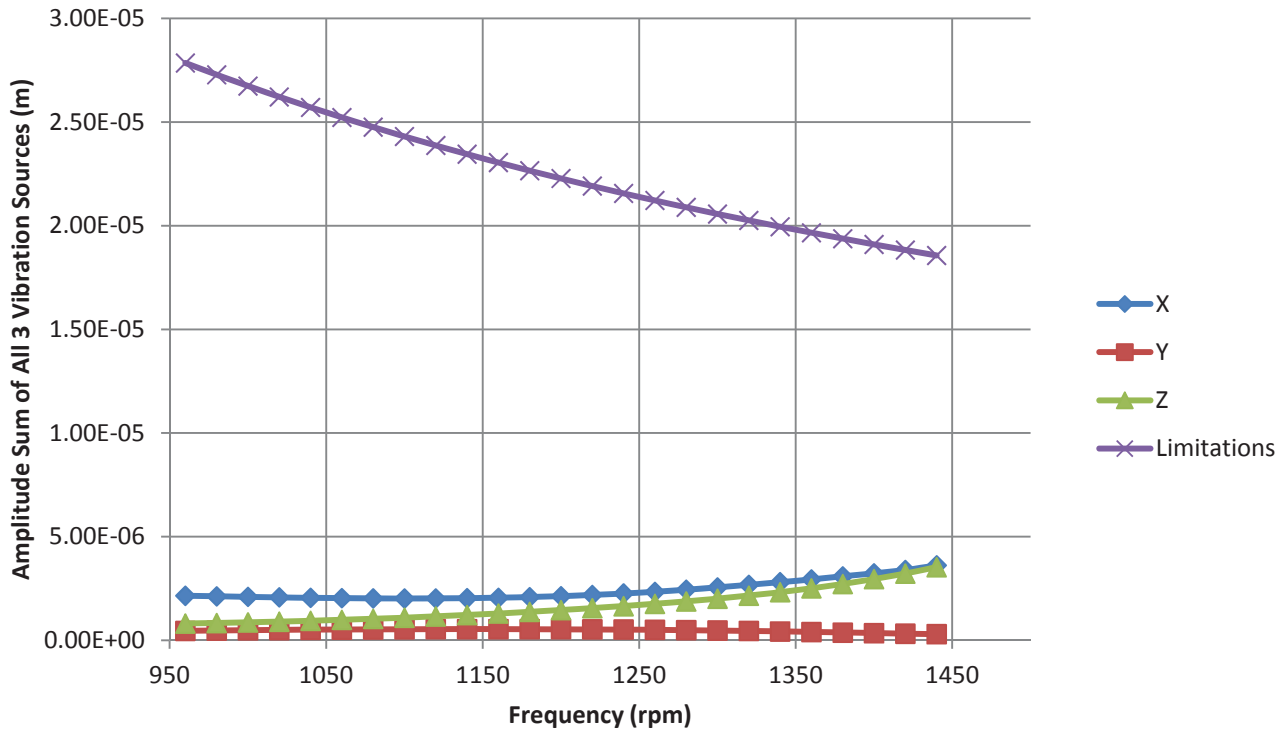
Translations at CG10001 Motor A (LB)



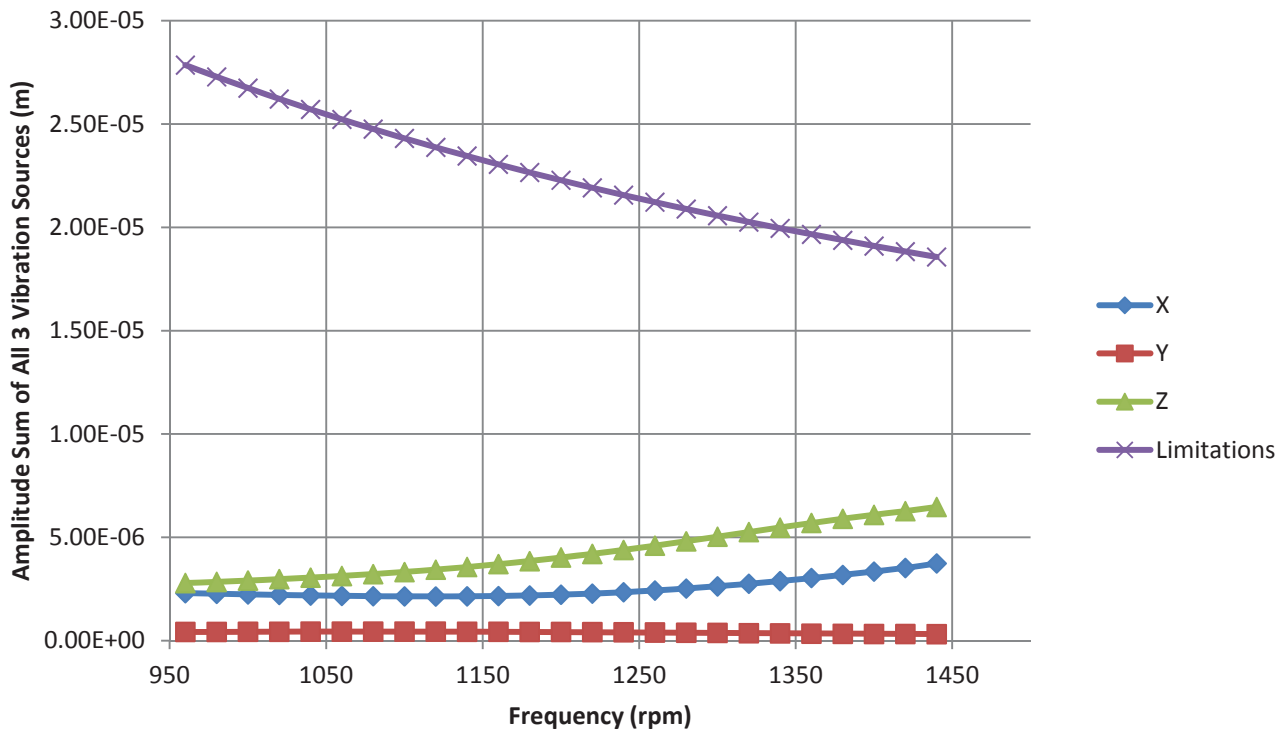
Translations at CG10001 Compressor B (LB)



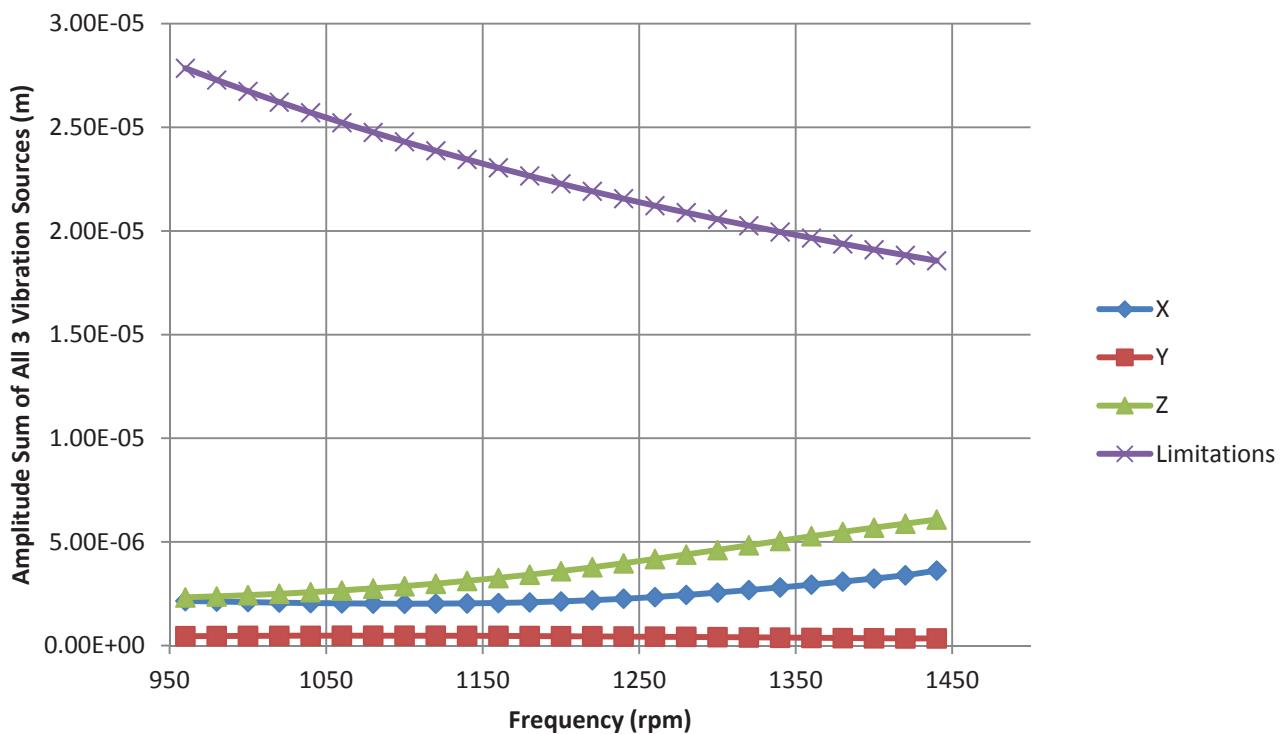
Translations at CG10001 Motor B (LB)



Translations at CG10001 Compressor C (LB)



Translations at CG10001 Motor C (LB)



```

*****
*
*           D Y N A 5   S I M U L A T I O N
*
*   Copyright (c) M. Novak, U.W.O. - 1999
*           RUN DATE - 2016/ 8/ 4
*           TIME    - 14:18:19
*           REVISION - 1999/04/23
*
*****

CG10001A LB 1200

                DATA ECHO
                *****

GRAVITATIONAL CONSTANT SET TO 9.81 m/s**2
*****

FREQUENCY UNITS : RPM
*****

LENGTH UNITS : m
*****

FORCE UNITS : N
*****

MASS UNITS : kg
*****

DAMPING SAFETY FACTOR - 2.00
*****

FOUNDATION TYPE - PILE
*****

RIGID CAP
*****

RECTANGULAR FOUNDATION - 11.500 m BY 6.000 m
*****

MASS CONSTANTS
*****

TOTAL MASS OF FOOTING ..... 4.1124E+05 kg
MASS MOMENT ABOUT X-AXIS ..... 1.7127E+06 kg.m**2
MASS MOMENT ABOUT Y-AXIS ..... 4.6937E+06 kg.m**2
MASS MOMENT ABOUT Z-AXIS ..... 5.3209E+06 kg.m**2
CROSS PRODUCT X-Y ..... 1.9995E+03 kg.m**2
CROSS PRODUCT X-Z ..... -2.3290E+04 kg.m**2
CROSS PRODUCT Y-Z ..... -1.8327E+04 kg.m**2

NUMBER OF LAYERS OR ELEMENTS - 3
*****

FILES ARE FIXED AT BUTT
*****

LOCATION OF PILES
*****

FILE      X      Y
NO.      LOCATION LOCATION
          m      m

```

1	-4.979	-2.241
2	-4.979	0.009
3	-4.979	2.259
4	-2.479	-2.241
5	-2.479	0.009
6	-2.479	2.259
7	0.021	-2.241
8	0.021	0.009
9	0.021	2.259
10	2.521	-2.241
11	2.521	0.009
12	2.521	2.259
13	5.021	-2.241
14	5.021	0.009
15	5.021	2.259

COORDINATES OF BASE CENTRE

X-COORD. OF BASE CENTRE REL. TO C.G. .. -0.021 m
Y-COORD. OF BASE CENTRE REL. TO C.G. .. -0.009 m
Z-COORD. OF BASE CENTRE REL. TO C.G. .. 1.590 m

PILE PROPERTIES

TOTAL LENGTH 20.250 m
UNIT WEIGHT 77000.000 N/m**3
STATIC LOAD ON PILE 5.580E+05 N
POISSON'S RATIO 0.300
MATERIAL DAMPING 0.010
COEFF. OF RIGIDITY 1.110
YOUNG'S MODULUS 2.000E+11 N/m**2

ELEMENT

LAYER NO.	LAYER DEPTH m	PILE RADIUS (X) m	PILE RADIUS (Y) m	CROSS SECTIONAL AREA m**2	MOMENT OF INERTIA ABOUT Y m**4	MOMENT OF INERTIA ABOUT X m**4	TORSIONAL CONSTANT m**4
1	0.250	0.303	0.303	0.021	9.31E-04	9.31E-04	1.86E-03
2	13.250	0.303	0.303	0.021	9.31E-04	9.31E-04	1.86E-03
3	6.750	0.303	0.303	0.021	9.31E-04	9.31E-04	1.86E-03

PILE TIP CONDITION : FLOATING (DEFAULT)

PILE-SOIL-PILE INTERACTION IS CONSIDERED

SOIL

SOIL CONSTANTS

LAYER NO.	SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
1	0.000	0.00	0.000	0.000
2	100.000	20000.00	0.350	0.040
3	200.000	21000.00	0.350	0.040

SOIL BELOW PILE

SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
2.00E+02	21000.00	0.350	0.040

EMBEDDED OPTION

NUMBER OF SIDE LAYERS - 1

LAYER NO.	THICKNESS OF LAYER m	SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
1	0.000	0.00	0.00	0.000	0.000

HARMONIC LOAD (QUADRATIC)

MAXIMUM FREQUENCY		1440.000	RPM
MINIMUM FREQUENCY		960.000	RPM
STEP FREQUENCY		20.000	RPM
FORCE IN X-DIRECTION		2.48E-01	kg.m
FORCE IN Y-DIRECTION		0.00E+00	kg.m
FORCE IN Z-DIRECTION		2.48E-01	kg.m
MOMENT ABOUT X-AXIS		-1.16E-01	kg.m**2
MOMENT ABOUT Y-AXIS		1.74E+00	kg.m**2
MOMENT ABOUT Z-AXIS		1.16E-01	kg.m**2

RESULTANT TRANSLATIONS REQUIRED AT 6 POINTS

COORDINATES OF POINTS :

POINT NO.	X m	Y m	Z m
1	-3.909	1.504	-2.342
2	-3.909	-0.991	-2.512
3	-0.229	1.504	-2.342
4	-0.229	-0.991	-2.512
5	3.451	1.504	-2.342
6	3.451	-0.991	-2.512

```

*****
*
*           D Y N A 5   S I M U L A T I O N
*
*   Copyright (c) M. Novak, U.W.O. - 1999
*   RUN DATE - 2016/ 8/ 4
*   TIME - 14:18:19
*   REVISION - 1999/04/23
*
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CG10001A LB 1200

RESULTS

FOOTING RESPONSE AMPLITUDES

FREQ. RPM	TRANS. X DIRECTION (m)	TRANS. Y DIRECTION (m)	VERTICAL DIRECTION (m)	ROT ABOUT X AXIS (rad)	ROT ABOUT Y AXIS (rad)	TORSIONAL Z AXIS (rad)
960.00	7.53E-07	5.02E-08	3.56E-07	6.27E-08	2.42E-07	2.60E-08
980.00	7.44E-07	5.05E-08	3.64E-07	6.45E-08	2.46E-07	2.59E-08
1000.00	7.37E-07	5.08E-08	3.74E-07	6.62E-08	2.51E-07	2.57E-08
1020.00	7.30E-07	5.10E-08	3.85E-07	6.77E-08	2.56E-07	2.56E-08
1040.00	7.23E-07	5.13E-08	3.97E-07	6.90E-08	2.62E-07	2.56E-08
1060.00	7.17E-07	5.15E-08	4.12E-07	7.01E-08	2.68E-07	2.55E-08
1080.00	7.13E-07	5.18E-08	4.28E-07	7.10E-08	2.75E-07	2.56E-08
1100.00	7.09E-07	5.22E-08	4.46E-07	7.16E-08	2.83E-07	2.56E-08
1120.00	7.07E-07	5.27E-08	4.66E-07	7.21E-08	2.92E-07	2.57E-08
1140.00	7.06E-07	5.32E-08	4.88E-07	7.24E-08	3.01E-07	2.59E-08
1160.00	7.07E-07	5.40E-08	5.13E-07	7.24E-08	3.12E-07	2.61E-08
1180.00	7.11E-07	5.49E-08	5.40E-07	7.22E-08	3.24E-07	2.64E-08
1200.00	7.17E-07	5.61E-08	5.69E-07	7.18E-08	3.37E-07	2.67E-08
1220.00	7.27E-07	5.75E-08	6.01E-07	7.12E-08	3.51E-07	2.70E-08
1240.00	7.41E-07	5.92E-08	6.36E-07	7.03E-08	3.65E-07	2.75E-08
1260.00	7.59E-07	6.13E-08	6.75E-07	6.91E-08	3.81E-07	2.79E-08
1280.00	7.83E-07	6.37E-08	7.17E-07	6.76E-08	3.96E-07	2.85E-08
1300.00	8.14E-07	6.64E-08	7.63E-07	6.59E-08	4.11E-07	2.91E-08
1320.00	8.53E-07	6.95E-08	8.13E-07	6.38E-08	4.25E-07	2.98E-08
1340.00	9.01E-07	7.28E-08	8.70E-07	6.16E-08	4.36E-07	3.05E-08
1360.00	9.59E-07	7.63E-08	9.32E-07	5.93E-08	4.44E-07	3.12E-08
1380.00	1.03E-06	7.99E-08	1.00E-06	5.70E-08	4.48E-07	3.19E-08
1400.00	1.11E-06	8.33E-08	1.08E-06	5.51E-08	4.46E-07	3.25E-08
1420.00	1.21E-06	8.65E-08	1.17E-06	5.36E-08	4.40E-07	3.30E-08
1440.00	1.31E-06	8.93E-08	1.27E-06	5.22E-08	4.31E-07	3.33E-08

MAXIMA OF FOOTING RESPONSE AMPLITUDES

				FREQ. RPM
MAX. TRANS. IN X-DIRECTION	-	1.313E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	8.933E-08	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	1.267E-06	m	1440.00
MAX. ROT. ABOUT X AXIS	-	7.240E-08	rad	1160.00
MAX. ROT. ABOUT Y AXIS	-	4.479E-07	rad	1380.00
MAX. ROT. ABOUT Z AXIS	-	3.328E-08	rad	1440.00

RESULTANT TRANSLATIONS (m)

POINT NO. 1				POINT NO. 2		
-----				-----		
FREQ. RPM	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION
-----	-----	-----	-----	-----	-----	-----
960.00	8.71E-07	1.96E-07	5.11E-07	8.39E-07	2.03E-07	6.67E-07
980.00	8.61E-07	2.01E-07	5.20E-07	8.32E-07	2.08E-07	6.81E-07
1000.00	8.52E-07	2.06E-07	5.29E-07	8.26E-07	2.13E-07	6.94E-07
1020.00	8.45E-07	2.10E-07	5.38E-07	8.21E-07	2.19E-07	7.06E-07
1040.00	8.39E-07	2.15E-07	5.48E-07	8.18E-07	2.24E-07	7.18E-07
1060.00	8.36E-07	2.19E-07	5.58E-07	8.18E-07	2.28E-07	7.30E-07
1080.00	8.35E-07	2.23E-07	5.70E-07	8.20E-07	2.32E-07	7.42E-07
1100.00	8.38E-07	2.27E-07	5.84E-07	8.26E-07	2.36E-07	7.55E-07
1120.00	8.44E-07	2.30E-07	5.99E-07	8.36E-07	2.39E-07	7.69E-07
1140.00	8.55E-07	2.32E-07	6.16E-07	8.51E-07	2.42E-07	7.85E-07
1160.00	8.71E-07	2.34E-07	6.35E-07	8.71E-07	2.43E-07	8.03E-07
1180.00	8.93E-07	2.35E-07	6.57E-07	8.96E-07	2.44E-07	8.24E-07
1200.00	9.20E-07	2.36E-07	6.80E-07	9.28E-07	2.45E-07	8.46E-07
1220.00	9.54E-07	2.35E-07	7.05E-07	9.65E-07	2.44E-07	8.70E-07
1240.00	9.93E-07	2.34E-07	7.31E-07	1.01E-06	2.43E-07	8.95E-07
1260.00	1.04E-06	2.32E-07	7.57E-07	1.05E-06	2.40E-07	9.21E-07
1280.00	1.09E-06	2.29E-07	7.81E-07	1.11E-06	2.37E-07	9.44E-07
1300.00	1.14E-06	2.25E-07	8.02E-07	1.16E-06	2.32E-07	9.62E-07
1320.00	1.20E-06	2.21E-07	8.15E-07	1.22E-06	2.27E-07	9.73E-07
1340.00	1.25E-06	2.16E-07	8.18E-07	1.27E-06	2.21E-07	9.72E-07
1360.00	1.31E-06	2.10E-07	8.06E-07	1.32E-06	2.15E-07	9.53E-07
1380.00	1.36E-06	2.04E-07	7.75E-07	1.37E-06	2.08E-07	9.14E-07
1400.00	1.41E-06	1.99E-07	7.23E-07	1.41E-06	2.01E-07	8.51E-07
1420.00	1.46E-06	1.93E-07	6.56E-07	1.45E-06	1.95E-07	7.69E-07
1440.00	1.54E-06	1.89E-07	5.99E-07	1.52E-06	1.90E-07	6.87E-07

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 1

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	1.541E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	2.355E-07	m	1200.00
MAX. TRANS. IN Z-DIRECTION	-	8.180E-07	m	1340.00
MAX. TRANS. IN HORIZ. PLANE	-	1.553E-06	m	1440.00

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 2

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	1.524E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	2.448E-07	m	1200.00
MAX. TRANS. IN Z-DIRECTION	-	9.731E-07	m	1320.00
MAX. TRANS. IN HORIZ. PLANE	-	1.536E-06	m	1440.00

RESULTANT TRANSLATIONS (m)

POINT NO. 3				POINT NO. 4		
-----				-----		
FREQ. RPM	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION
-----	-----	-----	-----	-----	-----	-----
960.00	8.71E-07	1.42E-07	3.93E-07	8.39E-07	1.52E-07	2.46E-07
980.00	8.61E-07	1.47E-07	4.00E-07	8.32E-07	1.57E-07	2.54E-07
1000.00	8.52E-07	1.51E-07	4.09E-07	8.26E-07	1.62E-07	2.63E-07
1020.00	8.45E-07	1.55E-07	4.18E-07	8.21E-07	1.66E-07	2.74E-07
1040.00	8.39E-07	1.59E-07	4.28E-07	8.18E-07	1.70E-07	2.87E-07
1060.00	8.36E-07	1.62E-07	4.40E-07	8.18E-07	1.73E-07	3.02E-07
1080.00	8.35E-07	1.65E-07	4.53E-07	8.20E-07	1.76E-07	3.18E-07
1100.00	8.38E-07	1.67E-07	4.67E-07	8.26E-07	1.78E-07	3.36E-07
1120.00	8.44E-07	1.68E-07	4.84E-07	8.36E-07	1.80E-07	3.56E-07
1140.00	8.55E-07	1.69E-07	5.02E-07	8.51E-07	1.81E-07	3.77E-07
1160.00	8.71E-07	1.70E-07	5.23E-07	8.71E-07	1.81E-07	4.00E-07
1180.00	8.93E-07	1.69E-07	5.46E-07	8.96E-07	1.81E-07	4.25E-07
1200.00	9.20E-07	1.68E-07	5.71E-07	9.28E-07	1.80E-07	4.52E-07
1220.00	9.54E-07	1.66E-07	5.99E-07	9.65E-07	1.77E-07	4.81E-07
1240.00	9.93E-07	1.63E-07	6.31E-07	1.01E-06	1.74E-07	5.12E-07
1260.00	1.04E-06	1.59E-07	6.66E-07	1.05E-06	1.70E-07	5.47E-07
1280.00	1.09E-06	1.55E-07	7.04E-07	1.11E-06	1.65E-07	5.85E-07
1300.00	1.14E-06	1.49E-07	7.47E-07	1.16E-06	1.59E-07	6.27E-07
1320.00	1.20E-06	1.42E-07	7.96E-07	1.22E-06	1.51E-07	6.74E-07
1340.00	1.25E-06	1.34E-07	8.50E-07	1.27E-06	1.43E-07	7.27E-07
1360.00	1.31E-06	1.25E-07	9.11E-07	1.32E-06	1.34E-07	7.88E-07
1380.00	1.36E-06	1.17E-07	9.80E-07	1.37E-06	1.24E-07	8.56E-07
1400.00	1.41E-06	1.08E-07	1.06E-06	1.41E-06	1.15E-07	9.35E-07
1420.00	1.46E-06	1.00E-07	1.15E-06	1.45E-06	1.07E-07	1.02E-06
1440.00	1.54E-06	9.42E-08	1.25E-06	1.52E-06	1.00E-07	1.13E-06

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 3

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	1.541E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	1.696E-07	m	1160.00
MAX. TRANS. IN Z-DIRECTION	-	1.250E-06	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.544E-06	m	1440.00

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 4

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	1.524E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	1.812E-07	m	1160.00
MAX. TRANS. IN Z-DIRECTION	-	1.125E-06	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.527E-06	m	1440.00

RESULTANT TRANSLATIONS (m)

POINT NO. 5				POINT NO. 6		
-----				-----		
FREQ. RPM	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION
-----	-----	-----	-----	-----	-----	-----
960.00	8.71E-07	1.43E-07	1.28E-06	8.39E-07	1.53E-07	1.12E-06
980.00	8.61E-07	1.44E-07	1.30E-06	8.32E-07	1.55E-07	1.14E-06
1000.00	8.52E-07	1.46E-07	1.33E-06	8.26E-07	1.57E-07	1.17E-06
1020.00	8.45E-07	1.47E-07	1.36E-06	8.21E-07	1.58E-07	1.20E-06
1040.00	8.39E-07	1.48E-07	1.39E-06	8.18E-07	1.59E-07	1.23E-06
1060.00	8.36E-07	1.48E-07	1.42E-06	8.18E-07	1.60E-07	1.26E-06
1080.00	8.35E-07	1.48E-07	1.46E-06	8.20E-07	1.60E-07	1.30E-06
1100.00	8.38E-07	1.48E-07	1.50E-06	8.26E-07	1.60E-07	1.35E-06
1120.00	8.44E-07	1.48E-07	1.55E-06	8.36E-07	1.60E-07	1.40E-06
1140.00	8.55E-07	1.47E-07	1.61E-06	8.51E-07	1.59E-07	1.46E-06
1160.00	8.71E-07	1.46E-07	1.67E-06	8.71E-07	1.58E-07	1.52E-06
1180.00	8.93E-07	1.44E-07	1.73E-06	8.96E-07	1.56E-07	1.59E-06
1200.00	9.20E-07	1.43E-07	1.81E-06	9.28E-07	1.54E-07	1.66E-06
1220.00	9.54E-07	1.40E-07	1.88E-06	9.65E-07	1.52E-07	1.74E-06
1240.00	9.93E-07	1.38E-07	1.97E-06	1.01E-06	1.49E-07	1.83E-06
1260.00	1.04E-06	1.35E-07	2.06E-06	1.05E-06	1.46E-07	1.92E-06
1280.00	1.09E-06	1.31E-07	2.15E-06	1.11E-06	1.43E-07	2.01E-06
1300.00	1.14E-06	1.28E-07	2.25E-06	1.16E-06	1.39E-07	2.11E-06
1320.00	1.20E-06	1.24E-07	2.34E-06	1.22E-06	1.34E-07	2.20E-06
1340.00	1.25E-06	1.20E-07	2.43E-06	1.27E-06	1.30E-07	2.29E-06
1360.00	1.31E-06	1.16E-07	2.52E-06	1.32E-06	1.26E-07	2.38E-06
1380.00	1.36E-06	1.14E-07	2.60E-06	1.37E-06	1.23E-07	2.46E-06
1400.00	1.41E-06	1.12E-07	2.67E-06	1.41E-06	1.21E-07	2.53E-06
1420.00	1.46E-06	1.11E-07	2.73E-06	1.45E-06	1.20E-07	2.60E-06
1440.00	1.54E-06	1.10E-07	2.79E-06	1.52E-06	1.19E-07	2.66E-06

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 5

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	1.541E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	1.484E-07	m	1080.00
MAX. TRANS. IN Z-DIRECTION	-	2.791E-06	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.545E-06	m	1440.00

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 6

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	1.524E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	1.601E-07	m	1080.00
MAX. TRANS. IN Z-DIRECTION	-	2.661E-06	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.528E-06	m	1440.00

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*           D Y N A 5   S I M U L A T I O N
*
*   Copyright (c) M. Novak, U.W.O. - 1999
*           RUN DATE - 2016/ 8/ 4
*           TIME    - 15:39:34
*           REVISION - 1999/04/23
*
*****

CG10001B LB 1200

                DATA ECHO
                *****

GRAVITATIONAL CONSTANT SET TO 9.81 m/s**2
*****

FREQUENCY UNITS : RPM
*****

LENGTH UNITS : m
*****

FORCE UNITS : N
*****

MASS UNITS : kg
*****

DAMPING SAFETY FACTOR - 2.00
*****

FOUNDATION TYPE - PILE
*****

RIGID CAP
*****

RECTANGULAR FOUNDATION - 11.500 m BY 6.000 m
*****

MASS CONSTANTS
*****

TOTAL MASS OF FOOTING ..... 4.1124E+05 kg
MASS MOMENT ABOUT X-AXIS ..... 1.7127E+06 kg.m**2
MASS MOMENT ABOUT Y-AXIS ..... 4.6937E+06 kg.m**2
MASS MOMENT ABOUT Z-AXIS ..... 5.3209E+06 kg.m**2
CROSS PRODUCT X-Y ..... 1.9995E+03 kg.m**2
CROSS PRODUCT X-Z ..... -2.3290E+04 kg.m**2
CROSS PRODUCT Y-Z ..... -1.8327E+04 kg.m**2

NUMBER OF LAYERS OR ELEMENTS - 3
*****

PILES ARE FIXED AT BUTT
*****

LOCATION OF PILES
*****

PILE      X      Y
NO.      LOCATION LOCATION
          m      m

```

1	-4.979	-2.241
2	-4.979	0.009
3	-4.979	2.259
4	-2.479	-2.241
5	-2.479	0.009
6	-2.479	2.259
7	0.021	-2.241
8	0.021	0.009
9	0.021	2.259
10	2.521	-2.241
11	2.521	0.009
12	2.521	2.259
13	5.021	-2.241
14	5.021	0.009
15	5.021	2.259

COORDINATES OF BASE CENTRE

X-COORD. OF BASE CENTRE REL. TO C.G. .. -0.021 m
Y-COORD. OF BASE CENTRE REL. TO C.G. .. -0.009 m
Z-COORD. OF BASE CENTRE REL. TO C.G. .. 1.590 m

PILE PROPERTIES

TOTAL LENGTH 20.250 m
UNIT WEIGHT 77000.000 N/m**3
STATIC LOAD ON PILE 5.580E+05 N
POISSON'S RATIO 0.300
MATERIAL DAMPING 0.010
COEFF. OF RIGIDITY 1.110
YOUNG'S MODULUS 2.000E+11 N/m**2

ELEMENT

LAYER NO.	LAYER DEPTH m	PILE RADIUS (X) m	PILE RADIUS (Y) m	CROSS SECTIONAL AREA m**2	MOMENT OF INERTIA ABOUT Y m**4	MOMENT OF INERTIA ABOUT X m**4	TORSIONAL CONSTANT m**4
1	0.250	0.303	0.303	0.021	9.31E-04	9.31E-04	1.86E-03
2	13.250	0.303	0.303	0.021	9.31E-04	9.31E-04	1.86E-03
3	6.750	0.303	0.303	0.021	9.31E-04	9.31E-04	1.86E-03

PILE TIP CONDITION : FLOATING (DEFAULT)

PILE-SOIL-PILE INTERACTION IS CONSIDERED

SOIL

SOIL CONSTANTS

LAYER NO.	SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
1	0.000	0.00	0.000	0.000
2	100.000	20000.00	0.350	0.040
3	200.000	21000.00	0.350	0.040

SOIL BELOW PILE

SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
-------------------------------	--------------------------	--------------------	---------------------

2.00E+02	21000.00	0.350	0.040
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EMBEDDED OPTION

NUMBER OF SIDE LAYERS - 1

LAYER NO.	THICKNESS OF LAYER m	SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
1	0.000	0.00	0.00	0.000	0.000

HARMONIC LOAD (QUADRATIC)

MAXIMUM FREQUENCY		1440.000	RPM
MINIMUM FREQUENCY		960.000	RPM
STEP FREQUENCY		20.000	RPM
FORCE IN X-DIRECTION		2.48E-01	kg.m
FORCE IN Y-DIRECTION		0.00E+00	kg.m
FORCE IN Z-DIRECTION		2.48E-01	kg.m
MOMENT ABOUT X-AXIS		-1.16E-01	kg.m**2
MOMENT ABOUT Y-AXIS		6.69E-01	kg.m**2
MOMENT ABOUT Z-AXIS		1.16E-01	kg.m**2

RESULTANT TRANSLATIONS REQUIRED AT 6 POINTS

COORDINATES OF POINTS :

POINT NO.	X m	Y m	Z m
1	-3.909	1.504	-2.342
2	-3.909	-0.991	-2.512
3	-0.229	1.504	-2.342
4	-0.229	-0.991	-2.512
5	3.451	1.504	-2.342
6	3.451	-0.991	-2.512

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*
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*   REVISION - 1999/04/23
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CG10001B LB 1200

RESULTS

FOOTING RESPONSE AMPLITUDES

FREQ. RPM	TRANS. X DIRECTION (m)	TRANS. Y DIRECTION (m)	VERTICAL DIRECTION (m)	ROT ABOUT X AXIS (rad)	ROT ABOUT Y AXIS (rad)	TORSIONAL Z AXIS (rad)
-----	-----	-----	-----	-----	-----	-----
960.00	6.60E-07	5.02E-08	3.61E-07	6.26E-08	9.12E-08	2.56E-08
980.00	6.54E-07	5.05E-08	3.69E-07	6.44E-08	9.34E-08	2.55E-08
1000.00	6.49E-07	5.07E-08	3.78E-07	6.61E-08	9.57E-08	2.53E-08
1020.00	6.44E-07	5.10E-08	3.89E-07	6.75E-08	9.81E-08	2.52E-08
1040.00	6.39E-07	5.12E-08	4.02E-07	6.88E-08	1.01E-07	2.51E-08
1060.00	6.34E-07	5.14E-08	4.16E-07	6.99E-08	1.04E-07	2.51E-08
1080.00	6.30E-07	5.17E-08	4.33E-07	7.08E-08	1.07E-07	2.51E-08
1100.00	6.27E-07	5.21E-08	4.51E-07	7.15E-08	1.10E-07	2.51E-08
1120.00	6.24E-07	5.25E-08	4.72E-07	7.19E-08	1.14E-07	2.52E-08
1140.00	6.23E-07	5.31E-08	4.94E-07	7.22E-08	1.18E-07	2.53E-08
1160.00	6.22E-07	5.39E-08	5.19E-07	7.22E-08	1.23E-07	2.55E-08
1180.00	6.22E-07	5.48E-08	5.46E-07	7.20E-08	1.28E-07	2.58E-08
1200.00	6.24E-07	5.59E-08	5.76E-07	7.16E-08	1.33E-07	2.60E-08
1220.00	6.28E-07	5.74E-08	6.09E-07	7.10E-08	1.39E-07	2.64E-08
1240.00	6.34E-07	5.91E-08	6.45E-07	7.01E-08	1.45E-07	2.67E-08
1260.00	6.42E-07	6.11E-08	6.83E-07	6.89E-08	1.51E-07	2.72E-08
1280.00	6.54E-07	6.35E-08	7.26E-07	6.74E-08	1.58E-07	2.77E-08
1300.00	6.69E-07	6.62E-08	7.73E-07	6.56E-08	1.63E-07	2.82E-08
1320.00	6.90E-07	6.92E-08	8.24E-07	6.36E-08	1.68E-07	2.88E-08
1340.00	7.16E-07	7.26E-08	8.81E-07	6.14E-08	1.72E-07	2.95E-08
1360.00	7.48E-07	7.60E-08	9.44E-07	5.91E-08	1.75E-07	3.01E-08
1380.00	7.87E-07	7.96E-08	1.01E-06	5.68E-08	1.75E-07	3.07E-08
1400.00	8.34E-07	8.30E-08	1.09E-06	5.49E-08	1.72E-07	3.13E-08
1420.00	8.85E-07	8.62E-08	1.18E-06	5.34E-08	1.68E-07	3.17E-08
1440.00	9.40E-07	8.90E-08	1.28E-06	5.20E-08	1.64E-07	3.20E-08

MAXIMA OF FOOTING RESPONSE AMPLITUDES

				FREQ. RPM
-----	-----	-----	-----	-----
MAX. TRANS. IN X-DIRECTION	-	9.403E-07	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	8.901E-08	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	1.281E-06	m	1440.00
MAX. ROT. ABOUT X AXIS	-	7.222E-08	rad	1160.00
MAX. ROT. ABOUT Y AXIS	-	1.748E-07	rad	1380.00
MAX. ROT. ABOUT Z AXIS	-	3.198E-08	rad	1440.00

RESULTANT TRANSLATIONS (m)

POINT NO. 1				POINT NO. 2		
-----				-----		
FREQ.	HORIZ. X	HORIZ. Y	VERTICAL	HORIZ. X	HORIZ. Y	VERTICAL
RPM	DIRECTION	DIRECTION	DIRECTION	DIRECTION	DIRECTION	DIRECTION

960.00	7.14E-07	1.95E-07	1.59E-07	6.56E-07	2.02E-07	1.55E-07
980.00	7.04E-07	2.00E-07	1.59E-07	6.46E-07	2.08E-07	1.65E-07
1000.00	6.94E-07	2.05E-07	1.58E-07	6.36E-07	2.13E-07	1.74E-07
1020.00	6.84E-07	2.10E-07	1.57E-07	6.27E-07	2.18E-07	1.81E-07
1040.00	6.75E-07	2.15E-07	1.56E-07	6.18E-07	2.23E-07	1.88E-07
1060.00	6.66E-07	2.19E-07	1.56E-07	6.10E-07	2.28E-07	1.94E-07
1080.00	6.59E-07	2.23E-07	1.57E-07	6.03E-07	2.32E-07	2.00E-07
1100.00	6.53E-07	2.26E-07	1.59E-07	5.97E-07	2.35E-07	2.06E-07
1120.00	6.48E-07	2.29E-07	1.62E-07	5.93E-07	2.38E-07	2.11E-07
1140.00	6.46E-07	2.31E-07	1.67E-07	5.92E-07	2.41E-07	2.17E-07
1160.00	6.45E-07	2.33E-07	1.74E-07	5.92E-07	2.42E-07	2.24E-07
1180.00	6.48E-07	2.34E-07	1.84E-07	5.95E-07	2.43E-07	2.32E-07
1200.00	6.54E-07	2.34E-07	1.96E-07	6.02E-07	2.43E-07	2.41E-07
1220.00	6.63E-07	2.34E-07	2.11E-07	6.12E-07	2.43E-07	2.52E-07
1240.00	6.76E-07	2.32E-07	2.31E-07	6.27E-07	2.41E-07	2.66E-07
1260.00	6.94E-07	2.30E-07	2.55E-07	6.45E-07	2.39E-07	2.82E-07
1280.00	7.17E-07	2.27E-07	2.86E-07	6.69E-07	2.35E-07	3.02E-07
1300.00	7.45E-07	2.23E-07	3.23E-07	6.97E-07	2.30E-07	3.27E-07
1320.00	7.78E-07	2.18E-07	3.69E-07	7.30E-07	2.25E-07	3.59E-07
1340.00	8.17E-07	2.13E-07	4.25E-07	7.68E-07	2.19E-07	3.99E-07
1360.00	8.62E-07	2.07E-07	4.93E-07	8.11E-07	2.12E-07	4.50E-07
1380.00	9.12E-07	2.02E-07	5.76E-07	8.59E-07	2.05E-07	5.15E-07
1400.00	9.68E-07	1.96E-07	6.75E-07	9.13E-07	1.98E-07	5.98E-07
1420.00	1.03E-06	1.90E-07	7.91E-07	9.72E-07	1.92E-07	7.01E-07
1440.00	1.10E-06	1.85E-07	9.27E-07	1.05E-06	1.86E-07	8.29E-07

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 1

				FREQ.
				RPM

MAX. TRANS. IN X-DIRECTION	-	1.105E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	2.342E-07	m	1200.00
MAX. TRANS. IN Z-DIRECTION	-	9.274E-07	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.120E-06	m	1440.00

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 2

				FREQ.
				RPM

MAX. TRANS. IN X-DIRECTION	-	1.045E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	2.435E-07	m	1200.00
MAX. TRANS. IN Z-DIRECTION	-	8.285E-07	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.062E-06	m	1440.00

RESULTANT TRANSLATIONS (m)

POINT NO. 3				POINT NO. 4		
-----				-----		
FREQ. RPM	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION
-----	-----	-----	-----	-----	-----	-----
960.00	7.14E-07	1.42E-07	4.31E-07	6.56E-07	1.52E-07	2.83E-07
980.00	7.04E-07	1.46E-07	4.40E-07	6.46E-07	1.57E-07	2.91E-07
1000.00	6.94E-07	1.51E-07	4.49E-07	6.36E-07	1.61E-07	3.01E-07
1020.00	6.84E-07	1.55E-07	4.59E-07	6.27E-07	1.66E-07	3.13E-07
1040.00	6.75E-07	1.58E-07	4.70E-07	6.18E-07	1.69E-07	3.26E-07
1060.00	6.66E-07	1.62E-07	4.83E-07	6.10E-07	1.73E-07	3.42E-07
1080.00	6.59E-07	1.64E-07	4.97E-07	6.03E-07	1.76E-07	3.59E-07
1100.00	6.53E-07	1.66E-07	5.13E-07	5.97E-07	1.78E-07	3.78E-07
1120.00	6.48E-07	1.68E-07	5.31E-07	5.93E-07	1.80E-07	3.99E-07
1140.00	6.46E-07	1.69E-07	5.51E-07	5.92E-07	1.80E-07	4.22E-07
1160.00	6.45E-07	1.69E-07	5.73E-07	5.92E-07	1.81E-07	4.47E-07
1180.00	6.48E-07	1.69E-07	5.98E-07	5.95E-07	1.80E-07	4.74E-07
1200.00	6.54E-07	1.68E-07	6.26E-07	6.02E-07	1.79E-07	5.03E-07
1220.00	6.63E-07	1.66E-07	6.56E-07	6.12E-07	1.77E-07	5.34E-07
1240.00	6.76E-07	1.63E-07	6.90E-07	6.27E-07	1.74E-07	5.68E-07
1260.00	6.94E-07	1.59E-07	7.27E-07	6.45E-07	1.70E-07	6.06E-07
1280.00	7.17E-07	1.54E-07	7.69E-07	6.69E-07	1.64E-07	6.46E-07
1300.00	7.45E-07	1.48E-07	8.15E-07	6.97E-07	1.58E-07	6.91E-07
1320.00	7.78E-07	1.41E-07	8.65E-07	7.30E-07	1.51E-07	7.41E-07
1340.00	8.17E-07	1.33E-07	9.22E-07	7.68E-07	1.42E-07	7.97E-07
1360.00	8.62E-07	1.25E-07	9.85E-07	8.11E-07	1.33E-07	8.60E-07
1380.00	9.12E-07	1.16E-07	1.06E-06	8.59E-07	1.24E-07	9.30E-07
1400.00	9.68E-07	1.08E-07	1.14E-06	9.13E-07	1.15E-07	1.01E-06
1420.00	1.03E-06	1.00E-07	1.23E-06	9.72E-07	1.07E-07	1.10E-06
1440.00	1.10E-06	9.39E-08	1.33E-06	1.05E-06	9.97E-08	1.20E-06

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 3

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	1.105E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	1.692E-07	m	1160.00
MAX. TRANS. IN Z-DIRECTION	-	1.327E-06	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.109E-06	m	1440.00

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 4

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	1.045E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	1.808E-07	m	1160.00
MAX. TRANS. IN Z-DIRECTION	-	1.202E-06	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.050E-06	m	1440.00

RESULTANT TRANSLATIONS (m)

POINT NO. 5				POINT NO. 6		
-----				-----		
FREQ. RPM	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION
-----	-----	-----	-----	-----	-----	-----
960.00	7.14E-07	1.41E-07	7.56E-07	6.56E-07	1.51E-07	6.02E-07
980.00	7.04E-07	1.42E-07	7.73E-07	6.46E-07	1.53E-07	6.16E-07
1000.00	6.94E-07	1.44E-07	7.91E-07	6.36E-07	1.55E-07	6.32E-07
1020.00	6.84E-07	1.45E-07	8.11E-07	6.27E-07	1.56E-07	6.51E-07
1040.00	6.75E-07	1.46E-07	8.32E-07	6.18E-07	1.57E-07	6.73E-07
1060.00	6.66E-07	1.46E-07	8.56E-07	6.10E-07	1.58E-07	6.98E-07
1080.00	6.59E-07	1.46E-07	8.82E-07	6.03E-07	1.58E-07	7.26E-07
1100.00	6.53E-07	1.46E-07	9.11E-07	5.97E-07	1.58E-07	7.57E-07
1120.00	6.48E-07	1.46E-07	9.43E-07	5.93E-07	1.58E-07	7.92E-07
1140.00	6.46E-07	1.45E-07	9.79E-07	5.92E-07	1.57E-07	8.30E-07
1160.00	6.45E-07	1.44E-07	1.02E-06	5.92E-07	1.56E-07	8.71E-07
1180.00	6.48E-07	1.42E-07	1.06E-06	5.95E-07	1.54E-07	9.16E-07
1200.00	6.54E-07	1.41E-07	1.11E-06	6.02E-07	1.52E-07	9.64E-07
1220.00	6.63E-07	1.38E-07	1.16E-06	6.12E-07	1.50E-07	1.02E-06
1240.00	6.76E-07	1.36E-07	1.21E-06	6.27E-07	1.47E-07	1.07E-06
1260.00	6.94E-07	1.32E-07	1.27E-06	6.45E-07	1.44E-07	1.13E-06
1280.00	7.17E-07	1.29E-07	1.33E-06	6.69E-07	1.40E-07	1.19E-06
1300.00	7.45E-07	1.25E-07	1.39E-06	6.97E-07	1.36E-07	1.25E-06
1320.00	7.78E-07	1.21E-07	1.46E-06	7.30E-07	1.31E-07	1.32E-06
1340.00	8.17E-07	1.17E-07	1.52E-06	7.68E-07	1.26E-07	1.38E-06
1360.00	8.62E-07	1.13E-07	1.59E-06	8.11E-07	1.22E-07	1.45E-06
1380.00	9.12E-07	1.09E-07	1.65E-06	8.59E-07	1.19E-07	1.51E-06
1400.00	9.68E-07	1.07E-07	1.71E-06	9.13E-07	1.16E-07	1.58E-06
1420.00	1.03E-06	1.07E-07	1.77E-06	9.72E-07	1.15E-07	1.64E-06
1440.00	1.10E-06	1.05E-07	1.84E-06	1.05E-06	1.14E-07	1.71E-06

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 5

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	1.105E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	1.465E-07	m	1080.00
MAX. TRANS. IN Z-DIRECTION	-	1.843E-06	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.110E-06	m	1440.00

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 6

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	1.045E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	1.582E-07	m	1080.00
MAX. TRANS. IN Z-DIRECTION	-	1.713E-06	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.051E-06	m	1440.00


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*           D Y N A 5   S I M U L A T I O N
*
*   Copyright (c)  M. Novak,  U.W.O.  - 1999
*           RUN DATE - 2016/ 8/ 4
*           TIME    - 16:32:45
*           REVISION - 1999/04/23
*
*****

CG10001C LB 1200

                DATA ECHO
                *****

GRAVITATIONAL CONSTANT SET TO 9.81 m/s**2
*****

FREQUENCY UNITS : RPM
*****

LENGTH UNITS : m
*****

FORCE UNITS : N
*****

MASS UNITS : kg
*****

DAMPING SAFETY FACTOR - 2.00
*****

FOUNDATION TYPE - PILE
*****

RIGID CAP
*****

RECTANGULAR FOUNDATION - 11.500 m BY 6.000 m
*****

MASS CONSTANTS
*****

TOTAL MASS OF FOOTING ..... 4.1124E+05 kg
MASS MOMENT ABOUT X-AXIS ..... 1.7127E+06 kg.m**2
MASS MOMENT ABOUT Y-AXIS ..... 4.6937E+06 kg.m**2
MASS MOMENT ABOUT Z-AXIS ..... 5.3209E+06 kg.m**2
CROSS PRODUCT X-Y ..... 1.9995E+03 kg.m**2
CROSS PRODUCT X-Z ..... -2.3290E+04 kg.m**2
CROSS PRODUCT Y-Z ..... -1.8327E+04 kg.m**2

NUMBER OF LAYERS OR ELEMENTS - 3
*****

FILES ARE FIXED AT BUTT
*****

LOCATION OF PILES
*****

PILE      X      Y
NO.      LOCATION LOCATION
          m      m

```

1	-4.979	-2.241
2	-4.979	0.009
3	-4.979	2.259
4	-2.479	-2.241
5	-2.479	0.009
6	-2.479	2.259
7	0.021	-2.241
8	0.021	0.009
9	0.021	2.259
10	2.521	-2.241
11	2.521	0.009
12	2.521	2.259
13	5.021	-2.241
14	5.021	0.009
15	5.021	2.259

COORDINATES OF BASE CENTRE

X-COORD. OF BASE CENTRE REL. TO C.G. .. -0.021 m
Y-COORD. OF BASE CENTRE REL. TO C.G. .. -0.009 m
Z-COORD. OF BASE CENTRE REL. TO C.G. .. 1.590 m

PILE PROPERTIES

TOTAL LENGTH 20.250 m
UNIT WEIGHT 77000.000 N/m**3
STATIC LOAD ON PILE 5.580E+05 N
POISSON'S RATIO 0.300
MATERIAL DAMPING 0.010
COEFF. OF RIGIDITY 1.110
YOUNG'S MODULUS 2.000E+11 N/m**2

ELEMENT

LAYER NO.	LAYER DEPTH m	PILE RADIUS (X) m	PILE RADIUS (Y) m	CROSS SECTIONAL AREA m**2	MOMENT OF INERTIA ABOUT Y m**4	MOMENT OF INERTIA ABOUT X m**4	TORSIONAL CONSTANT m**4
1	0.250	0.303	0.303	0.021	9.31E-04	9.31E-04	1.86E-03
2	13.250	0.303	0.303	0.021	9.31E-04	9.31E-04	1.86E-03
3	6.750	0.303	0.303	0.021	9.31E-04	9.31E-04	1.86E-03

PILE TIP CONDITION : FLOATING (DEFAULT)

PILE-SOIL-PILE INTERACTION IS CONSIDERED

SOIL

SOIL CONSTANTS

LAYER NO.	SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
1	0.000	0.00	0.000	0.000
2	100.000	20000.00	0.350	0.040
3	200.000	21000.00	0.350	0.040

SOIL BELOW PILE

SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
2.00E+02	21000.00	0.350	0.040

EMBEDDED OPTION

NUMBER OF SIDE LAYERS - 1

LAYER NO.	THICKNESS OF LAYER m	SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
1	0.000	0.00	0.00	0.000	0.000

HARMONIC LOAD (QUADRATIC)

MAXIMUM FREQUENCY		1440.000	RPM
MINIMUM FREQUENCY		960.000	RPM
STEP FREQUENCY		20.000	RPM
FORCE IN X-DIRECTION		2.48E-01	kg.m
FORCE IN Y-DIRECTION		0.00E+00	kg.m
FORCE IN Z-DIRECTION		2.48E-01	kg.m
MOMENT ABOUT X-AXIS		-1.16E-01	kg.m**2
MOMENT ABOUT Y-AXIS		6.69E-01	kg.m**2
MOMENT ABOUT Z-AXIS		1.16E-01	kg.m**2

RESULTANT TRANSLATIONS REQUIRED AT 6 POINTS

COORDINATES OF POINTS :

POINT NO.	X m	Y m	Z m
1	-3.909	1.504	-2.342
2	-3.909	-0.991	-2.512
3	-0.229	1.504	-2.342
4	-0.229	-0.991	-2.512
5	3.451	1.504	-2.342
6	3.451	-0.991	-2.512

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*
*           D Y N A 5   S I M U L A T I O N
*
*   Copyright (c) M. Novak, U.W.O. - 1999
*   RUN DATE - 2016/ 8/ 4
*   TIME      - 16:32:45
*   REVISION  - 1999/04/23
*
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CG10001C LB 1200

RESULTS

FOOTING RESPONSE AMPLITUDES

FREQ. RPM	TRANS. X DIRECTION (m)	TRANS. Y DIRECTION (m)	VERTICAL DIRECTION (m)	ROT ABOUT X AXIS (rad)	ROT ABOUT Y AXIS (rad)	TORSIONAL Z AXIS (rad)
960.00	6.60E-07	5.02E-08	3.61E-07	6.26E-08	9.12E-08	2.56E-08
980.00	6.54E-07	5.05E-08	3.69E-07	6.44E-08	9.34E-08	2.55E-08
1000.00	6.49E-07	5.07E-08	3.78E-07	6.61E-08	9.57E-08	2.53E-08
1020.00	6.44E-07	5.10E-08	3.89E-07	6.75E-08	9.81E-08	2.52E-08
1040.00	6.39E-07	5.12E-08	4.02E-07	6.88E-08	1.01E-07	2.51E-08
1060.00	6.34E-07	5.14E-08	4.16E-07	6.99E-08	1.04E-07	2.51E-08
1080.00	6.30E-07	5.17E-08	4.33E-07	7.08E-08	1.07E-07	2.51E-08
1100.00	6.27E-07	5.21E-08	4.51E-07	7.15E-08	1.10E-07	2.51E-08
1120.00	6.24E-07	5.25E-08	4.72E-07	7.19E-08	1.14E-07	2.52E-08
1140.00	6.23E-07	5.31E-08	4.94E-07	7.22E-08	1.18E-07	2.53E-08
1160.00	6.22E-07	5.39E-08	5.19E-07	7.22E-08	1.23E-07	2.55E-08
1180.00	6.22E-07	5.48E-08	5.46E-07	7.20E-08	1.28E-07	2.58E-08
1200.00	6.24E-07	5.59E-08	5.76E-07	7.16E-08	1.33E-07	2.60E-08
1220.00	6.28E-07	5.74E-08	6.09E-07	7.10E-08	1.39E-07	2.64E-08
1240.00	6.34E-07	5.91E-08	6.45E-07	7.01E-08	1.45E-07	2.67E-08
1260.00	6.42E-07	6.11E-08	6.83E-07	6.89E-08	1.51E-07	2.72E-08
1280.00	6.54E-07	6.35E-08	7.26E-07	6.74E-08	1.58E-07	2.77E-08
1300.00	6.69E-07	6.62E-08	7.73E-07	6.56E-08	1.63E-07	2.82E-08
1320.00	6.90E-07	6.92E-08	8.24E-07	6.36E-08	1.68E-07	2.88E-08
1340.00	7.16E-07	7.26E-08	8.81E-07	6.14E-08	1.72E-07	2.95E-08
1360.00	7.48E-07	7.60E-08	9.44E-07	5.91E-08	1.75E-07	3.01E-08
1380.00	7.87E-07	7.96E-08	1.01E-06	5.68E-08	1.75E-07	3.07E-08
1400.00	8.34E-07	8.30E-08	1.09E-06	5.49E-08	1.72E-07	3.13E-08
1420.00	8.85E-07	8.62E-08	1.18E-06	5.34E-08	1.68E-07	3.17E-08
1440.00	9.40E-07	8.90E-08	1.28E-06	5.20E-08	1.64E-07	3.20E-08

MAXIMA OF FOOTING RESPONSE AMPLITUDES

				FREQ. RPM
MAX. TRANS. IN X-DIRECTION	-	9.403E-07	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	8.901E-08	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	1.281E-06	m	1440.00
MAX. ROT. ABOUT X AXIS	-	7.222E-08	rad	1160.00
MAX. ROT. ABOUT Y AXIS	-	1.748E-07	rad	1380.00
MAX. ROT. ABOUT Z AXIS	-	3.198E-08	rad	1440.00

RESULTANT TRANSLATIONS (m)

POINT NO. 1				POINT NO. 2		
-----				-----		
FREQ.	HORIZ. X	HORIZ. Y	VERTICAL	HORIZ. X	HORIZ. Y	VERTICAL
RPM	DIRECTION	DIRECTION	DIRECTION	DIRECTION	DIRECTION	DIRECTION

960.00	7.14E-07	1.95E-07	1.59E-07	6.56E-07	2.02E-07	1.55E-07
980.00	7.04E-07	2.00E-07	1.59E-07	6.46E-07	2.08E-07	1.65E-07
1000.00	6.94E-07	2.05E-07	1.58E-07	6.36E-07	2.13E-07	1.74E-07
1020.00	6.84E-07	2.10E-07	1.57E-07	6.27E-07	2.18E-07	1.81E-07
1040.00	6.75E-07	2.15E-07	1.56E-07	6.18E-07	2.23E-07	1.88E-07
1060.00	6.66E-07	2.19E-07	1.56E-07	6.10E-07	2.28E-07	1.94E-07
1080.00	6.59E-07	2.23E-07	1.57E-07	6.03E-07	2.32E-07	2.00E-07
1100.00	6.53E-07	2.26E-07	1.59E-07	5.97E-07	2.35E-07	2.06E-07
1120.00	6.48E-07	2.29E-07	1.62E-07	5.93E-07	2.38E-07	2.11E-07
1140.00	6.46E-07	2.31E-07	1.67E-07	5.92E-07	2.41E-07	2.17E-07
1160.00	6.45E-07	2.33E-07	1.74E-07	5.92E-07	2.42E-07	2.24E-07
1180.00	6.48E-07	2.34E-07	1.84E-07	5.95E-07	2.43E-07	2.32E-07
1200.00	6.54E-07	2.34E-07	1.96E-07	6.02E-07	2.43E-07	2.41E-07
1220.00	6.63E-07	2.34E-07	2.11E-07	6.12E-07	2.43E-07	2.52E-07
1240.00	6.76E-07	2.32E-07	2.31E-07	6.27E-07	2.41E-07	2.66E-07
1260.00	6.94E-07	2.30E-07	2.55E-07	6.45E-07	2.39E-07	2.82E-07
1280.00	7.17E-07	2.27E-07	2.86E-07	6.69E-07	2.35E-07	3.02E-07
1300.00	7.45E-07	2.23E-07	3.23E-07	6.97E-07	2.30E-07	3.27E-07
1320.00	7.78E-07	2.18E-07	3.69E-07	7.30E-07	2.25E-07	3.59E-07
1340.00	8.17E-07	2.13E-07	4.25E-07	7.68E-07	2.19E-07	3.99E-07
1360.00	8.62E-07	2.07E-07	4.93E-07	8.11E-07	2.12E-07	4.50E-07
1380.00	9.12E-07	2.02E-07	5.76E-07	8.59E-07	2.05E-07	5.15E-07
1400.00	9.68E-07	1.96E-07	6.75E-07	9.13E-07	1.98E-07	5.98E-07
1420.00	1.03E-06	1.90E-07	7.91E-07	9.72E-07	1.92E-07	7.01E-07
1440.00	1.10E-06	1.85E-07	9.27E-07	1.05E-06	1.86E-07	8.29E-07

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 1

				FREQ.
				RPM

MAX. TRANS. IN X-DIRECTION	-	1.105E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	2.342E-07	m	1200.00
MAX. TRANS. IN Z-DIRECTION	-	9.274E-07	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.120E-06	m	1440.00

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 2

				FREQ.
				RPM

MAX. TRANS. IN X-DIRECTION	-	1.045E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	2.435E-07	m	1200.00
MAX. TRANS. IN Z-DIRECTION	-	8.285E-07	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.062E-06	m	1440.00

RESULTANT TRANSLATIONS (m)

POINT NO. 3				POINT NO. 4		
-----				-----		
FREQ. RPM	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION
-----	-----	-----	-----	-----	-----	-----
960.00	7.14E-07	1.42E-07	4.31E-07	6.56E-07	1.52E-07	2.83E-07
980.00	7.04E-07	1.46E-07	4.40E-07	6.46E-07	1.57E-07	2.91E-07
1000.00	6.94E-07	1.51E-07	4.49E-07	6.36E-07	1.61E-07	3.01E-07
1020.00	6.84E-07	1.55E-07	4.59E-07	6.27E-07	1.66E-07	3.13E-07
1040.00	6.75E-07	1.58E-07	4.70E-07	6.18E-07	1.69E-07	3.26E-07
1060.00	6.66E-07	1.62E-07	4.83E-07	6.10E-07	1.73E-07	3.42E-07
1080.00	6.59E-07	1.64E-07	4.97E-07	6.03E-07	1.76E-07	3.59E-07
1100.00	6.53E-07	1.66E-07	5.13E-07	5.97E-07	1.78E-07	3.78E-07
1120.00	6.48E-07	1.68E-07	5.31E-07	5.93E-07	1.80E-07	3.99E-07
1140.00	6.46E-07	1.69E-07	5.51E-07	5.92E-07	1.80E-07	4.22E-07
1160.00	6.45E-07	1.69E-07	5.73E-07	5.92E-07	1.81E-07	4.47E-07
1180.00	6.48E-07	1.69E-07	5.98E-07	5.95E-07	1.80E-07	4.74E-07
1200.00	6.54E-07	1.68E-07	6.26E-07	6.02E-07	1.79E-07	5.03E-07
1220.00	6.63E-07	1.66E-07	6.56E-07	6.12E-07	1.77E-07	5.34E-07
1240.00	6.76E-07	1.63E-07	6.90E-07	6.27E-07	1.74E-07	5.68E-07
1260.00	6.94E-07	1.59E-07	7.27E-07	6.45E-07	1.70E-07	6.06E-07
1280.00	7.17E-07	1.54E-07	7.69E-07	6.69E-07	1.64E-07	6.46E-07
1300.00	7.45E-07	1.48E-07	8.15E-07	6.97E-07	1.58E-07	6.91E-07
1320.00	7.78E-07	1.41E-07	8.65E-07	7.30E-07	1.51E-07	7.41E-07
1340.00	8.17E-07	1.33E-07	9.22E-07	7.68E-07	1.42E-07	7.97E-07
1360.00	8.62E-07	1.25E-07	9.85E-07	8.11E-07	1.33E-07	8.60E-07
1380.00	9.12E-07	1.16E-07	1.06E-06	8.59E-07	1.24E-07	9.30E-07
1400.00	9.68E-07	1.08E-07	1.14E-06	9.13E-07	1.15E-07	1.01E-06
1420.00	1.03E-06	1.00E-07	1.23E-06	9.72E-07	1.07E-07	1.10E-06
1440.00	1.10E-06	9.39E-08	1.33E-06	1.05E-06	9.97E-08	1.20E-06

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 3

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	1.105E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	1.692E-07	m	1160.00
MAX. TRANS. IN Z-DIRECTION	-	1.327E-06	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.109E-06	m	1440.00

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 4

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	1.045E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	1.808E-07	m	1160.00
MAX. TRANS. IN Z-DIRECTION	-	1.202E-06	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.050E-06	m	1440.00

RESULTANT TRANSLATIONS (m)

POINT NO. 5				POINT NO. 6		
-----				-----		
FREQ. RPM	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION
-----	-----	-----	-----	-----	-----	-----
960.00	7.14E-07	1.41E-07	7.56E-07	6.56E-07	1.51E-07	6.02E-07
980.00	7.04E-07	1.42E-07	7.73E-07	6.46E-07	1.53E-07	6.16E-07
1000.00	6.94E-07	1.44E-07	7.91E-07	6.36E-07	1.55E-07	6.32E-07
1020.00	6.84E-07	1.45E-07	8.11E-07	6.27E-07	1.56E-07	6.51E-07
1040.00	6.75E-07	1.46E-07	8.32E-07	6.18E-07	1.57E-07	6.73E-07
1060.00	6.66E-07	1.46E-07	8.56E-07	6.10E-07	1.58E-07	6.98E-07
1080.00	6.59E-07	1.46E-07	8.82E-07	6.03E-07	1.58E-07	7.26E-07
1100.00	6.53E-07	1.46E-07	9.11E-07	5.97E-07	1.58E-07	7.57E-07
1120.00	6.48E-07	1.46E-07	9.43E-07	5.93E-07	1.58E-07	7.92E-07
1140.00	6.46E-07	1.45E-07	9.79E-07	5.92E-07	1.57E-07	8.30E-07
1160.00	6.45E-07	1.44E-07	1.02E-06	5.92E-07	1.56E-07	8.71E-07
1180.00	6.48E-07	1.42E-07	1.06E-06	5.95E-07	1.54E-07	9.16E-07
1200.00	6.54E-07	1.41E-07	1.11E-06	6.02E-07	1.52E-07	9.64E-07
1220.00	6.63E-07	1.38E-07	1.16E-06	6.12E-07	1.50E-07	1.02E-06
1240.00	6.76E-07	1.36E-07	1.21E-06	6.27E-07	1.47E-07	1.07E-06
1260.00	6.94E-07	1.32E-07	1.27E-06	6.45E-07	1.44E-07	1.13E-06
1280.00	7.17E-07	1.29E-07	1.33E-06	6.69E-07	1.40E-07	1.19E-06
1300.00	7.45E-07	1.25E-07	1.39E-06	6.97E-07	1.36E-07	1.25E-06
1320.00	7.78E-07	1.21E-07	1.46E-06	7.30E-07	1.31E-07	1.32E-06
1340.00	8.17E-07	1.17E-07	1.52E-06	7.68E-07	1.26E-07	1.38E-06
1360.00	8.62E-07	1.13E-07	1.59E-06	8.11E-07	1.22E-07	1.45E-06
1380.00	9.12E-07	1.09E-07	1.65E-06	8.59E-07	1.19E-07	1.51E-06
1400.00	9.68E-07	1.07E-07	1.71E-06	9.13E-07	1.16E-07	1.58E-06
1420.00	1.03E-06	1.07E-07	1.77E-06	9.72E-07	1.15E-07	1.64E-06
1440.00	1.10E-06	1.05E-07	1.84E-06	1.05E-06	1.14E-07	1.71E-06

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 5

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	1.105E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	1.465E-07	m	1080.00
MAX. TRANS. IN Z-DIRECTION	-	1.843E-06	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.110E-06	m	1440.00

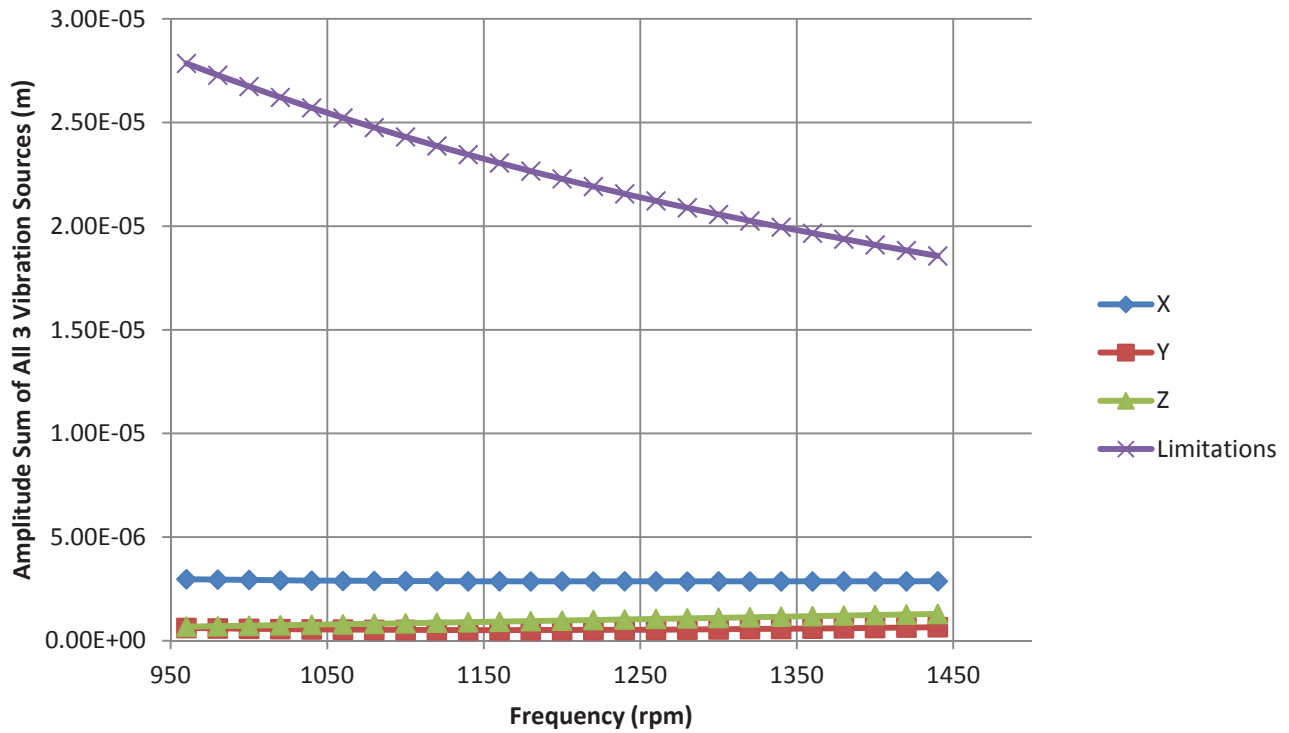
MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 6

				FREQ. RPM

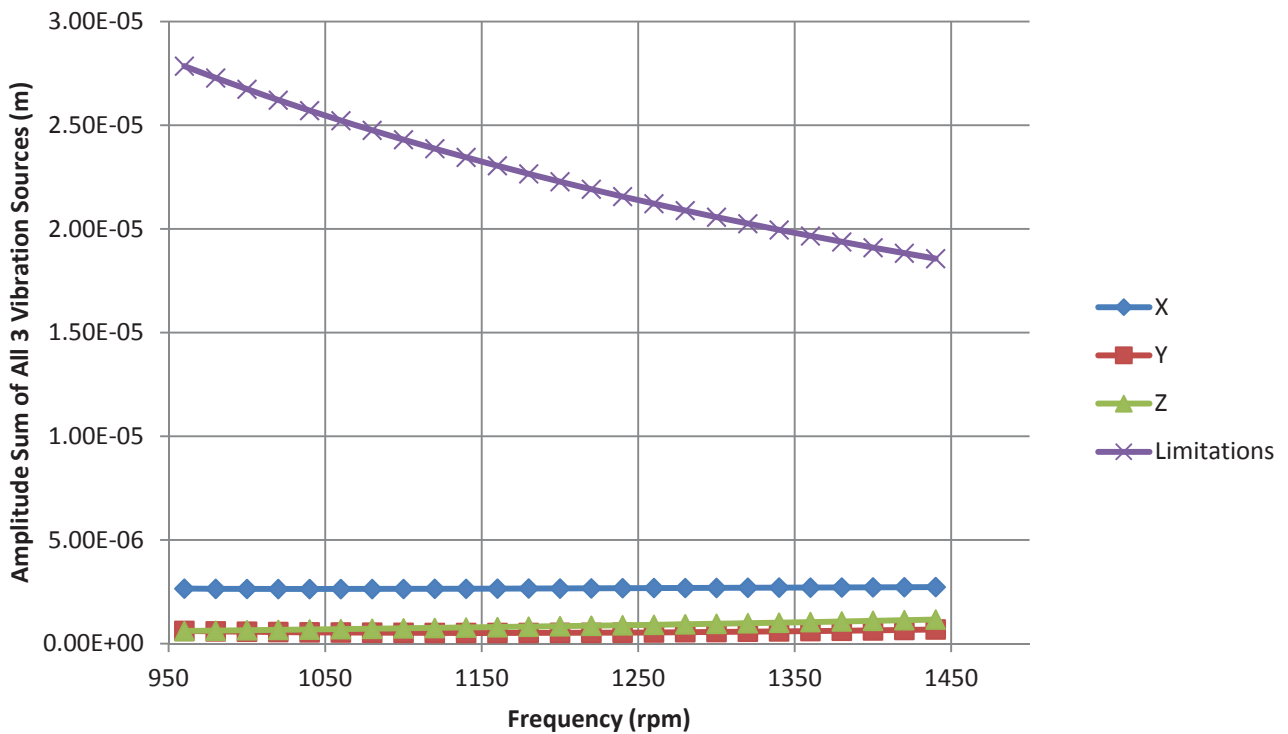
MAX. TRANS. IN X-DIRECTION	-	1.045E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	1.582E-07	m	1080.00
MAX. TRANS. IN Z-DIRECTION	-	1.713E-06	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.051E-06	m	1440.00

13.6 DYNA 5 Upper Bound Input & Output for Combined Foundation

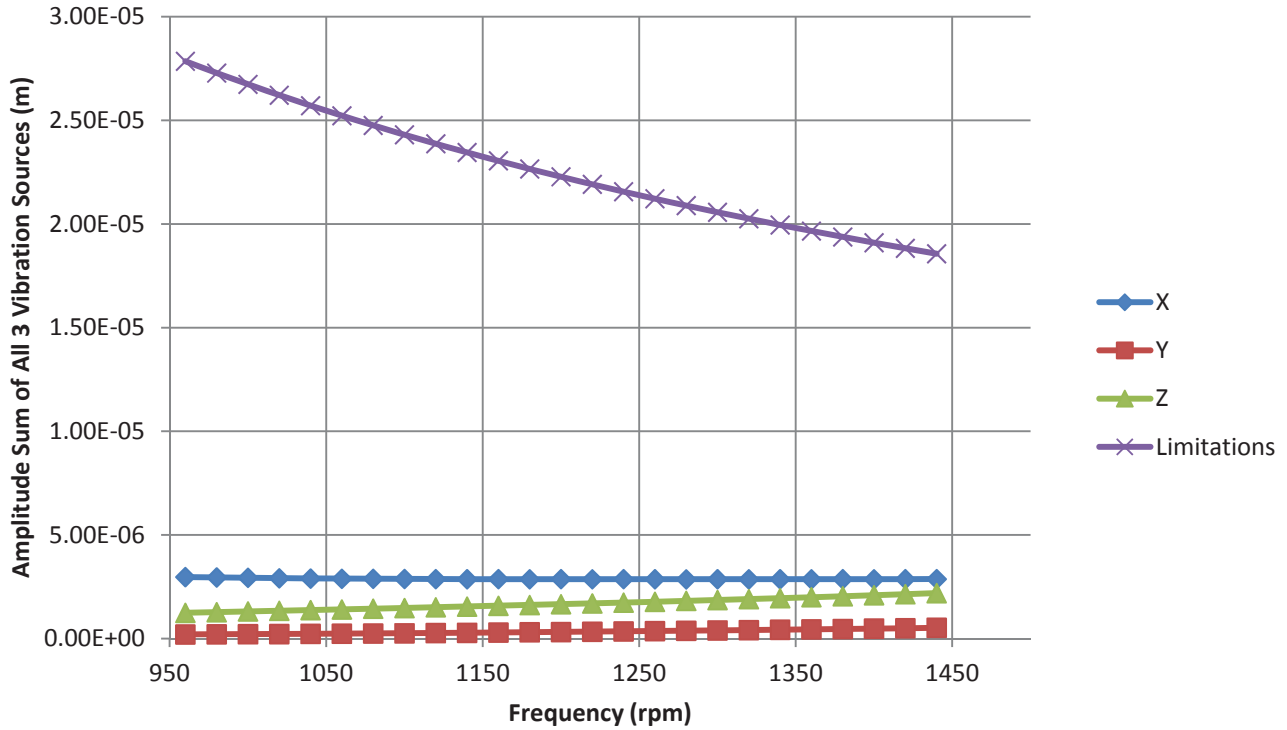
Translations at CG10001 Compressor A (UB)



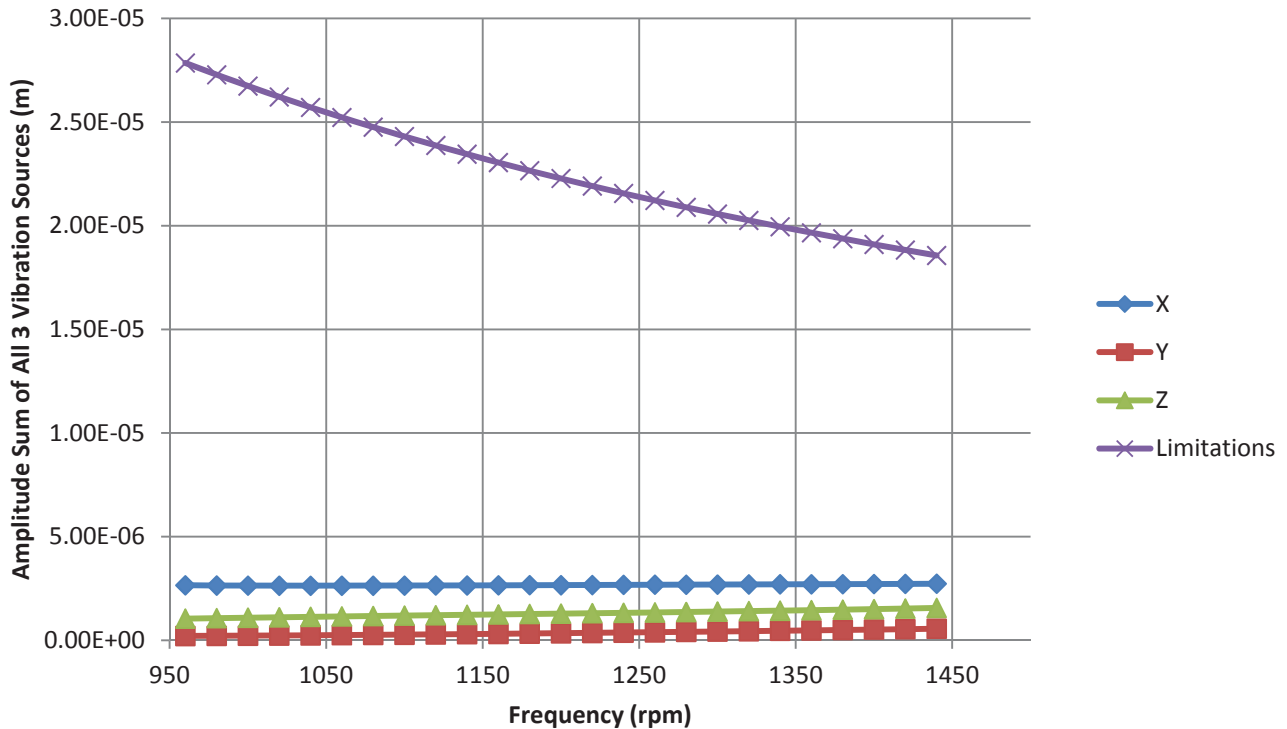
Translations at CG10001 Motor A (UB)



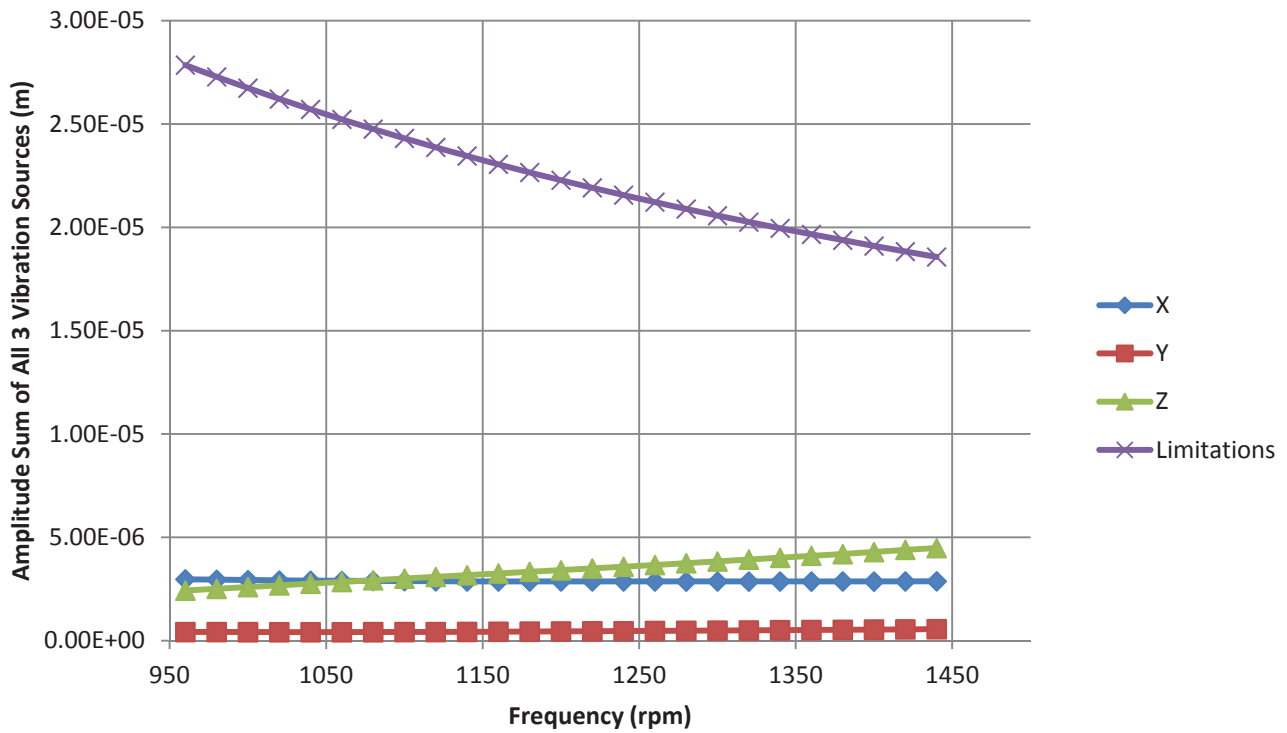
Translations at CG10001 Compressor B (UB)



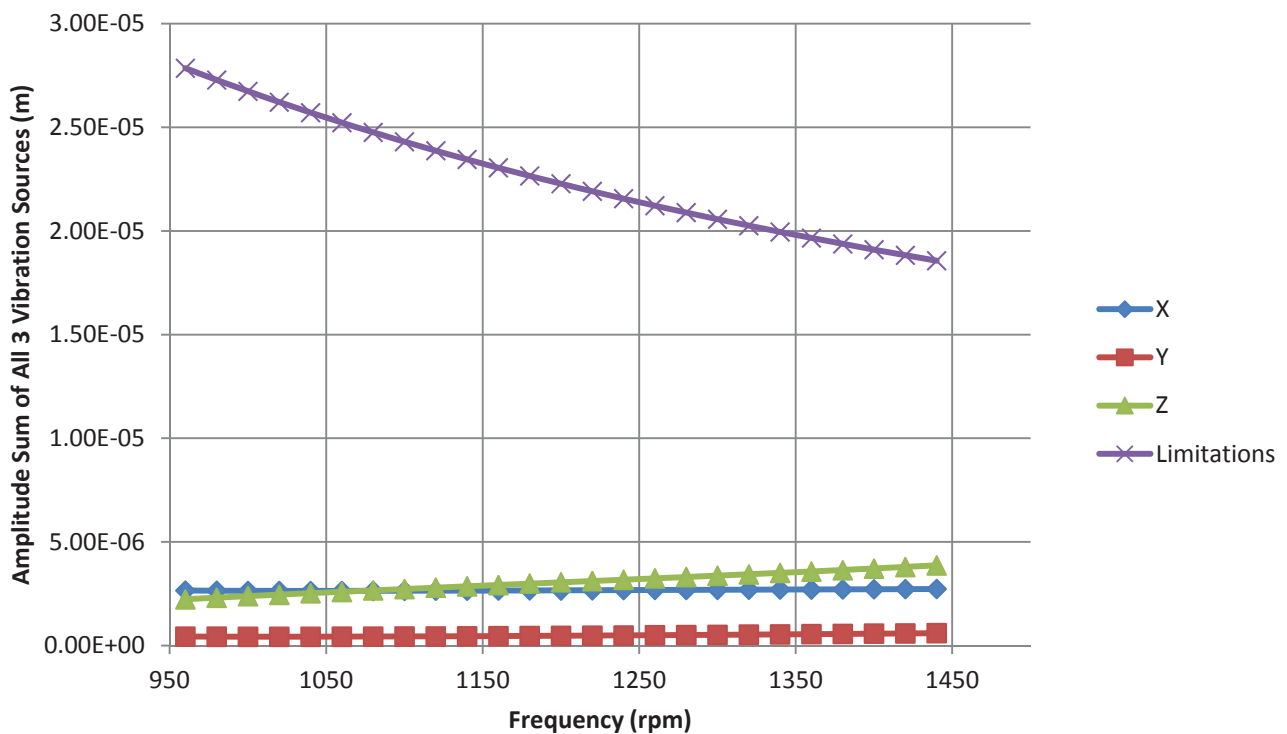
Translations at CG10001 Motor B (UB)



Translations at CG10001 Compressor C (UB)



Translations at CG10001 Motor C (UB)



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*
*   Copyright (c) M. Novak, U.W.O. - 1999
*           RUN DATE - 2016/ 8/ 9
*           TIME    - 10:27:12
*           REVISION - 1999/04/23
*
*****

CG10001A UB 1200

                DATA ECHO
                *****

GRAVITATIONAL CONSTANT SET TO 9.81 m/s**2
*****

FREQUENCY UNITS : RPM
*****

LENGTH UNITS : m
*****

FORCE UNITS : N
*****

MASS UNITS : kg
*****

DAMPING SAFETY FACTOR - 2.00
*****

FOUNDATION TYPE - PILE
*****

RIGID CAP
*****

RECTANGULAR FOUNDATION - 11.500 m BY 6.000 m
*****

MASS CONSTANTS
*****

TOTAL MASS OF FOOTING ..... 4.1124E+05 kg
MASS MOMENT ABOUT X-AXIS ..... 1.7127E+06 kg.m**2
MASS MOMENT ABOUT Y-AXIS ..... 4.6937E+06 kg.m**2
MASS MOMENT ABOUT Z-AXIS ..... 5.3209E+06 kg.m**2
CROSS PRODUCT X-Y ..... 1.9995E+03 kg.m**2
CROSS PRODUCT X-Z ..... -2.3290E+04 kg.m**2
CROSS PRODUCT Y-Z ..... -1.8327E+04 kg.m**2

NUMBER OF LAYERS OR ELEMENTS - 3
*****

FILES ARE FIXED AT BUTT
*****

LOCATION OF PILES
*****

PILE          X          Y
NO.    LOCATION LOCATION
          m          m

```

1	-4.979	-2.241
2	-4.979	0.009
3	-4.979	2.259
4	-2.479	-2.241
5	-2.479	0.009
6	-2.479	2.259
7	0.021	-2.241
8	0.021	0.009
9	0.021	2.259
10	2.521	-2.241
11	2.521	0.009
12	2.521	2.259
13	5.021	-2.241
14	5.021	0.009
15	5.021	2.259

COORDINATES OF BASE CENTRE

X-COORD. OF BASE CENTRE REL. TO C.G. .. -0.021 m
Y-COORD. OF BASE CENTRE REL. TO C.G. .. -0.009 m
Z-COORD. OF BASE CENTRE REL. TO C.G. .. 1.590 m

PILE PROPERTIES

TOTAL LENGTH 20.250 m
UNIT WEIGHT 77000.000 N/m**3
STATIC LOAD ON PILE 5.580E+05 N
POISSON'S RATIO 0.300
MATERIAL DAMPING 0.010
COEFF. OF RIGIDITY 1.110
YOUNG'S MODULUS 2.000E+11 N/m**2

ELEMENT

LAYER NO.	LAYER DEPTH m	PILE RADIUS (X) m	PILE RADIUS (Y) m	CROSS SECTIONAL AREA m**2	MOMENT OF INERTIA ABOUT Y m**4	MOMENT OF INERTIA ABOUT X m**4	TORSIONAL CONSTANT m**4
1	0.250	0.305	0.305	0.024	1.06E-03	1.06E-03	2.13E-03
2	13.250	0.305	0.305	0.024	1.06E-03	1.06E-03	2.13E-03
3	6.750	0.305	0.305	0.024	1.06E-03	1.06E-03	2.13E-03

PILE TIP CONDITION : FLOATING (DEFAULT)

PILE-SOIL-PILE INTERACTION IS CONSIDERED

SOIL

SOIL CONSTANTS

LAYER NO.	SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
1	0.000	0.00	0.000	0.000
2	200.000	20000.00	0.350	0.040
3	400.000	21000.00	0.350	0.040

SOIL BELOW PILE

SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
-------------------------------	--------------------------	--------------------	---------------------

4.00E+02	21000.00	0.350	0.040
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EMBEDDED OPTION

NUMBER OF SIDE LAYERS - 1

LAYER NO.	THICKNESS OF LAYER m	SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
1	0.000	0.00	0.00	0.000	0.000

HARMONIC LOAD (QUADRATIC)

MAXIMUM FREQUENCY		1440.000	RPM
MINIMUM FREQUENCY		960.000	RPM
STEP FREQUENCY		20.000	RPM
FORCE IN X-DIRECTION		2.48E-01	kg.m
FORCE IN Y-DIRECTION		0.00E+00	kg.m
FORCE IN Z-DIRECTION		2.48E-01	kg.m
MOMENT ABOUT X-AXIS		-1.16E-01	kg.m**2
MOMENT ABOUT Y-AXIS		1.74E+00	kg.m**2
MOMENT ABOUT Z-AXIS		1.16E-01	kg.m**2

RESULTANT TRANSLATIONS REQUIRED AT 6 POINTS

COORDINATES OF POINTS :

POINT NO.	X m	Y m	Z m
1	-3.909	1.504	-2.342
2	-3.909	-0.991	-2.512
3	-0.229	1.504	-2.342
4	-0.229	-0.991	-2.512
5	3.451	1.504	-2.342
6	3.451	-0.991	-2.512

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*
*           D Y N A 5   S I M U L A T I O N
*
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*   RUN DATE - 2016/ 8/ 9
*   TIME - 10:27:12
*   REVISION - 1999/04/23
*
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CG10001A UB 1200

RESULTS

FOOTING RESPONSE AMPLITUDES

FREQ. RPM	TRANS. X DIRECTION (m)	TRANS. Y DIRECTION (m)	VERTICAL DIRECTION (m)	ROT ABOUT X AXIS (rad)	ROT ABOUT Y AXIS (rad)	TORSIONAL Z AXIS (rad)
-----	-----	-----	-----	-----	-----	-----
960.00	1.04E-06	4.91E-08	3.97E-07	2.97E-08	1.87E-07	4.50E-08
980.00	1.03E-06	4.96E-08	4.07E-07	3.12E-08	1.96E-07	4.33E-08
1000.00	1.02E-06	5.01E-08	4.16E-07	3.29E-08	2.04E-07	4.18E-08
1020.00	1.02E-06	5.07E-08	4.26E-07	3.46E-08	2.12E-07	4.05E-08
1040.00	1.02E-06	5.12E-08	4.35E-07	3.63E-08	2.20E-07	3.93E-08
1060.00	1.01E-06	5.19E-08	4.45E-07	3.82E-08	2.28E-07	3.82E-08
1080.00	1.01E-06	5.25E-08	4.54E-07	4.00E-08	2.36E-07	3.72E-08
1100.00	1.00E-06	5.32E-08	4.63E-07	4.20E-08	2.44E-07	3.64E-08
1120.00	1.00E-06	5.39E-08	4.73E-07	4.40E-08	2.51E-07	3.56E-08
1140.00	9.98E-07	5.47E-08	4.82E-07	4.61E-08	2.59E-07	3.49E-08
1160.00	9.95E-07	5.54E-08	4.91E-07	4.82E-08	2.66E-07	3.42E-08
1180.00	9.91E-07	5.62E-08	5.01E-07	5.05E-08	2.74E-07	3.36E-08
1200.00	9.88E-07	5.70E-08	5.11E-07	5.27E-08	2.81E-07	3.31E-08
1220.00	9.85E-07	5.78E-08	5.20E-07	5.51E-08	2.88E-07	3.26E-08
1240.00	9.81E-07	5.86E-08	5.30E-07	5.75E-08	2.95E-07	3.22E-08
1260.00	9.78E-07	5.94E-08	5.41E-07	6.00E-08	3.02E-07	3.17E-08
1280.00	9.74E-07	6.03E-08	5.51E-07	6.25E-08	3.09E-07	3.13E-08
1300.00	9.71E-07	6.11E-08	5.62E-07	6.51E-08	3.16E-07	3.10E-08
1320.00	9.67E-07	6.19E-08	5.73E-07	6.78E-08	3.24E-07	3.06E-08
1340.00	9.63E-07	6.27E-08	5.84E-07	7.05E-08	3.31E-07	3.03E-08
1360.00	9.59E-07	6.35E-08	5.96E-07	7.32E-08	3.38E-07	3.00E-08
1380.00	9.55E-07	6.43E-08	6.08E-07	7.60E-08	3.45E-07	2.98E-08
1400.00	9.51E-07	6.50E-08	6.21E-07	7.88E-08	3.53E-07	2.95E-08
1420.00	9.47E-07	6.57E-08	6.34E-07	8.16E-08	3.60E-07	2.93E-08
1440.00	9.43E-07	6.64E-08	6.47E-07	8.44E-08	3.68E-07	2.91E-08

MAXIMA OF FOOTING RESPONSE AMPLITUDES

				FREQ. RPM
-----	-----	-----	-----	-----
MAX. TRANS. IN X-DIRECTION	-	1.037E-06	m	960.00
MAX. TRANS. IN Y-DIRECTION	-	6.636E-08	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	6.475E-07	m	1440.00
MAX. ROT. ABOUT X AXIS	-	8.438E-08	rad	1440.00
MAX. ROT. ABOUT Y AXIS	-	3.681E-07	rad	1440.00
MAX. ROT. ABOUT Z AXIS	-	4.504E-08	rad	960.00

RESULTANT TRANSLATIONS (m)

POINT NO. 1				POINT NO. 2		
-----				-----		
FREQ. RPM	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION
-----	-----	-----	-----	-----	-----	-----
960.00	1.07E-06	2.09E-07	3.04E-07	9.68E-07	2.08E-07	3.77E-07
980.00	1.07E-06	2.01E-07	3.23E-07	9.69E-07	2.01E-07	4.00E-07
1000.00	1.07E-06	1.95E-07	3.43E-07	9.72E-07	1.94E-07	4.23E-07
1020.00	1.06E-06	1.89E-07	3.62E-07	9.75E-07	1.89E-07	4.46E-07
1040.00	1.06E-06	1.84E-07	3.80E-07	9.80E-07	1.84E-07	4.69E-07
1060.00	1.06E-06	1.81E-07	3.98E-07	9.85E-07	1.81E-07	4.91E-07
1080.00	1.07E-06	1.77E-07	4.16E-07	9.92E-07	1.78E-07	5.13E-07
1100.00	1.07E-06	1.75E-07	4.33E-07	9.98E-07	1.76E-07	5.35E-07
1120.00	1.07E-06	1.73E-07	4.50E-07	1.01E-06	1.74E-07	5.56E-07
1140.00	1.07E-06	1.72E-07	4.67E-07	1.01E-06	1.73E-07	5.78E-07
1160.00	1.08E-06	1.71E-07	4.82E-07	1.02E-06	1.73E-07	5.98E-07
1180.00	1.08E-06	1.71E-07	4.98E-07	1.03E-06	1.73E-07	6.19E-07
1200.00	1.09E-06	1.72E-07	5.13E-07	1.03E-06	1.74E-07	6.39E-07
1220.00	1.09E-06	1.73E-07	5.27E-07	1.04E-06	1.76E-07	6.59E-07
1240.00	1.10E-06	1.75E-07	5.41E-07	1.05E-06	1.78E-07	6.79E-07
1260.00	1.10E-06	1.77E-07	5.55E-07	1.06E-06	1.81E-07	6.99E-07
1280.00	1.10E-06	1.80E-07	5.68E-07	1.06E-06	1.85E-07	7.18E-07
1300.00	1.11E-06	1.83E-07	5.82E-07	1.07E-06	1.88E-07	7.38E-07
1320.00	1.11E-06	1.87E-07	5.94E-07	1.08E-06	1.93E-07	7.58E-07
1340.00	1.12E-06	1.91E-07	6.07E-07	1.09E-06	1.98E-07	7.77E-07
1360.00	1.12E-06	1.96E-07	6.20E-07	1.09E-06	2.03E-07	7.97E-07
1380.00	1.13E-06	2.01E-07	6.33E-07	1.10E-06	2.09E-07	8.17E-07
1400.00	1.13E-06	2.07E-07	6.45E-07	1.11E-06	2.15E-07	8.37E-07
1420.00	1.14E-06	2.12E-07	6.58E-07	1.12E-06	2.21E-07	8.58E-07
1440.00	1.15E-06	2.18E-07	6.71E-07	1.13E-06	2.28E-07	8.78E-07

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 1

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	1.146E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	2.183E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	6.707E-07	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.167E-06	m	1440.00

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 2

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	1.128E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	2.278E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	8.779E-07	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.151E-06	m	1440.00

RESULTANT TRANSLATIONS (m)

POINT NO. 3				POINT NO. 4		
-----				-----		
FREQ. RPM	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION
-----	-----	-----	-----	-----	-----	-----
960.00	1.07E-06	6.73E-08	3.94E-07	9.68E-07	7.05E-08	3.29E-07
980.00	1.07E-06	6.91E-08	4.04E-07	9.69E-07	7.27E-08	3.36E-07
1000.00	1.07E-06	7.13E-08	4.14E-07	9.72E-07	7.52E-08	3.42E-07
1020.00	1.06E-06	7.37E-08	4.24E-07	9.75E-07	7.79E-08	3.48E-07
1040.00	1.06E-06	7.64E-08	4.34E-07	9.80E-07	8.10E-08	3.54E-07
1060.00	1.06E-06	7.93E-08	4.44E-07	9.85E-07	8.43E-08	3.60E-07
1080.00	1.07E-06	8.25E-08	4.55E-07	9.92E-07	8.79E-08	3.66E-07
1100.00	1.07E-06	8.59E-08	4.65E-07	9.98E-07	9.17E-08	3.71E-07
1120.00	1.07E-06	8.96E-08	4.76E-07	1.01E-06	9.57E-08	3.77E-07
1140.00	1.07E-06	9.35E-08	4.87E-07	1.01E-06	1.00E-07	3.82E-07
1160.00	1.08E-06	9.76E-08	4.98E-07	1.02E-06	1.04E-07	3.88E-07
1180.00	1.08E-06	1.02E-07	5.09E-07	1.03E-06	1.09E-07	3.93E-07
1200.00	1.09E-06	1.06E-07	5.21E-07	1.03E-06	1.14E-07	3.98E-07
1220.00	1.09E-06	1.11E-07	5.33E-07	1.04E-06	1.19E-07	4.04E-07
1240.00	1.10E-06	1.16E-07	5.45E-07	1.05E-06	1.25E-07	4.09E-07
1260.00	1.10E-06	1.21E-07	5.58E-07	1.06E-06	1.30E-07	4.15E-07
1280.00	1.10E-06	1.26E-07	5.71E-07	1.06E-06	1.36E-07	4.21E-07
1300.00	1.11E-06	1.32E-07	5.85E-07	1.07E-06	1.42E-07	4.27E-07
1320.00	1.11E-06	1.37E-07	5.98E-07	1.08E-06	1.48E-07	4.33E-07
1340.00	1.12E-06	1.43E-07	6.13E-07	1.09E-06	1.54E-07	4.40E-07
1360.00	1.12E-06	1.49E-07	6.27E-07	1.09E-06	1.61E-07	4.47E-07
1380.00	1.13E-06	1.55E-07	6.42E-07	1.10E-06	1.67E-07	4.54E-07
1400.00	1.13E-06	1.61E-07	6.58E-07	1.11E-06	1.74E-07	4.62E-07
1420.00	1.14E-06	1.68E-07	6.74E-07	1.12E-06	1.80E-07	4.71E-07
1440.00	1.15E-06	1.74E-07	6.90E-07	1.13E-06	1.87E-07	4.80E-07

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 3

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	1.146E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	1.738E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	6.903E-07	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.159E-06	m	1440.00

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 4

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	1.128E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	1.872E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	4.797E-07	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.144E-06	m	1440.00

RESULTANT TRANSLATIONS (m)

POINT NO. 5				POINT NO. 6		
-----				-----		
FREQ. RPM	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION
-----	-----	-----	-----	-----	-----	-----
960.00	1.07E-06	1.43E-07	1.08E-06	9.68E-07	1.46E-07	1.01E-06
980.00	1.07E-06	1.41E-07	1.12E-06	9.69E-07	1.45E-07	1.05E-06
1000.00	1.07E-06	1.40E-07	1.16E-06	9.72E-07	1.44E-07	1.09E-06
1020.00	1.06E-06	1.40E-07	1.20E-06	9.75E-07	1.44E-07	1.12E-06
1040.00	1.06E-06	1.39E-07	1.24E-06	9.80E-07	1.45E-07	1.16E-06
1060.00	1.06E-06	1.40E-07	1.28E-06	9.85E-07	1.45E-07	1.19E-06
1080.00	1.07E-06	1.40E-07	1.32E-06	9.92E-07	1.46E-07	1.23E-06
1100.00	1.07E-06	1.41E-07	1.36E-06	9.98E-07	1.48E-07	1.26E-06
1120.00	1.07E-06	1.43E-07	1.40E-06	1.01E-06	1.50E-07	1.30E-06
1140.00	1.07E-06	1.44E-07	1.44E-06	1.01E-06	1.52E-07	1.33E-06
1160.00	1.08E-06	1.46E-07	1.48E-06	1.02E-06	1.54E-07	1.36E-06
1180.00	1.08E-06	1.48E-07	1.52E-06	1.03E-06	1.56E-07	1.40E-06
1200.00	1.09E-06	1.51E-07	1.55E-06	1.03E-06	1.59E-07	1.43E-06
1220.00	1.09E-06	1.53E-07	1.59E-06	1.04E-06	1.62E-07	1.46E-06
1240.00	1.10E-06	1.56E-07	1.63E-06	1.05E-06	1.65E-07	1.49E-06
1260.00	1.10E-06	1.59E-07	1.67E-06	1.06E-06	1.68E-07	1.53E-06
1280.00	1.10E-06	1.62E-07	1.71E-06	1.06E-06	1.72E-07	1.56E-06
1300.00	1.11E-06	1.65E-07	1.75E-06	1.07E-06	1.75E-07	1.59E-06
1320.00	1.11E-06	1.68E-07	1.79E-06	1.08E-06	1.79E-07	1.62E-06
1340.00	1.12E-06	1.71E-07	1.83E-06	1.09E-06	1.83E-07	1.66E-06
1360.00	1.12E-06	1.75E-07	1.87E-06	1.09E-06	1.87E-07	1.69E-06
1380.00	1.13E-06	1.78E-07	1.91E-06	1.10E-06	1.91E-07	1.72E-06
1400.00	1.13E-06	1.82E-07	1.95E-06	1.11E-06	1.95E-07	1.76E-06
1420.00	1.14E-06	1.85E-07	2.00E-06	1.12E-06	1.99E-07	1.80E-06
1440.00	1.15E-06	1.89E-07	2.04E-06	1.13E-06	2.03E-07	1.83E-06

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 5

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	1.146E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	1.887E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	2.043E-06	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.161E-06	m	1440.00

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 6

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	1.128E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	2.027E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	1.833E-06	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	1.146E-06	m	1440.00

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*           D Y N A 5   S I M U L A T I O N
*
*   Copyright (c) M. Novak, U.W.O. - 1999
*           RUN DATE - 2016/ 8/ 9
*           TIME    - 10:41:23
*           REVISION - 1999/04/23
*
*****

CG10001B UB 1200

                DATA ECHO
                *****

GRAVITATIONAL CONSTANT SET TO 9.81 m/s**2
*****

FREQUENCY UNITS : RPM
*****

LENGTH UNITS : m
*****

FORCE UNITS : N
*****

MASS UNITS : kg
*****

DAMPING SAFETY FACTOR - 2.00
*****

FOUNDATION TYPE - PILE
*****

RIGID CAP
*****

RECTANGULAR FOUNDATION - 11.500 m BY 6.000 m
*****

MASS CONSTANTS
*****

TOTAL MASS OF FOOTING ..... 4.1124E+05 kg
MASS MOMENT ABOUT X-AXIS ..... 1.7127E+06 kg.m**2
MASS MOMENT ABOUT Y-AXIS ..... 4.6937E+06 kg.m**2
MASS MOMENT ABOUT Z-AXIS ..... 5.3209E+06 kg.m**2
CROSS PRODUCT X-Y ..... 1.9995E+03 kg.m**2
CROSS PRODUCT X-Z ..... -2.3290E+04 kg.m**2
CROSS PRODUCT Y-Z ..... -1.8327E+04 kg.m**2

NUMBER OF LAYERS OR ELEMENTS - 3
*****

FILES ARE FIXED AT BUTT
*****

LOCATION OF PILES
*****

PILE          X          Y
NO.    LOCATION LOCATION
          m          m

```

1	-4.979	-2.241
2	-4.979	0.009
3	-4.979	2.259
4	-2.479	-2.241
5	-2.479	0.009
6	-2.479	2.259
7	0.021	-2.241
8	0.021	0.009
9	0.021	2.259
10	2.521	-2.241
11	2.521	0.009
12	2.521	2.259
13	5.021	-2.241
14	5.021	0.009
15	5.021	2.259

COORDINATES OF BASE CENTRE

X-COORD. OF BASE CENTRE REL. TO C.G. .. -0.021 m
Y-COORD. OF BASE CENTRE REL. TO C.G. .. -0.009 m
Z-COORD. OF BASE CENTRE REL. TO C.G. .. 1.590 m

PILE PROPERTIES

TOTAL LENGTH 20.250 m
UNIT WEIGHT 77000.000 N/m**3
STATIC LOAD ON PILE 5.580E+05 N
POISSON'S RATIO 0.300
MATERIAL DAMPING 0.010
COEFF. OF RIGIDITY 1.110
YOUNG'S MODULUS 2.000E+11 N/m**2

ELEMENT

LAYER NO.	LAYER DEPTH m	PILE RADIUS (X) m	PILE RADIUS (Y) m	CROSS SECTIONAL AREA m**2	MOMENT OF INERTIA ABOUT Y m**4	MOMENT OF INERTIA ABOUT X m**4	TORSIONAL CONSTANT m**4
1	0.250	0.305	0.305	0.024	1.06E-03	1.06E-03	2.13E-03
2	13.250	0.305	0.305	0.024	1.06E-03	1.06E-03	2.13E-03
3	6.750	0.305	0.305	0.024	1.06E-03	1.06E-03	2.13E-03

PILE TIP CONDITION : FLOATING (DEFAULT)

PILE-SOIL-PILE INTERACTION IS CONSIDERED

SOIL

SOIL CONSTANTS

LAYER NO.	SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
1	0.000	0.00	0.000	0.000
2	200.000	20000.00	0.350	0.040
3	400.000	21000.00	0.350	0.040

SOIL BELOW PILE

SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
-------------------------------	--------------------------	--------------------	---------------------

4.00E+02	21000.00	0.350	0.040
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EMBEDDED OPTION

NUMBER OF SIDE LAYERS - 1

LAYER NO.	THICKNESS OF LAYER m	SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
--------------	----------------------------	-------------------------------	--------------------------	--------------------	---------------------

1	0.000	0.00	0.00	0.000	0.000
---	-------	------	------	-------	-------

HARMONIC LOAD (QUADRATIC)

MAXIMUM FREQUENCY		1440.000	RPM
MINIMUM FREQUENCY		960.000	RPM
STEP FREQUENCY		20.000	RPM
FORCE IN X-DIRECTION		2.48E-01	kg.m
FORCE IN Y-DIRECTION		0.00E+00	kg.m
FORCE IN Z-DIRECTION		2.48E-01	kg.m
MOMENT ABOUT X-AXIS		-1.16E-01	kg.m**2
MOMENT ABOUT Y-AXIS		6.69E-01	kg.m**2
MOMENT ABOUT Z-AXIS		1.16E-01	kg.m**2

RESULTANT TRANSLATIONS REQUIRED AT 6 POINTS

COORDINATES OF POINTS :

POINT NO.	X m	Y m	Z m
1	-3.909	1.504	-2.342
2	-3.909	-0.991	-2.512
3	-0.229	1.504	-2.342
4	-0.229	-0.991	-2.512
5	3.451	1.504	-2.342
6	3.451	-0.991	-2.512

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*
*           D Y N A 5   S I M U L A T I O N
*
*   Copyright (c)  M. Novak,  U.W.O.  - 1999
*   RUN DATE  - 2016/ 8/ 9
*   TIME      - 10:41:23
*   REVISION  - 1999/04/23
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CG10001B UB 1200

RESULTS

FOOTING RESPONSE AMPLITUDES

FREQ. RPM	TRANS. X DIRECTION (m)	TRANS. Y DIRECTION (m)	VERTICAL DIRECTION (m)	ROT ABOUT X AXIS (rad)	ROT ABOUT Y AXIS (rad)	TORSIONAL Z AXIS (rad)
960.00	8.56E-07	4.91E-08	4.01E-07	2.96E-08	6.92E-08	4.42E-08
980.00	8.49E-07	4.95E-08	4.11E-07	3.12E-08	7.18E-08	4.25E-08
1000.00	8.42E-07	5.00E-08	4.20E-07	3.28E-08	7.45E-08	4.10E-08
1020.00	8.36E-07	5.06E-08	4.30E-07	3.45E-08	7.72E-08	3.96E-08
1040.00	8.31E-07	5.11E-08	4.40E-07	3.63E-08	7.98E-08	3.84E-08
1060.00	8.26E-07	5.18E-08	4.49E-07	3.81E-08	8.25E-08	3.74E-08
1080.00	8.22E-07	5.24E-08	4.59E-07	4.00E-08	8.51E-08	3.64E-08
1100.00	8.18E-07	5.31E-08	4.68E-07	4.19E-08	8.76E-08	3.56E-08
1120.00	8.14E-07	5.38E-08	4.78E-07	4.39E-08	9.02E-08	3.48E-08
1140.00	8.10E-07	5.45E-08	4.87E-07	4.60E-08	9.27E-08	3.41E-08
1160.00	8.07E-07	5.53E-08	4.97E-07	4.81E-08	9.52E-08	3.35E-08
1180.00	8.03E-07	5.61E-08	5.07E-07	5.04E-08	9.77E-08	3.29E-08
1200.00	8.00E-07	5.69E-08	5.16E-07	5.26E-08	1.00E-07	3.24E-08
1220.00	7.97E-07	5.77E-08	5.26E-07	5.50E-08	1.03E-07	3.19E-08
1240.00	7.94E-07	5.85E-08	5.36E-07	5.74E-08	1.05E-07	3.14E-08
1260.00	7.92E-07	5.93E-08	5.47E-07	5.99E-08	1.07E-07	3.10E-08
1280.00	7.89E-07	6.02E-08	5.57E-07	6.24E-08	1.10E-07	3.06E-08
1300.00	7.86E-07	6.10E-08	5.68E-07	6.50E-08	1.12E-07	3.03E-08
1320.00	7.83E-07	6.18E-08	5.79E-07	6.76E-08	1.15E-07	2.99E-08
1340.00	7.81E-07	6.26E-08	5.91E-07	7.03E-08	1.17E-07	2.96E-08
1360.00	7.78E-07	6.34E-08	6.03E-07	7.31E-08	1.20E-07	2.94E-08
1380.00	7.75E-07	6.41E-08	6.15E-07	7.58E-08	1.22E-07	2.91E-08
1400.00	7.72E-07	6.49E-08	6.28E-07	7.86E-08	1.25E-07	2.88E-08
1420.00	7.69E-07	6.56E-08	6.41E-07	8.14E-08	1.28E-07	2.86E-08
1440.00	7.67E-07	6.62E-08	6.55E-07	8.42E-08	1.31E-07	2.84E-08

MAXIMA OF FOOTING RESPONSE AMPLITUDES

				FREQ. RPM
MAX. TRANS. IN X-DIRECTION	-	8.561E-07	m	960.00
MAX. TRANS. IN Y-DIRECTION	-	6.623E-08	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	6.550E-07	m	1440.00
MAX. ROT. ABOUT X AXIS	-	8.422E-08	rad	1440.00
MAX. ROT. ABOUT Y AXIS	-	1.308E-07	rad	1440.00
MAX. ROT. ABOUT Z AXIS	-	4.415E-08	rad	960.00

RESULTANT TRANSLATIONS (m)

POINT NO. 1				POINT NO. 2		
-----				-----		
FREQ. RPM	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION
-----	-----	-----	-----	-----	-----	-----
960.00	9.50E-07	2.06E-07	1.90E-07	8.44E-07	2.06E-07	1.17E-07
980.00	9.41E-07	1.98E-07	1.91E-07	8.38E-07	1.98E-07	1.14E-07
1000.00	9.33E-07	1.92E-07	1.93E-07	8.34E-07	1.92E-07	1.12E-07
1020.00	9.25E-07	1.87E-07	1.95E-07	8.30E-07	1.87E-07	1.09E-07
1040.00	9.19E-07	1.82E-07	1.97E-07	8.27E-07	1.82E-07	1.07E-07
1060.00	9.14E-07	1.78E-07	2.00E-07	8.25E-07	1.79E-07	1.05E-07
1080.00	9.09E-07	1.75E-07	2.03E-07	8.22E-07	1.76E-07	1.03E-07
1100.00	9.05E-07	1.73E-07	2.06E-07	8.21E-07	1.74E-07	1.02E-07
1120.00	9.01E-07	1.72E-07	2.11E-07	8.19E-07	1.73E-07	1.01E-07
1140.00	8.98E-07	1.71E-07	2.15E-07	8.18E-07	1.72E-07	1.00E-07
1160.00	8.95E-07	1.70E-07	2.20E-07	8.17E-07	1.72E-07	1.00E-07
1180.00	8.92E-07	1.70E-07	2.25E-07	8.16E-07	1.73E-07	1.00E-07
1200.00	8.90E-07	1.71E-07	2.31E-07	8.15E-07	1.74E-07	1.01E-07
1220.00	8.87E-07	1.73E-07	2.37E-07	8.14E-07	1.76E-07	1.02E-07
1240.00	8.85E-07	1.75E-07	2.43E-07	8.14E-07	1.78E-07	1.03E-07
1260.00	8.83E-07	1.77E-07	2.50E-07	8.13E-07	1.81E-07	1.05E-07
1280.00	8.80E-07	1.80E-07	2.56E-07	8.12E-07	1.85E-07	1.08E-07
1300.00	8.78E-07	1.83E-07	2.63E-07	8.10E-07	1.89E-07	1.10E-07
1320.00	8.76E-07	1.87E-07	2.70E-07	8.09E-07	1.93E-07	1.14E-07
1340.00	8.73E-07	1.92E-07	2.78E-07	8.08E-07	1.98E-07	1.18E-07
1360.00	8.71E-07	1.96E-07	2.85E-07	8.07E-07	2.03E-07	1.22E-07
1380.00	8.69E-07	2.02E-07	2.92E-07	8.05E-07	2.09E-07	1.27E-07
1400.00	8.66E-07	2.07E-07	2.99E-07	8.04E-07	2.15E-07	1.32E-07
1420.00	8.64E-07	2.13E-07	3.06E-07	8.03E-07	2.22E-07	1.37E-07
1440.00	8.61E-07	2.19E-07	3.13E-07	8.01E-07	2.29E-07	1.43E-07

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 1

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	9.504E-07	m	960.00
MAX. TRANS. IN Y-DIRECTION	-	2.190E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	3.135E-07	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	9.724E-07	m	960.00

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 2

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	8.438E-07	m	960.00
MAX. TRANS. IN Y-DIRECTION	-	2.286E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	1.431E-07	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	8.685E-07	m	960.00

RESULTANT TRANSLATIONS (m)

POINT NO. 3				POINT NO. 4		
-----				-----		
FREQ. RPM	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION
-----	-----	-----	-----	-----	-----	-----
960.00	9.50E-07	6.72E-08	4.25E-07	8.44E-07	7.04E-08	3.60E-07
980.00	9.41E-07	6.90E-08	4.36E-07	8.38E-07	7.26E-08	3.68E-07
1000.00	9.33E-07	7.12E-08	4.48E-07	8.34E-07	7.51E-08	3.75E-07
1020.00	9.25E-07	7.36E-08	4.59E-07	8.30E-07	7.78E-08	3.83E-07
1040.00	9.19E-07	7.63E-08	4.71E-07	8.27E-07	8.09E-08	3.90E-07
1060.00	9.14E-07	7.92E-08	4.82E-07	8.25E-07	8.42E-08	3.98E-07
1080.00	9.09E-07	8.24E-08	4.94E-07	8.22E-07	8.78E-08	4.05E-07
1100.00	9.05E-07	8.59E-08	5.06E-07	8.21E-07	9.16E-08	4.12E-07
1120.00	9.01E-07	8.95E-08	5.18E-07	8.19E-07	9.56E-08	4.19E-07
1140.00	8.98E-07	9.34E-08	5.30E-07	8.18E-07	9.99E-08	4.25E-07
1160.00	8.95E-07	9.75E-08	5.43E-07	8.17E-07	1.04E-07	4.32E-07
1180.00	8.92E-07	1.02E-07	5.55E-07	8.16E-07	1.09E-07	4.39E-07
1200.00	8.90E-07	1.06E-07	5.68E-07	8.15E-07	1.14E-07	4.46E-07
1220.00	8.87E-07	1.11E-07	5.82E-07	8.14E-07	1.19E-07	4.53E-07
1240.00	8.85E-07	1.16E-07	5.95E-07	8.14E-07	1.25E-07	4.59E-07
1260.00	8.83E-07	1.21E-07	6.09E-07	8.13E-07	1.30E-07	4.66E-07
1280.00	8.80E-07	1.26E-07	6.24E-07	8.12E-07	1.36E-07	4.74E-07
1300.00	8.78E-07	1.32E-07	6.38E-07	8.10E-07	1.42E-07	4.81E-07
1320.00	8.76E-07	1.37E-07	6.53E-07	8.09E-07	1.48E-07	4.89E-07
1340.00	8.73E-07	1.43E-07	6.69E-07	8.08E-07	1.54E-07	4.96E-07
1360.00	8.71E-07	1.49E-07	6.85E-07	8.07E-07	1.60E-07	5.05E-07
1380.00	8.69E-07	1.55E-07	7.01E-07	8.05E-07	1.67E-07	5.13E-07
1400.00	8.66E-07	1.61E-07	7.18E-07	8.04E-07	1.74E-07	5.22E-07
1420.00	8.64E-07	1.67E-07	7.35E-07	8.03E-07	1.80E-07	5.32E-07
1440.00	8.61E-07	1.74E-07	7.53E-07	8.01E-07	1.87E-07	5.43E-07

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 3

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	9.504E-07	m	960.00
MAX. TRANS. IN Y-DIRECTION	-	1.736E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	7.526E-07	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	9.528E-07	m	960.00

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 4

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	8.438E-07	m	960.00
MAX. TRANS. IN Y-DIRECTION	-	1.870E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	5.426E-07	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	8.467E-07	m	960.00

RESULTANT TRANSLATIONS (m)

POINT NO. 5				POINT NO. 6		
-----				-----		
FREQ. RPM	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION
-----	-----	-----	-----	-----	-----	-----
960.00	9.50E-07	1.39E-07	6.74E-07	8.44E-07	1.43E-07	6.12E-07
980.00	9.41E-07	1.38E-07	6.96E-07	8.38E-07	1.42E-07	6.30E-07
1000.00	9.33E-07	1.37E-07	7.17E-07	8.34E-07	1.41E-07	6.48E-07
1020.00	9.25E-07	1.36E-07	7.38E-07	8.30E-07	1.41E-07	6.65E-07
1040.00	9.19E-07	1.36E-07	7.60E-07	8.27E-07	1.41E-07	6.82E-07
1060.00	9.14E-07	1.36E-07	7.81E-07	8.25E-07	1.42E-07	6.99E-07
1080.00	9.09E-07	1.37E-07	8.02E-07	8.22E-07	1.43E-07	7.16E-07
1100.00	9.05E-07	1.38E-07	8.23E-07	8.21E-07	1.44E-07	7.32E-07
1120.00	9.01E-07	1.39E-07	8.44E-07	8.19E-07	1.46E-07	7.49E-07
1140.00	8.98E-07	1.41E-07	8.65E-07	8.18E-07	1.48E-07	7.65E-07
1160.00	8.95E-07	1.43E-07	8.87E-07	8.17E-07	1.50E-07	7.80E-07
1180.00	8.92E-07	1.45E-07	9.08E-07	8.16E-07	1.53E-07	7.96E-07
1200.00	8.90E-07	1.47E-07	9.30E-07	8.15E-07	1.55E-07	8.12E-07
1220.00	8.87E-07	1.50E-07	9.52E-07	8.14E-07	1.58E-07	8.27E-07
1240.00	8.85E-07	1.52E-07	9.74E-07	8.14E-07	1.61E-07	8.43E-07
1260.00	8.83E-07	1.55E-07	9.96E-07	8.13E-07	1.65E-07	8.58E-07
1280.00	8.80E-07	1.58E-07	1.02E-06	8.12E-07	1.68E-07	8.74E-07
1300.00	8.78E-07	1.61E-07	1.04E-06	8.10E-07	1.72E-07	8.90E-07
1320.00	8.76E-07	1.64E-07	1.07E-06	8.09E-07	1.75E-07	9.06E-07
1340.00	8.73E-07	1.68E-07	1.09E-06	8.08E-07	1.79E-07	9.23E-07
1360.00	8.71E-07	1.71E-07	1.12E-06	8.07E-07	1.83E-07	9.40E-07
1380.00	8.69E-07	1.74E-07	1.14E-06	8.05E-07	1.87E-07	9.58E-07
1400.00	8.66E-07	1.78E-07	1.17E-06	8.04E-07	1.91E-07	9.76E-07
1420.00	8.64E-07	1.81E-07	1.20E-06	8.03E-07	1.95E-07	9.96E-07
1440.00	8.61E-07	1.85E-07	1.22E-06	8.01E-07	1.99E-07	1.02E-06

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 5

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	9.504E-07	m	960.00
MAX. TRANS. IN Y-DIRECTION	-	1.848E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	1.224E-06	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	9.606E-07	m	960.00

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 6

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	8.438E-07	m	960.00
MAX. TRANS. IN Y-DIRECTION	-	1.988E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	1.016E-06	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	8.558E-07	m	960.00

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*
*   Copyright (c) M. Novak, U.W.O. - 1999
*           RUN DATE - 2016/ 8/ 9
*           TIME    - 10:47:54
*           REVISION - 1999/04/23
*
*****

CG10001C UB 1200

                DATA ECHO
                *****

GRAVITATIONAL CONSTANT SET TO 9.81 m/s**2
*****

FREQUENCY UNITS : RPM
*****

LENGTH UNITS : m
*****

FORCE UNITS : N
*****

MASS UNITS : kg
*****

DAMPING SAFETY FACTOR - 2.00
*****

FOUNDATION TYPE - PILE
*****

RIGID CAP
*****

RECTANGULAR FOUNDATION - 11.500 m BY 6.000 m
*****

MASS CONSTANTS
*****

TOTAL MASS OF FOOTING ..... 4.1124E+05 kg
MASS MOMENT ABOUT X-AXIS ..... 1.7127E+06 kg.m**2
MASS MOMENT ABOUT Y-AXIS ..... 4.6937E+06 kg.m**2
MASS MOMENT ABOUT Z-AXIS ..... 5.3209E+06 kg.m**2
CROSS PRODUCT X-Y ..... 1.9995E+03 kg.m**2
CROSS PRODUCT X-Z ..... -2.3290E+04 kg.m**2
CROSS PRODUCT Y-Z ..... -1.8327E+04 kg.m**2

NUMBER OF LAYERS OR ELEMENTS - 3
*****

FILES ARE FIXED AT BUTT
*****

LOCATION OF PILES
*****

PILE          X          Y
NO.    LOCATION LOCATION
          m          m

```

1	-4.979	-2.241
2	-4.979	0.009
3	-4.979	2.259
4	-2.479	-2.241
5	-2.479	0.009
6	-2.479	2.259
7	0.021	-2.241
8	0.021	0.009
9	0.021	2.259
10	2.521	-2.241
11	2.521	0.009
12	2.521	2.259
13	5.021	-2.241
14	5.021	0.009
15	5.021	2.259

COORDINATES OF BASE CENTRE

X-COORD. OF BASE CENTRE REL. TO C.G. .. -0.021 m
Y-COORD. OF BASE CENTRE REL. TO C.G. .. -0.009 m
Z-COORD. OF BASE CENTRE REL. TO C.G. .. 1.590 m

PILE PROPERTIES

TOTAL LENGTH 20.250 m
UNIT WEIGHT 77000.000 N/m**3
STATIC LOAD ON PILE 5.580E+05 N
POISSON'S RATIO 0.300
MATERIAL DAMPING 0.010
COEFF. OF RIGIDITY 1.110
YOUNG'S MODULUS 2.000E+11 N/m**2

ELEMENT

LAYER NO.	LAYER DEPTH m	PILE RADIUS (X) m	PILE RADIUS (Y) m	CROSS SECTIONAL AREA m**2	MOMENT OF INERTIA ABOUT Y m**4	MOMENT OF INERTIA ABOUT X m**4	TORSIONAL CONSTANT m**4
1	0.250	0.305	0.305	0.024	1.06E-03	1.06E-03	2.13E-03
2	13.250	0.305	0.305	0.024	1.06E-03	1.06E-03	2.13E-03
3	6.750	0.305	0.305	0.024	1.06E-03	1.06E-03	2.13E-03

PILE TIP CONDITION : FLOATING (DEFAULT)

PILE-SOIL-PILE INTERACTION IS CONSIDERED

SOIL

SOIL CONSTANTS

LAYER NO.	SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
1	0.000	0.00	0.000	0.000
2	200.000	20000.00	0.350	0.040
3	400.000	21000.00	0.350	0.040

SOIL BELOW PILE

SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
-------------------------------	--------------------------	--------------------	---------------------

4.00E+02	21000.00	0.350	0.040
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EMBEDDED OPTION

NUMBER OF SIDE LAYERS - 1

LAYER NO.	THICKNESS OF LAYER m	SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
--------------	----------------------------	-------------------------------	--------------------------	--------------------	---------------------

1	0.000	0.00	0.00	0.000	0.000
---	-------	------	------	-------	-------

HARMONIC LOAD (QUADRATIC)

MAXIMUM FREQUENCY		1440.000	RPM
MINIMUM FREQUENCY		960.000	RPM
STEP FREQUENCY		20.000	RPM
FORCE IN X-DIRECTION		2.48E-01	kg.m
FORCE IN Y-DIRECTION		0.00E+00	kg.m
FORCE IN Z-DIRECTION		2.48E-01	kg.m
MOMENT ABOUT X-AXIS		-1.16E-01	kg.m**2
MOMENT ABOUT Y-AXIS		6.69E-01	kg.m**2
MOMENT ABOUT Z-AXIS		1.16E-01	kg.m**2

RESULTANT TRANSLATIONS REQUIRED AT 6 POINTS

COORDINATES OF POINTS :

POINT NO.	X m	Y m	Z m
1	-3.909	1.504	-2.342
2	-3.909	-0.991	-2.512
3	-0.229	1.504	-2.342
4	-0.229	-0.991	-2.512
5	3.451	1.504	-2.342
6	3.451	-0.991	-2.512

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*
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*   REVISION - 1999/04/23
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CG10001C UB 1200

RESULTS

FOOTING RESPONSE AMPLITUDES

FREQ. RPM	TRANS. X DIRECTION (m)	TRANS. Y DIRECTION (m)	VERTICAL DIRECTION (m)	ROT ABOUT X AXIS (rad)	ROT ABOUT Y AXIS (rad)	TORSIONAL Z AXIS (rad)
-----	-----	-----	-----	-----	-----	-----
960.00	8.56E-07	4.91E-08	4.01E-07	2.96E-08	6.92E-08	4.42E-08
980.00	8.49E-07	4.95E-08	4.11E-07	3.12E-08	7.18E-08	4.25E-08
1000.00	8.42E-07	5.00E-08	4.20E-07	3.28E-08	7.45E-08	4.10E-08
1020.00	8.36E-07	5.06E-08	4.30E-07	3.45E-08	7.72E-08	3.96E-08
1040.00	8.31E-07	5.11E-08	4.40E-07	3.63E-08	7.98E-08	3.84E-08
1060.00	8.26E-07	5.18E-08	4.49E-07	3.81E-08	8.25E-08	3.74E-08
1080.00	8.22E-07	5.24E-08	4.59E-07	4.00E-08	8.51E-08	3.64E-08
1100.00	8.18E-07	5.31E-08	4.68E-07	4.19E-08	8.76E-08	3.56E-08
1120.00	8.14E-07	5.38E-08	4.78E-07	4.39E-08	9.02E-08	3.48E-08
1140.00	8.10E-07	5.45E-08	4.87E-07	4.60E-08	9.27E-08	3.41E-08
1160.00	8.07E-07	5.53E-08	4.97E-07	4.81E-08	9.52E-08	3.35E-08
1180.00	8.03E-07	5.61E-08	5.07E-07	5.04E-08	9.77E-08	3.29E-08
1200.00	8.00E-07	5.69E-08	5.16E-07	5.26E-08	1.00E-07	3.24E-08
1220.00	7.97E-07	5.77E-08	5.26E-07	5.50E-08	1.03E-07	3.19E-08
1240.00	7.94E-07	5.85E-08	5.36E-07	5.74E-08	1.05E-07	3.14E-08
1260.00	7.92E-07	5.93E-08	5.47E-07	5.99E-08	1.07E-07	3.10E-08
1280.00	7.89E-07	6.02E-08	5.57E-07	6.24E-08	1.10E-07	3.06E-08
1300.00	7.86E-07	6.10E-08	5.68E-07	6.50E-08	1.12E-07	3.03E-08
1320.00	7.83E-07	6.18E-08	5.79E-07	6.76E-08	1.15E-07	2.99E-08
1340.00	7.81E-07	6.26E-08	5.91E-07	7.03E-08	1.17E-07	2.96E-08
1360.00	7.78E-07	6.34E-08	6.03E-07	7.31E-08	1.20E-07	2.94E-08
1380.00	7.75E-07	6.41E-08	6.15E-07	7.58E-08	1.22E-07	2.91E-08
1400.00	7.72E-07	6.49E-08	6.28E-07	7.86E-08	1.25E-07	2.88E-08
1420.00	7.69E-07	6.56E-08	6.41E-07	8.14E-08	1.28E-07	2.86E-08
1440.00	7.67E-07	6.62E-08	6.55E-07	8.42E-08	1.31E-07	2.84E-08

MAXIMA OF FOOTING RESPONSE AMPLITUDES

				FREQ. RPM
-----	-----	-----	-----	-----
MAX. TRANS. IN X-DIRECTION	-	8.561E-07	m	960.00
MAX. TRANS. IN Y-DIRECTION	-	6.623E-08	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	6.550E-07	m	1440.00
MAX. ROT. ABOUT X AXIS	-	8.422E-08	rad	1440.00
MAX. ROT. ABOUT Y AXIS	-	1.308E-07	rad	1440.00
MAX. ROT. ABOUT Z AXIS	-	4.415E-08	rad	960.00

RESULTANT TRANSLATIONS (m)

POINT NO. 1				POINT NO. 2		
-----				-----		
FREQ. RPM	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION
-----	-----	-----	-----	-----	-----	-----
960.00	9.50E-07	2.06E-07	1.90E-07	8.44E-07	2.06E-07	1.17E-07
980.00	9.41E-07	1.98E-07	1.91E-07	8.38E-07	1.98E-07	1.14E-07
1000.00	9.33E-07	1.92E-07	1.93E-07	8.34E-07	1.92E-07	1.12E-07
1020.00	9.25E-07	1.87E-07	1.95E-07	8.30E-07	1.87E-07	1.09E-07
1040.00	9.19E-07	1.82E-07	1.97E-07	8.27E-07	1.82E-07	1.07E-07
1060.00	9.14E-07	1.78E-07	2.00E-07	8.25E-07	1.79E-07	1.05E-07
1080.00	9.09E-07	1.75E-07	2.03E-07	8.22E-07	1.76E-07	1.03E-07
1100.00	9.05E-07	1.73E-07	2.06E-07	8.21E-07	1.74E-07	1.02E-07
1120.00	9.01E-07	1.72E-07	2.11E-07	8.19E-07	1.73E-07	1.01E-07
1140.00	8.98E-07	1.71E-07	2.15E-07	8.18E-07	1.72E-07	1.00E-07
1160.00	8.95E-07	1.70E-07	2.20E-07	8.17E-07	1.72E-07	1.00E-07
1180.00	8.92E-07	1.70E-07	2.25E-07	8.16E-07	1.73E-07	1.00E-07
1200.00	8.90E-07	1.71E-07	2.31E-07	8.15E-07	1.74E-07	1.01E-07
1220.00	8.87E-07	1.73E-07	2.37E-07	8.14E-07	1.76E-07	1.02E-07
1240.00	8.85E-07	1.75E-07	2.43E-07	8.14E-07	1.78E-07	1.03E-07
1260.00	8.83E-07	1.77E-07	2.50E-07	8.13E-07	1.81E-07	1.05E-07
1280.00	8.80E-07	1.80E-07	2.56E-07	8.12E-07	1.85E-07	1.08E-07
1300.00	8.78E-07	1.83E-07	2.63E-07	8.10E-07	1.89E-07	1.10E-07
1320.00	8.76E-07	1.87E-07	2.70E-07	8.09E-07	1.93E-07	1.14E-07
1340.00	8.73E-07	1.92E-07	2.78E-07	8.08E-07	1.98E-07	1.18E-07
1360.00	8.71E-07	1.96E-07	2.85E-07	8.07E-07	2.03E-07	1.22E-07
1380.00	8.69E-07	2.02E-07	2.92E-07	8.05E-07	2.09E-07	1.27E-07
1400.00	8.66E-07	2.07E-07	2.99E-07	8.04E-07	2.15E-07	1.32E-07
1420.00	8.64E-07	2.13E-07	3.06E-07	8.03E-07	2.22E-07	1.37E-07
1440.00	8.61E-07	2.19E-07	3.13E-07	8.01E-07	2.29E-07	1.43E-07

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 1

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	9.504E-07	m	960.00
MAX. TRANS. IN Y-DIRECTION	-	2.190E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	3.135E-07	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	9.724E-07	m	960.00

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 2

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	8.438E-07	m	960.00
MAX. TRANS. IN Y-DIRECTION	-	2.286E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	1.431E-07	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	8.685E-07	m	960.00

RESULTANT TRANSLATIONS (m)

POINT NO. 3				POINT NO. 4		
-----				-----		
FREQ. RPM	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION
-----	-----	-----	-----	-----	-----	-----
960.00	9.50E-07	6.72E-08	4.25E-07	8.44E-07	7.04E-08	3.60E-07
980.00	9.41E-07	6.90E-08	4.36E-07	8.38E-07	7.26E-08	3.68E-07
1000.00	9.33E-07	7.12E-08	4.48E-07	8.34E-07	7.51E-08	3.75E-07
1020.00	9.25E-07	7.36E-08	4.59E-07	8.30E-07	7.78E-08	3.83E-07
1040.00	9.19E-07	7.63E-08	4.71E-07	8.27E-07	8.09E-08	3.90E-07
1060.00	9.14E-07	7.92E-08	4.82E-07	8.25E-07	8.42E-08	3.98E-07
1080.00	9.09E-07	8.24E-08	4.94E-07	8.22E-07	8.78E-08	4.05E-07
1100.00	9.05E-07	8.59E-08	5.06E-07	8.21E-07	9.16E-08	4.12E-07
1120.00	9.01E-07	8.95E-08	5.18E-07	8.19E-07	9.56E-08	4.19E-07
1140.00	8.98E-07	9.34E-08	5.30E-07	8.18E-07	9.99E-08	4.25E-07
1160.00	8.95E-07	9.75E-08	5.43E-07	8.17E-07	1.04E-07	4.32E-07
1180.00	8.92E-07	1.02E-07	5.55E-07	8.16E-07	1.09E-07	4.39E-07
1200.00	8.90E-07	1.06E-07	5.68E-07	8.15E-07	1.14E-07	4.46E-07
1220.00	8.87E-07	1.11E-07	5.82E-07	8.14E-07	1.19E-07	4.53E-07
1240.00	8.85E-07	1.16E-07	5.95E-07	8.14E-07	1.25E-07	4.59E-07
1260.00	8.83E-07	1.21E-07	6.09E-07	8.13E-07	1.30E-07	4.66E-07
1280.00	8.80E-07	1.26E-07	6.24E-07	8.12E-07	1.36E-07	4.74E-07
1300.00	8.78E-07	1.32E-07	6.38E-07	8.10E-07	1.42E-07	4.81E-07
1320.00	8.76E-07	1.37E-07	6.53E-07	8.09E-07	1.48E-07	4.89E-07
1340.00	8.73E-07	1.43E-07	6.69E-07	8.08E-07	1.54E-07	4.96E-07
1360.00	8.71E-07	1.49E-07	6.85E-07	8.07E-07	1.60E-07	5.05E-07
1380.00	8.69E-07	1.55E-07	7.01E-07	8.05E-07	1.67E-07	5.13E-07
1400.00	8.66E-07	1.61E-07	7.18E-07	8.04E-07	1.74E-07	5.22E-07
1420.00	8.64E-07	1.67E-07	7.35E-07	8.03E-07	1.80E-07	5.32E-07
1440.00	8.61E-07	1.74E-07	7.53E-07	8.01E-07	1.87E-07	5.43E-07

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 3

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	9.504E-07	m	960.00
MAX. TRANS. IN Y-DIRECTION	-	1.736E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	7.526E-07	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	9.528E-07	m	960.00

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 4

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	8.438E-07	m	960.00
MAX. TRANS. IN Y-DIRECTION	-	1.870E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	5.426E-07	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	8.467E-07	m	960.00

RESULTANT TRANSLATIONS (m)

POINT NO. 5				POINT NO. 6		
-----				-----		
FREQ. RPM	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION
-----	-----	-----	-----	-----	-----	-----
960.00	9.50E-07	1.39E-07	6.74E-07	8.44E-07	1.43E-07	6.12E-07
980.00	9.41E-07	1.38E-07	6.96E-07	8.38E-07	1.42E-07	6.30E-07
1000.00	9.33E-07	1.37E-07	7.17E-07	8.34E-07	1.41E-07	6.48E-07
1020.00	9.25E-07	1.36E-07	7.38E-07	8.30E-07	1.41E-07	6.65E-07
1040.00	9.19E-07	1.36E-07	7.60E-07	8.27E-07	1.41E-07	6.82E-07
1060.00	9.14E-07	1.36E-07	7.81E-07	8.25E-07	1.42E-07	6.99E-07
1080.00	9.09E-07	1.37E-07	8.02E-07	8.22E-07	1.43E-07	7.16E-07
1100.00	9.05E-07	1.38E-07	8.23E-07	8.21E-07	1.44E-07	7.32E-07
1120.00	9.01E-07	1.39E-07	8.44E-07	8.19E-07	1.46E-07	7.49E-07
1140.00	8.98E-07	1.41E-07	8.65E-07	8.18E-07	1.48E-07	7.65E-07
1160.00	8.95E-07	1.43E-07	8.87E-07	8.17E-07	1.50E-07	7.80E-07
1180.00	8.92E-07	1.45E-07	9.08E-07	8.16E-07	1.53E-07	7.96E-07
1200.00	8.90E-07	1.47E-07	9.30E-07	8.15E-07	1.55E-07	8.12E-07
1220.00	8.87E-07	1.50E-07	9.52E-07	8.14E-07	1.58E-07	8.27E-07
1240.00	8.85E-07	1.52E-07	9.74E-07	8.14E-07	1.61E-07	8.43E-07
1260.00	8.83E-07	1.55E-07	9.96E-07	8.13E-07	1.65E-07	8.58E-07
1280.00	8.80E-07	1.58E-07	1.02E-06	8.12E-07	1.68E-07	8.74E-07
1300.00	8.78E-07	1.61E-07	1.04E-06	8.10E-07	1.72E-07	8.90E-07
1320.00	8.76E-07	1.64E-07	1.07E-06	8.09E-07	1.75E-07	9.06E-07
1340.00	8.73E-07	1.68E-07	1.09E-06	8.08E-07	1.79E-07	9.23E-07
1360.00	8.71E-07	1.71E-07	1.12E-06	8.07E-07	1.83E-07	9.40E-07
1380.00	8.69E-07	1.74E-07	1.14E-06	8.05E-07	1.87E-07	9.58E-07
1400.00	8.66E-07	1.78E-07	1.17E-06	8.04E-07	1.91E-07	9.76E-07
1420.00	8.64E-07	1.81E-07	1.20E-06	8.03E-07	1.95E-07	9.96E-07
1440.00	8.61E-07	1.85E-07	1.22E-06	8.01E-07	1.99E-07	1.02E-06

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 5

				FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	9.504E-07	m	960.00
MAX. TRANS. IN Y-DIRECTION	-	1.848E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	1.224E-06	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	9.606E-07	m	960.00

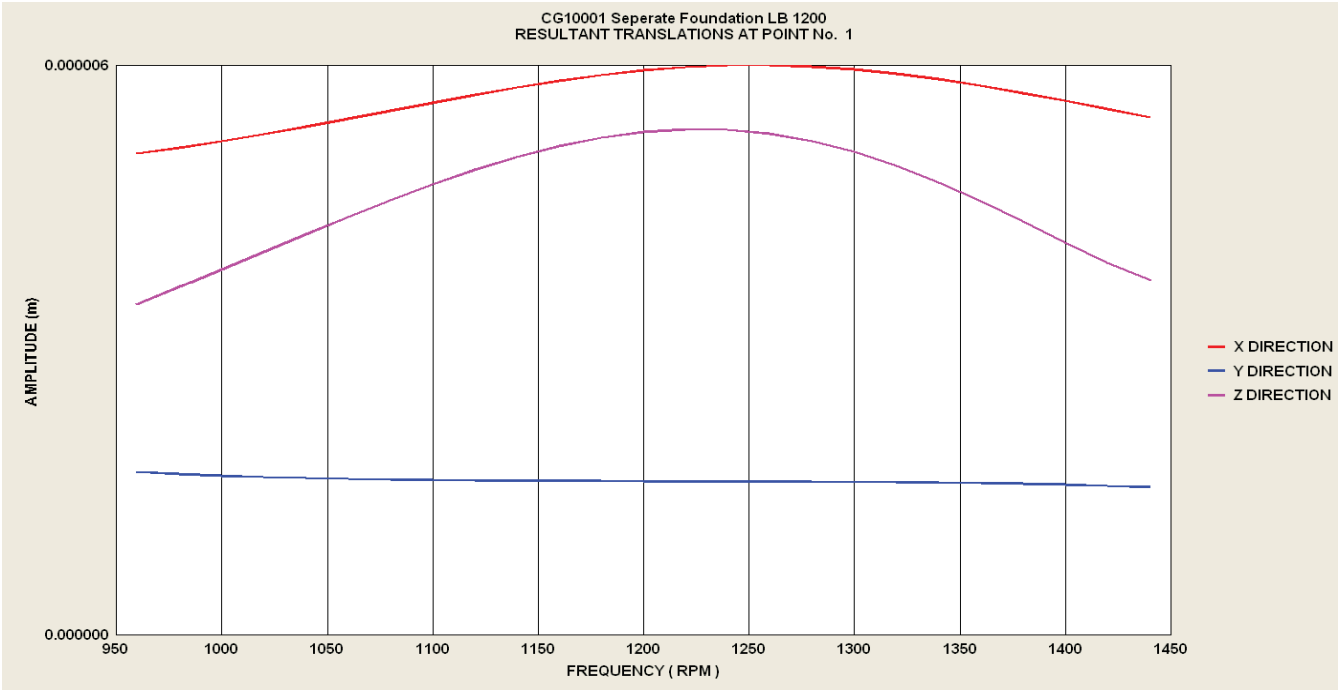
MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 6

				FREQ. RPM

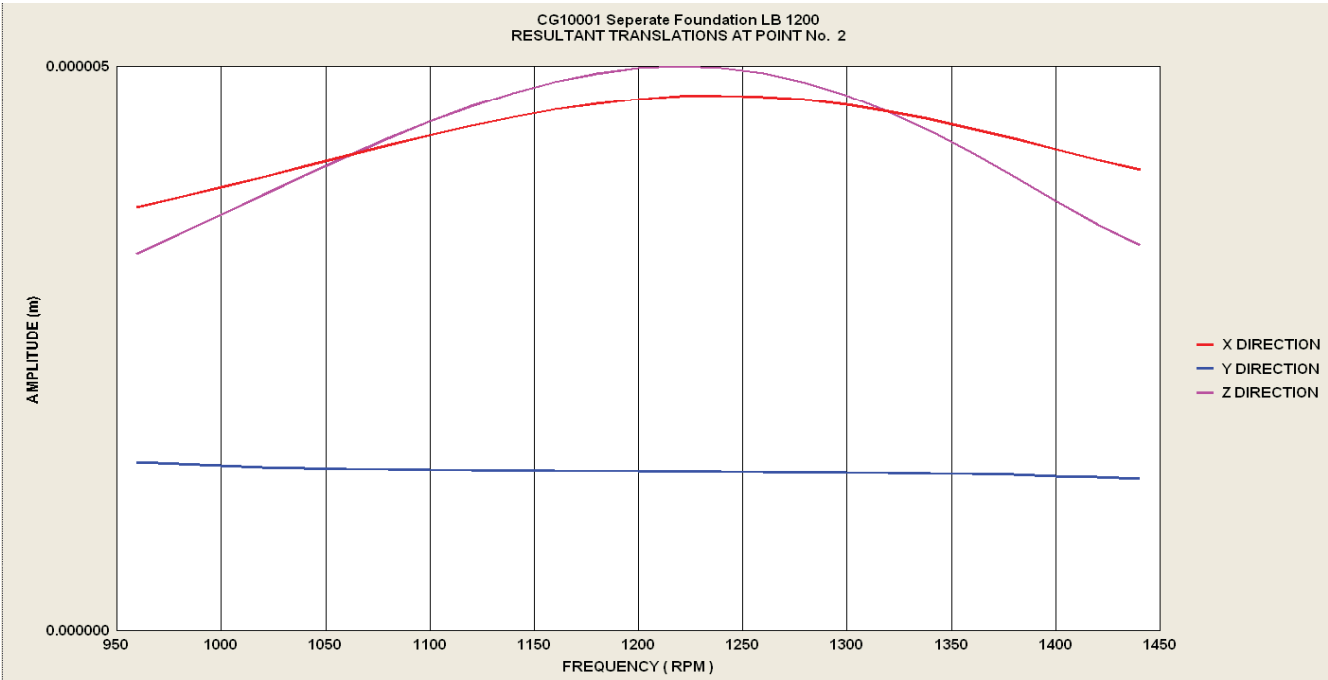
MAX. TRANS. IN X-DIRECTION	-	8.438E-07	m	960.00
MAX. TRANS. IN Y-DIRECTION	-	1.988E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	1.016E-06	m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	8.558E-07	m	960.00

13.7 DYNA 5 Lower Bound Input & Output for Separated Foundation

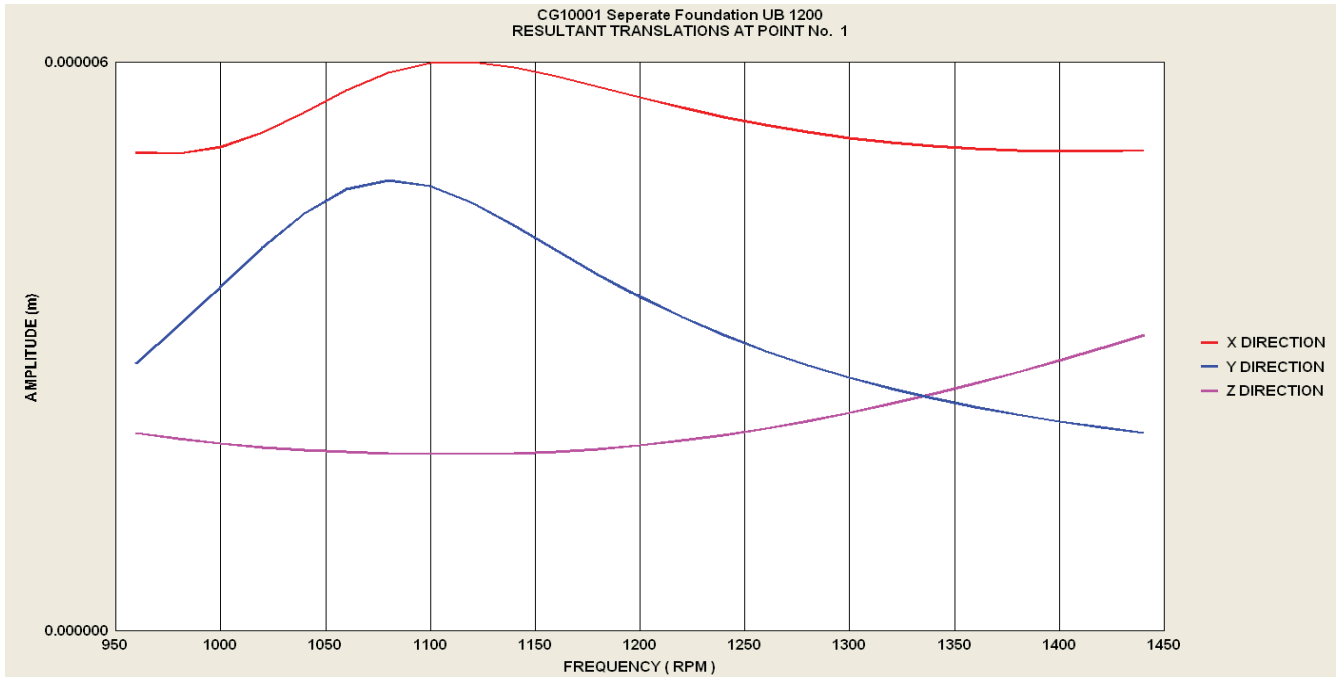
Vibration Amplitude at Compressor CG Analyzing Each
Foundation as Separate Foundation blocks
Lower Bound



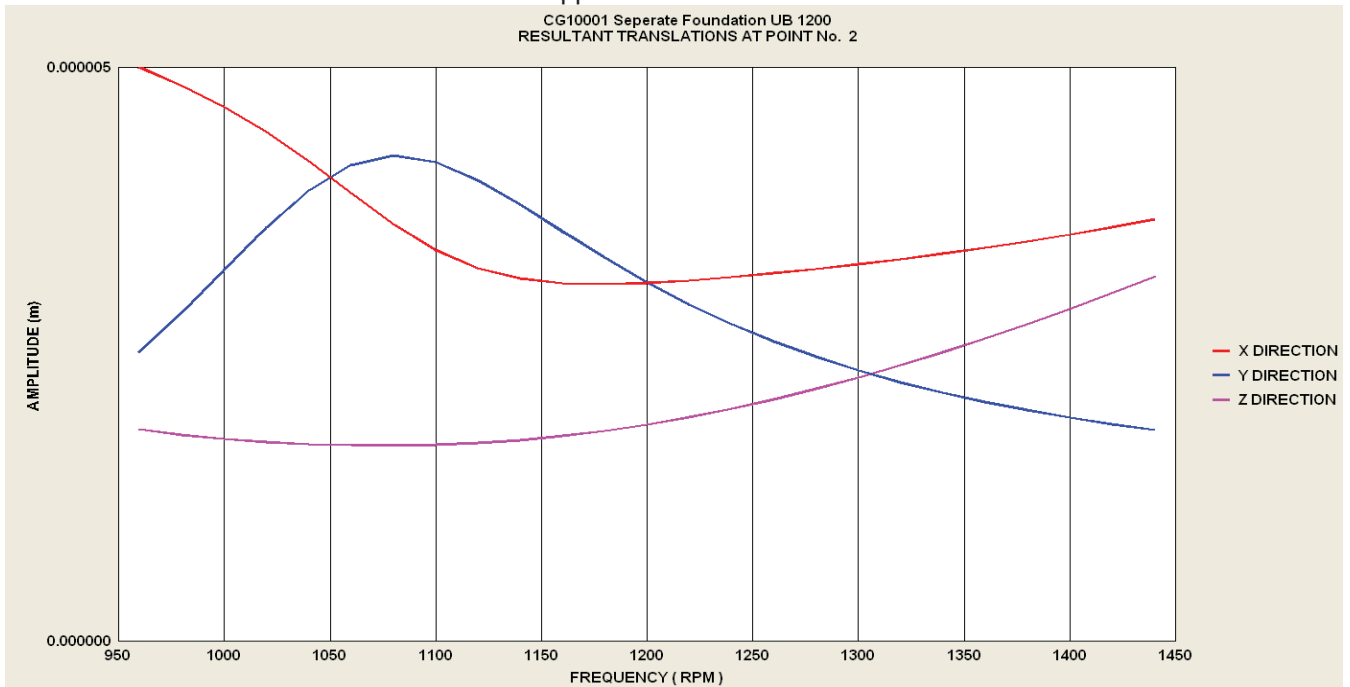
Vibration Amplitude at Motor CG Analyzing Each
Foundation as Separate foundation blocks
Lower Bound



Vibration Amplitude at Compressor CG Analyzing Each
Foundation as Separate Foundation blocks
Up Bound



Vibration Amplitude at Motor CG Analyzing Each
Foundation as Separate foundation blocks
Upper Bound



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*
*   Copyright (c) M. Novak, U.W.O. - 1999
*           RUN DATE - 2016/ 8/ 4
*           TIME    - 13:34:23
*           REVISION - 1999/04/23
*
*****

CG10001 Seperate Foundation LB 1200

                DATA ECHO
                *****

GRAVITATIONAL CONSTANT SET TO 9.81 m/s**2
*****

FREQUENCY UNITS : RPM
*****

LENGTH UNITS : m
*****

FORCE UNITS : N
*****

MASS UNITS : kg
*****

DAMPING SAFETY FACTOR - 2.00
*****

FOUNDATION TYPE - PILE
*****

RIGID CAP
*****

RECTANGULAR FOUNDATION - 3.900 m BY 6.000 m
*****

MASS CONSTANTS
*****

TOTAL MASS OF FOOTING ..... 1.3795E+05 kg
MASS MOMENT ABOUT X-AXIS ..... 5.7468E+05 kg.m**2
MASS MOMENT ABOUT Y-AXIS ..... 2.8588E+05 kg.m**2
MASS MOMENT ABOUT Z-AXIS ..... 4.9637E+05 kg.m**2
CROSS PRODUCT X-Y ..... 5.9685E+02 kg.m**2
CROSS PRODUCT X-Z ..... -1.4389E+03 kg.m**2
CROSS PRODUCT Y-Z ..... -6.1181E+03 kg.m**2

NUMBER OF LAYERS OR ELEMENTS - 3
*****

FILES ARE FIXED AT BUTT
*****

LOCATION OF PILES
*****

PILE          X          Y
NO.    LOCATION LOCATION
          m          m

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1	-1.244	-2.241
2	-1.244	0.009
3	-1.244	2.259
4	1.256	-2.241
5	1.256	0.009
6	1.256	2.259

COORDINATES OF BASE CENTRE

X-COORD. OF BASE CENTRE REL. TO C.G.	..	0.044	m
Y-COORD. OF BASE CENTRE REL. TO C.G.	..	-0.009	m
Z-COORD. OF BASE CENTRE REL. TO C.G.	..	1.583	m

PILE PROPERTIES

TOTAL LENGTH		20.250	m
UNIT WEIGHT		77000.000	N/m**3
STATIC LOAD ON PILE		5.580E+05	N
POISSON'S RATIO		0.300	
MATERIAL DAMPING		0.010	
COEFF. OF RIGIDITY		1.110	
YOUNG'S MODULUS		2.000E+11	N/m**2

ELEMENT

LAYER NO.	LAYER DEPTH m	PILE RADIUS (X) m	PILE RADIUS (Y) m	CROSS SECTIONAL AREA m**2	MOMENT OF INERTIA ABOUT Y m**4	MOMENT OF INERTIA ABOUT X m**4	TORSIONAL CONSTANT m**4
1	0.250	0.303	0.303	0.021	9.31E-04	9.31E-04	1.86E-03
2	13.250	0.303	0.303	0.021	9.31E-04	9.31E-04	1.86E-03
3	6.750	0.303	0.303	0.021	9.31E-04	9.31E-04	1.86E-03

PILE TIP CONDITION : FLOATING (DEFAULT)

PILE-SOIL-PILE INTERACTION IS CONSIDERED

SOIL

SOIL CONSTANTS

LAYER NO.	SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
1	0.000	0.00	0.000	0.000
2	100.000	20000.00	0.350	0.040
3	200.000	21000.00	0.350	0.040

SOIL BELOW PILE

SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
2.00E+02	21000.00	0.350	0.040

EMBEDDED OPTION

NUMBER OF SIDE LAYERS - 1

LAYER NO.	THICKNESS OF LAYER m	SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
1	0.000	0.00	0.00	0.000	0.000

HARMONIC LOAD (QUADRATIC)

MAXIMUM FREQUENCY		1440.000	RPM
MINIMUM FREQUENCY		960.000	RPM
STEP FREQUENCY		20.000	RPM
FORCE IN X-DIRECTION		2.48E-01	kg.m
FORCE IN Y-DIRECTION		0.00E+00	kg.m
FORCE IN Z-DIRECTION		2.48E-01	kg.m
MOMENT ABOUT X-AXIS		-1.16E-01	kg.m**2
MOMENT ABOUT Y-AXIS		8.10E-01	kg.m**2
MOMENT ABOUT Z-AXIS		1.16E-01	kg.m**2

RESULTANT TRANSLATIONS REQUIRED AT 2 POINTS

COORDINATES OF POINTS :

POINT NO.	X m	Y m	Z m
1	-3.909	1.504	-2.342
2	-3.909	-0.991	-2.512

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*
*           D Y N A 5   S I M U L A T I O N
*
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*   RUN DATE - 2016/ 8/ 4
*   TIME - 13:34:23
*   REVISION - 1999/04/23
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CG10001 Seperate Foundation LB 1200

RESULTS

FOOTING RESPONSE AMPLITUDES

FREQ. RPM	TRANS. X DIRECTION (m)	TRANS. Y DIRECTION (m)	VERTICAL DIRECTION (m)	ROT ABOUT X AXIS (rad)	ROT ABOUT Y AXIS (rad)	TORSIONAL Z AXIS (rad)
960.00	3.44E-06	1.61E-07	1.24E-06	1.42E-07	1.17E-06	3.76E-07
980.00	3.45E-06	1.63E-07	1.26E-06	1.47E-07	1.22E-06	3.69E-07
1000.00	3.47E-06	1.66E-07	1.27E-06	1.52E-07	1.27E-06	3.63E-07
1020.00	3.48E-06	1.68E-07	1.29E-06	1.57E-07	1.31E-06	3.58E-07
1040.00	3.50E-06	1.70E-07	1.31E-06	1.62E-07	1.36E-06	3.54E-07
1060.00	3.52E-06	1.72E-07	1.33E-06	1.65E-07	1.40E-06	3.50E-07
1080.00	3.54E-06	1.75E-07	1.35E-06	1.69E-07	1.44E-06	3.46E-07
1100.00	3.56E-06	1.77E-07	1.37E-06	1.72E-07	1.48E-06	3.43E-07
1120.00	3.58E-06	1.80E-07	1.39E-06	1.74E-07	1.51E-06	3.41E-07
1140.00	3.60E-06	1.83E-07	1.41E-06	1.76E-07	1.54E-06	3.39E-07
1160.00	3.63E-06	1.86E-07	1.44E-06	1.76E-07	1.56E-06	3.37E-07
1180.00	3.65E-06	1.90E-07	1.47E-06	1.77E-07	1.58E-06	3.35E-07
1200.00	3.67E-06	1.94E-07	1.50E-06	1.76E-07	1.58E-06	3.35E-07
1220.00	3.70E-06	1.98E-07	1.54E-06	1.75E-07	1.58E-06	3.34E-07
1240.00	3.72E-06	2.03E-07	1.58E-06	1.73E-07	1.58E-06	3.34E-07
1260.00	3.74E-06	2.08E-07	1.62E-06	1.71E-07	1.56E-06	3.35E-07
1280.00	3.77E-06	2.14E-07	1.67E-06	1.69E-07	1.53E-06	3.36E-07
1300.00	3.79E-06	2.20E-07	1.73E-06	1.66E-07	1.50E-06	3.37E-07
1320.00	3.82E-06	2.25E-07	1.80E-06	1.64E-07	1.46E-06	3.39E-07
1340.00	3.85E-06	2.31E-07	1.87E-06	1.62E-07	1.41E-06	3.41E-07
1360.00	3.88E-06	2.36E-07	1.96E-06	1.61E-07	1.36E-06	3.43E-07
1380.00	3.91E-06	2.41E-07	2.05E-06	1.60E-07	1.30E-06	3.44E-07
1400.00	3.94E-06	2.45E-07	2.16E-06	1.61E-07	1.24E-06	3.46E-07
1420.00	3.97E-06	2.48E-07	2.28E-06	1.63E-07	1.19E-06	3.47E-07
1440.00	4.00E-06	2.50E-07	2.42E-06	1.66E-07	1.16E-06	3.47E-07

MAXIMA OF FOOTING RESPONSE AMPLITUDES

				FREQ. RPM
MAX. TRANS. IN X-DIRECTION	-	4.000E-06	m	1440.00
MAX. TRANS. IN Y-DIRECTION	-	2.498E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	2.424E-06	m	1440.00
MAX. ROT. ABOUT X AXIS	-	1.766E-07	rad	1180.00
MAX. ROT. ABOUT Y AXIS	-	1.585E-06	rad	1200.00
MAX. ROT. ABOUT Z AXIS	-	3.756E-07	rad	960.00

RESULTANT TRANSLATIONS (m)

POINT NO. 1				POINT NO. 2		
-----				-----		
FREQ.	HORIZ. X	HORIZ. Y	VERTICAL	HORIZ. X	HORIZ. Y	VERTICAL
RPM	DIRECTION	DIRECTION	DIRECTION	DIRECTION	DIRECTION	DIRECTION
-----	-----	-----	-----	-----	-----	-----
960.00	4.66E-06	1.58E-06	3.20E-06	3.97E-06	1.58E-06	3.53E-06
980.00	4.71E-06	1.56E-06	3.37E-06	4.05E-06	1.56E-06	3.71E-06
1000.00	4.78E-06	1.55E-06	3.54E-06	4.15E-06	1.55E-06	3.89E-06
1020.00	4.84E-06	1.53E-06	3.71E-06	4.24E-06	1.53E-06	4.08E-06
1040.00	4.92E-06	1.52E-06	3.88E-06	4.34E-06	1.52E-06	4.26E-06
1060.00	4.99E-06	1.51E-06	4.05E-06	4.44E-06	1.52E-06	4.43E-06
1080.00	5.07E-06	1.51E-06	4.21E-06	4.54E-06	1.51E-06	4.60E-06
1100.00	5.15E-06	1.50E-06	4.36E-06	4.64E-06	1.51E-06	4.76E-06
1120.00	5.22E-06	1.50E-06	4.50E-06	4.72E-06	1.50E-06	4.91E-06
1140.00	5.30E-06	1.50E-06	4.62E-06	4.81E-06	1.50E-06	5.03E-06
1160.00	5.36E-06	1.49E-06	4.73E-06	4.88E-06	1.50E-06	5.14E-06
1180.00	5.42E-06	1.49E-06	4.81E-06	4.93E-06	1.49E-06	5.22E-06
1200.00	5.46E-06	1.49E-06	4.87E-06	4.98E-06	1.49E-06	5.27E-06
1220.00	5.49E-06	1.49E-06	4.89E-06	5.00E-06	1.49E-06	5.29E-06
1240.00	5.51E-06	1.49E-06	4.89E-06	5.01E-06	1.49E-06	5.27E-06
1260.00	5.51E-06	1.48E-06	4.85E-06	5.00E-06	1.49E-06	5.22E-06
1280.00	5.50E-06	1.48E-06	4.78E-06	4.97E-06	1.48E-06	5.13E-06
1300.00	5.47E-06	1.48E-06	4.67E-06	4.92E-06	1.48E-06	5.01E-06
1320.00	5.43E-06	1.48E-06	4.54E-06	4.86E-06	1.48E-06	4.86E-06
1340.00	5.37E-06	1.48E-06	4.38E-06	4.78E-06	1.48E-06	4.67E-06
1360.00	5.31E-06	1.47E-06	4.20E-06	4.70E-06	1.47E-06	4.47E-06
1380.00	5.24E-06	1.47E-06	4.00E-06	4.60E-06	1.46E-06	4.25E-06
1400.00	5.17E-06	1.46E-06	3.80E-06	4.50E-06	1.45E-06	4.02E-06
1420.00	5.09E-06	1.45E-06	3.60E-06	4.41E-06	1.44E-06	3.80E-06
1440.00	5.01E-06	1.43E-06	3.44E-06	4.31E-06	1.42E-06	3.61E-06

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 1

				FREQ.
				RPM

MAX. TRANS. IN X-DIRECTION	-	5.508E-06	m	1260.00
MAX. TRANS. IN Y-DIRECTION	-	1.581E-06	m	960.00
MAX. TRANS. IN Z-DIRECTION	-	4.893E-06	m	1220.00
MAX. TRANS. IN HORIZ. PLANE	-	5.705E-06	m	1260.00

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 2

				FREQ.
				RPM

MAX. TRANS. IN X-DIRECTION	-	5.010E-06	m	1240.00
MAX. TRANS. IN Y-DIRECTION	-	1.581E-06	m	960.00
MAX. TRANS. IN Z-DIRECTION	-	5.287E-06	m	1220.00
MAX. TRANS. IN HORIZ. PLANE	-	5.227E-06	m	1240.00


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*           D Y N A 5   S I M U L A T I O N
*
*   Copyright (c)  M. Novak,  U.W.O.  - 1999
*           RUN DATE - 2016/ 8/ 8
*           TIME    - 9:41:15
*           REVISION - 1999/04/23
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CG10001 Seperate Foundation UB 1200

                DATA ECHO
                *****

GRAVITATIONAL CONSTANT SET TO 9.81 m/s**2
*****

FREQUENCY UNITS : RPM
*****

LENGTH UNITS : m
*****

FORCE UNITS : N
*****

MASS UNITS : kg
*****

DAMPING SAFETY FACTOR - 2.00
*****

FOUNDATION TYPE - PILE
*****

RIGID CAP
*****

RECTANGULAR FOUNDATION - 3.900 m BY 6.000 m
*****

MASS CONSTANTS
*****

TOTAL MASS OF FOOTING ..... 1.3795E+05 kg
MASS MOMENT ABOUT X-AXIS ..... 5.7468E+05 kg.m**2
MASS MOMENT ABOUT Y-AXIS ..... 2.8588E+05 kg.m**2
MASS MOMENT ABOUT Z-AXIS ..... 4.9637E+05 kg.m**2
CROSS PRODUCT X-Y ..... 5.9685E+02 kg.m**2
CROSS PRODUCT X-Z ..... -1.4389E+03 kg.m**2
CROSS PRODUCT Y-Z ..... -6.1181E+03 kg.m**2

NUMBER OF LAYERS OR ELEMENTS - 3
*****

FILES ARE FIXED AT BUTT
*****

LOCATION OF PILES
*****

PILE          X          Y
NO.    LOCATION LOCATION
          m          m

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1	-1.244	-2.241
2	-1.244	0.009
3	-1.244	2.259
4	1.256	-2.241
5	1.256	0.009
6	1.256	2.259

COORDINATES OF BASE CENTRE

X-COORD. OF BASE CENTRE REL. TO C.G.	..	0.044	m
Y-COORD. OF BASE CENTRE REL. TO C.G.	..	-0.009	m
Z-COORD. OF BASE CENTRE REL. TO C.G.	..	1.583	m

PILE PROPERTIES

TOTAL LENGTH		20.250	m
UNIT WEIGHT		77000.000	N/m**3
STATIC LOAD ON PILE		5.580E+05	N
POISSON'S RATIO		0.300	
MATERIAL DAMPING		0.010	
COEFF. OF RIGIDITY		1.110	
YOUNG'S MODULUS		2.000E+11	N/m**2

ELEMENT

LAYER NO.	LAYER DEPTH m	PILE RADIUS (X) m	PILE RADIUS (Y) m	CROSS SECTIONAL AREA m**2	MOMENT OF INERTIA ABOUT Y m**4	MOMENT OF INERTIA ABOUT X m**4	TORSIONAL CONSTANT m**4
1	0.250	0.305	0.305	0.024	1.06E-03	1.06E-03	2.13E-03
2	13.250	0.305	0.305	0.024	1.06E-03	1.06E-03	2.13E-03
3	6.750	0.305	0.305	0.024	1.06E-03	1.06E-03	2.13E-03

PILE TIP CONDITION : FLOATING (DEFAULT)

PILE-SOIL-PILE INTERACTION IS CONSIDERED

SOIL

SOIL CONSTANTS

LAYER NO.	SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
1	0.000	0.00	0.000	0.000
2	200.000	20000.00	0.350	0.040
3	400.000	21000.00	0.350	0.040

SOIL BELOW PILE

SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
4.00E+02	21000.00	0.350	0.040

EMBEDDED OPTION

NUMBER OF SIDE LAYERS - 1

LAYER NO.	THICKNESS OF LAYER m	SHEAR WAVE VELOCITY m/s	UNIT WEIGHT N/m**3	POISSON'S RATIO	MATERIAL DAMPING
1	0.000	0.00	0.00	0.000	0.000

HARMONIC LOAD (QUADRATIC)

MAXIMUM FREQUENCY		1440.000	RPM
MINIMUM FREQUENCY		960.000	RPM
STEP FREQUENCY		20.000	RPM
FORCE IN X-DIRECTION		2.48E-01	kg.m
FORCE IN Y-DIRECTION		0.00E+00	kg.m
FORCE IN Z-DIRECTION		2.48E-01	kg.m
MOMENT ABOUT X-AXIS		-1.16E-01	kg.m**2
MOMENT ABOUT Y-AXIS		8.10E-01	kg.m**2
MOMENT ABOUT Z-AXIS		1.16E-01	kg.m**2

RESULTANT TRANSLATIONS REQUIRED AT 2 POINTS

COORDINATES OF POINTS :

POINT NO.	X m	Y m	Z m
1	-3.909	1.504	-2.342
2	-3.909	-0.991	-2.512

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CG10001 Seperate Foundation UB 1200

RESULTS

FOOTING RESPONSE AMPLITUDES

FREQ. RPM	TRANS. X DIRECTION (m)	TRANS. Y DIRECTION (m)	VERTICAL DIRECTION (m)	ROT ABOUT X AXIS (rad)	ROT ABOUT Y AXIS (rad)	TORSIONAL Z AXIS (rad)
960.00	4.21E-06	1.66E-07	9.41E-07	5.70E-08	5.64E-07	6.47E-07
980.00	4.14E-06	1.65E-07	9.82E-07	6.02E-08	5.69E-07	7.34E-07
1000.00	4.07E-06	1.64E-07	1.02E-06	6.36E-08	5.79E-07	8.27E-07
1020.00	4.02E-06	1.63E-07	1.07E-06	6.72E-08	5.91E-07	9.20E-07
1040.00	3.97E-06	1.63E-07	1.11E-06	7.10E-08	6.07E-07	1.00E-06
1060.00	3.93E-06	1.63E-07	1.15E-06	7.52E-08	6.25E-07	1.06E-06
1080.00	3.89E-06	1.63E-07	1.19E-06	7.96E-08	6.45E-07	1.08E-06
1100.00	3.85E-06	1.63E-07	1.24E-06	8.43E-08	6.67E-07	1.06E-06
1120.00	3.82E-06	1.64E-07	1.28E-06	8.91E-08	6.91E-07	1.03E-06
1140.00	3.80E-06	1.64E-07	1.33E-06	9.40E-08	7.17E-07	9.71E-07
1160.00	3.78E-06	1.65E-07	1.37E-06	9.90E-08	7.46E-07	9.13E-07
1180.00	3.76E-06	1.67E-07	1.42E-06	1.04E-07	7.76E-07	8.55E-07
1200.00	3.74E-06	1.68E-07	1.46E-06	1.09E-07	8.08E-07	8.02E-07
1220.00	3.73E-06	1.69E-07	1.51E-06	1.15E-07	8.42E-07	7.54E-07
1240.00	3.72E-06	1.71E-07	1.56E-06	1.20E-07	8.78E-07	7.11E-07
1260.00	3.71E-06	1.73E-07	1.60E-06	1.26E-07	9.15E-07	6.73E-07
1280.00	3.70E-06	1.75E-07	1.65E-06	1.32E-07	9.54E-07	6.40E-07
1300.00	3.70E-06	1.77E-07	1.70E-06	1.38E-07	9.94E-07	6.11E-07
1320.00	3.69E-06	1.79E-07	1.74E-06	1.44E-07	1.04E-06	5.85E-07
1340.00	3.69E-06	1.81E-07	1.79E-06	1.50E-07	1.08E-06	5.62E-07
1360.00	3.69E-06	1.83E-07	1.84E-06	1.57E-07	1.12E-06	5.41E-07
1380.00	3.69E-06	1.86E-07	1.88E-06	1.64E-07	1.17E-06	5.23E-07
1400.00	3.69E-06	1.88E-07	1.93E-06	1.70E-07	1.22E-06	5.06E-07
1420.00	3.70E-06	1.90E-07	1.98E-06	1.77E-07	1.26E-06	4.91E-07
1440.00	3.70E-06	1.93E-07	2.03E-06	1.84E-07	1.31E-06	4.78E-07

MAXIMA OF FOOTING RESPONSE AMPLITUDES

				FREQ. RPM
MAX. TRANS. IN X-DIRECTION	-	4.205E-06	m	960.00
MAX. TRANS. IN Y-DIRECTION	-	1.930E-07	m	1440.00
MAX. TRANS. IN Z-DIRECTION	-	2.026E-06	m	1440.00
MAX. ROT. ABOUT X AXIS	-	1.843E-07	rad	1440.00
MAX. ROT. ABOUT Y AXIS	-	1.312E-06	rad	1440.00
MAX. ROT. ABOUT Z AXIS	-	1.078E-06	rad	1080.00

RESULTANT TRANSLATIONS (m)

	POINT NO. 1			POINT NO. 2		
	-----			-----		
FREQ. RPM	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION	HORIZ. X DIRECTION	HORIZ. Y DIRECTION	VERTICAL DIRECTION
-----	-----	-----	-----	-----	-----	-----
960.00	4.65E-06	2.60E-06	1.92E-06	5.18E-06	2.61E-06	1.92E-06
980.00	4.63E-06	2.96E-06	1.86E-06	5.01E-06	2.97E-06	1.87E-06
1000.00	4.70E-06	3.34E-06	1.82E-06	4.82E-06	3.35E-06	1.83E-06
1020.00	4.84E-06	3.73E-06	1.78E-06	4.59E-06	3.74E-06	1.80E-06
1040.00	5.04E-06	4.06E-06	1.75E-06	4.33E-06	4.07E-06	1.78E-06
1060.00	5.25E-06	4.29E-06	1.73E-06	4.04E-06	4.30E-06	1.77E-06
1080.00	5.42E-06	4.38E-06	1.72E-06	3.76E-06	4.38E-06	1.77E-06
1100.00	5.51E-06	4.32E-06	1.71E-06	3.53E-06	4.33E-06	1.78E-06
1120.00	5.52E-06	4.16E-06	1.71E-06	3.36E-06	4.16E-06	1.80E-06
1140.00	5.47E-06	3.94E-06	1.72E-06	3.27E-06	3.94E-06	1.82E-06
1160.00	5.38E-06	3.70E-06	1.74E-06	3.23E-06	3.70E-06	1.86E-06
1180.00	5.28E-06	3.46E-06	1.77E-06	3.22E-06	3.46E-06	1.91E-06
1200.00	5.18E-06	3.24E-06	1.80E-06	3.23E-06	3.24E-06	1.96E-06
1220.00	5.08E-06	3.04E-06	1.85E-06	3.25E-06	3.04E-06	2.03E-06
1240.00	4.99E-06	2.87E-06	1.90E-06	3.28E-06	2.86E-06	2.11E-06
1260.00	4.91E-06	2.71E-06	1.97E-06	3.32E-06	2.71E-06	2.19E-06
1280.00	4.84E-06	2.58E-06	2.04E-06	3.36E-06	2.57E-06	2.29E-06
1300.00	4.79E-06	2.45E-06	2.12E-06	3.40E-06	2.45E-06	2.39E-06
1320.00	4.74E-06	2.35E-06	2.21E-06	3.45E-06	2.34E-06	2.50E-06
1340.00	4.70E-06	2.25E-06	2.30E-06	3.50E-06	2.25E-06	2.62E-06
1360.00	4.68E-06	2.17E-06	2.40E-06	3.55E-06	2.16E-06	2.74E-06
1380.00	4.66E-06	2.10E-06	2.51E-06	3.61E-06	2.09E-06	2.87E-06
1400.00	4.66E-06	2.03E-06	2.63E-06	3.67E-06	2.02E-06	3.01E-06
1420.00	4.66E-06	1.97E-06	2.75E-06	3.74E-06	1.97E-06	3.15E-06
1440.00	4.67E-06	1.92E-06	2.87E-06	3.81E-06	1.92E-06	3.30E-06

MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 1

			FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	5.520E-06 m	1120.00
MAX. TRANS. IN Y-DIRECTION	-	4.376E-06 m	1080.00
MAX. TRANS. IN Z-DIRECTION	-	2.871E-06 m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	7.002E-06 m	1100.00

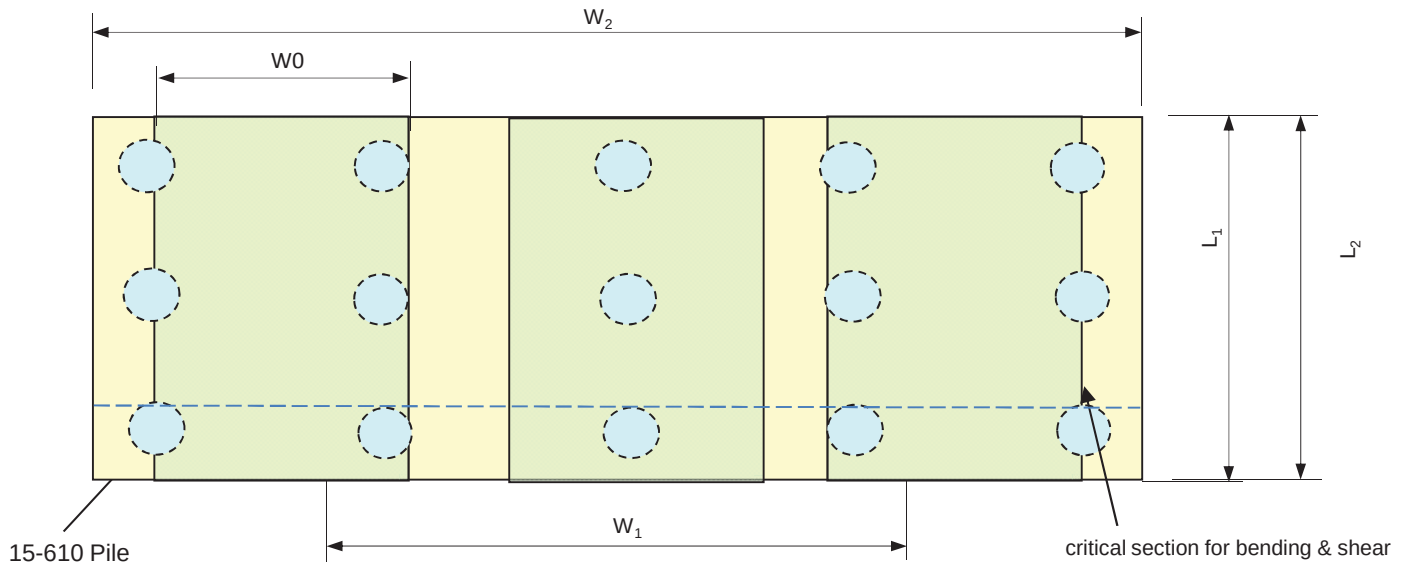
MAXIMUM RESULTANT TRANSLATIONS AT POINT NO. 2

			FREQ. RPM

MAX. TRANS. IN X-DIRECTION	-	5.177E-06 m	960.00
MAX. TRANS. IN Y-DIRECTION	-	4.383E-06 m	1080.00
MAX. TRANS. IN Z-DIRECTION	-	3.296E-06 m	1440.00
MAX. TRANS. IN HORIZ. PLANE	-	5.942E-06 m	1040.00

Unit 10: 10-CG-10001A/B/C

14. Static Check of Pile & FDN



Pedestal width, W ₁ =	7500	mm	W ₁ = 3*W ₀
Pedestal Lth, L ₁ =	6000	mm	
Pile cap Lth, L ₂ =	6000	mm	
Pile cap width, W ₂ =	11500	mm	
Pedestal Lth, L ₃ =	0	mm	
Pedestal width, W ₃ =	0	mm	
Pile cap ht, h =	900.0	mm	
Pedestal Ht, hp =	2100.0	mm (include grout)	

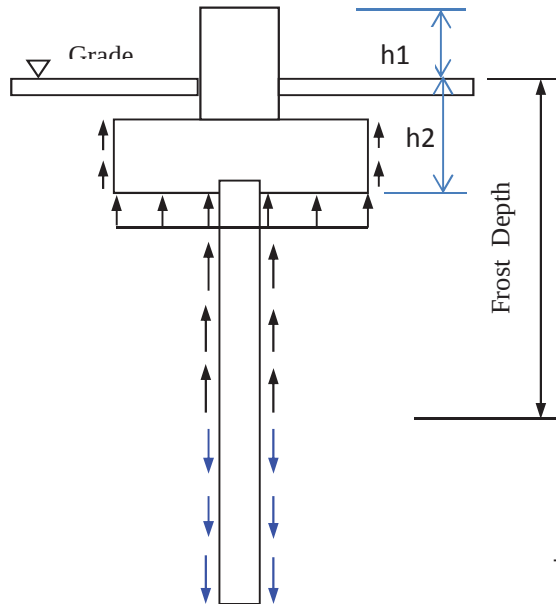
Diameter of the pile, d _p =	610	mm
No. of piles, n =	15	

Static pile Vertical loads

Weight of pile cap	= (6*11.5*0.9* 25) /15 =	104	kN
Weight of pedestal+grout	= (6*7.5*2.1+0*0*2.1)*25 /15 =	158	kN
Weight of equipment	= 22416*9.81/1000/15 =	25	kN
Live loads and soil surcharge	= (11.5*6-7.5*6-0*0)*4.8/15 =	39.7	kN
static pump torques	= (24+22.5+1.5)/4/5 =	2.40	kN
Total factored load:	= 1.25*(103.5+157.5+25)+ 1.5 * (2.4+39.7)=	421	kN per pile

Unit 10: 10-CG-10001A/B/C

Pile Frost Heave Loads



Adfreeze on Pile:	100	kPa
Adfreeze on Pile Cap:	0	kPa
Frost Heave:	60	kPa
α_T =	1	(Soil Report)
Steel Pile Dia.:	610	mm
Pile Circumference:	1.92	m
Pile Area:	0.29	m ²
Depth of Frost Penetration:	3	m
Pile Cap Area:	69.0	m ²
Pile Cap Perimeter:	35.0	m
Pile Cap Selfweight:	1553	kN
Pedestal Perimeter:	19.5	m
Pedstal Selfweight:	2363	kN
U/S base PI to T.O. grade, h1 =	1050.0	mm
T.o. Grade to U/S void form, h2 =	1925.0	mm

Frost Uplift per Pile: (operating)

Depth of Freezing of Pile	= 3 - 1.925 =	1.08	m
Adfreeze on Pile	= 1 * 1.92 * 100 * 1.08 =	207.4	kN
Adfreeze on Pile Cap	= 1 * 0 * 35 * 0.9 / 15 =	0.0	kN
Adfreeze on Pedestal	= 1 * 0 * 19.5 * (2.1 - 1.05) / 15 =	0.0	kN
Frost Heave on Pile Cap	= 1 * 60 * (69 - 0.293 * 15) / 15 =	258.4	kN
Pedestal & Pile Cap Selfweight	= 0.9 * (2363 + 1552.5) / 15 =	-234.9	kN
Frost Uplift per Pile:		230.9	kN

Frost Uplift per Pile: (construction)

Depth of Freezing of Pile		3.00	m
Adfreeze on Pile	= 1 * 1.92 * 100 * 3 =	576.0	kN
Frost Uplift per Pile:		576.0	kN

Pile Capacity

Pile compression capacity (20.0m embed)		1271	kN	
Pile frost uplift capacity (20m embed)	= 0.8 * (10.9 * 1.92 * 28 + 3 * 1.92 * 48 + 5 * 1.92 * 60) =	1151	kN	Operating
Pile frost uplift capacity (20m embed)	= 0.8 * (9 * 1.92 * 28 + 3 * 1.92 * 48 + 5 * 1.92 * 60) =	1069	kN	Construction

Pile Geo Capacity Check:

compression	= 420.65 / 1271 =	0.33
Frost uplift (operating)	= 230.9 / 1150.8 =	0.20
Frost uplift (construction)	= 576 / 1069.1 =	0.54

Unit 10: 10-CG-10001A/B/C

Estimated foundation settlement:

Pile spacing = $\frac{2250}{610} = 3.7 \times D$
 Group factor assuming 3x3 pile config. At 3.0D = 2.1
 Reference settlement = 6.0 mm
 Settlement = $420.65 / 1.25 / 1271 \times 6 \times 2.1 = 3.3 \text{ mm}$

Design of Pile Cap

a1. Check of flexural: (operating case)

Rebar size, $d_r =$	25 mm	
Dist. of critical section, $L =$	600 mm	$\phi_s = 0.85$
Depth of the pile cap, $h =$	900 mm	$\phi_c = 0.60$
Clear cover, $c =$	50 mm	$\lambda = 1$
Pile Embedment length, $E_p =$	100 mm	$f_c' = 30 \text{ MPa}$
$d = h - c - E_p - d_r =$	725 mm	$f_y = 400 \text{ MPa}$

Width of critical section, $b = L_2 = 6000 \text{ mm}$
 $\alpha_1 = 0.85 - 0.0015 f_c' = 0.805 \geq 0.67$
 $\beta_1 = 0.97 - 0.0025 f_c' = 0.895 \geq 0.67$

Critical section at mid of pile cap:

Max Factored Vertical loads on Pile 1, $R_1 = 420.7 \text{ kN}$
 Factored shear on critical section, $V_f = 3 \times R_1 = 1262.0 \text{ kN}$
 Factored moment on critical section, $M_f = 3 \times R_1 \times L = 757.2 \text{ kN-m}$

Check deep beam action

if $d \times V_f / M_f > 1.0$ $0.725 \times 1261.95 / 757.17 = 1.21 > 1.0$ **Deep Beam**

use $z = 0.4 (d + M_f / V_f)$ if $d \times V_f / M_f \geq 1.0$ $Z = 0.4 \times (0.725 + 757.17 / 1261.95) = 0.53 \text{ m}$

use $z = 1.2 \times (M_f / V_f)$ if $d \times V_f / M_f \geq 2.0$ $Z = 1.2 \times (757.17 / 1261.95) = \text{N/A}$

Reinforcement in pile cap, $A_s = 15250 \text{ mm}^2$ (No.25 @ 200mm)

$M_{r1} = A_s \times \phi_s \times f_y \times Z$ $M_{r1} = 2748 \text{ kN-m}$

$M_{cr} = 0.6 \cdot \lambda \cdot \sqrt{f_c'} \cdot \frac{b \cdot h^2}{6}$ $M_{cr} = 2661.9 \text{ kN-mm}$
 $1.2 M_{cr} = 3194.3 \text{ kN-mm}$

$M_r = \text{Max}(1.2 M_{cr}, M_{r1}) = 3194.3 \text{ kN-mm}$

Since $1.3 M_f = 984.3 \text{ kN-mm} < M_r$

Min. reinforcement requirement can be waived

Stress Ratio, $M_f / M_r = 0.24$

A23.3-04 [10.5.1]

Unit 10: 10-CG-10001A/B/C

a2. Check of flexural: (frost heave)

Critical section at mid of pile cap:

Max Factored tension loads on Pile cap per pile, $R' = 323.1 \text{ kN}$ Heave & adfreeze on pile cap

Factored shear on critical section, $V_f = 3 * R' = 969.3 \text{ kN}$

Factored moment on critical section, $M_f = 3 * R' * L = 581.6 \text{ kN-m}$

$d_{top} = h - c - d_r / 2 = 837.5 \text{ mm}$

Check deep beam action

if $d_{top} * V_f / M_f > 1.0$ $0.8375 * 969.27 / 581.562 = 1.40 > 1.0$ **Deep Beam**

use $z = 0.4 (d + M_f / V_f)$ if $d * V_f / M_f \geq 1.0$ $Z = 0.4 * (0 + 581.562 / 969.27) = 0.24m$

use $z = 1.2 * (M_f / V_f)$ if $d * V_f / M_f \geq 2.0$ $Z = 1.2 * (581.562 / 969.27) = \text{N/A}$

Reinforcement in pile cap, $A_s = 15250 \text{ mm}^2$ (No.25 @ 200mm)

$M_{r1} = A_s * \phi_s * f_y * Z$

$M_r = 1244 \text{ kN-m}$

Stress Ratio, $M_f / M_r = 0.47$

b. Check for one way Shear:

b.1 One-way shear by Pile 1 to 5:

$R_2 = 420.7 \text{ kN}$

$d_v = \max(0.9d, 0.72h) = 653 \text{ mm}$

$\beta = \frac{230}{(1000 + d_v)}$

$\beta = 0.139$

$v_c = \beta \cdot \lambda \cdot \phi_c \cdot \sqrt{f_c'}$

$v_c = 0.46 \text{ N/mm}^2$

width of critical section, $b = 6000.0 \text{ mm}$

$V_r = V_c = v_c * b * d_v = 1790.7 \text{ kN}$

$V_f = R_2 * 3 = 1262.0 \text{ kN}$

Stress Ratio, $V_f / V_r = 0.70$

A23.3-04 Definition

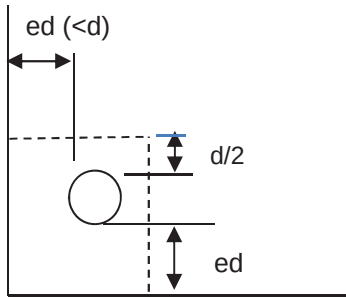
A23.3-04 [11.3.6.3]

A23.3-04 [13.3.4]

A23.3-04 [15.5.3]

Unit 10: 10-CG-10001A/B/C

b.2 One-way shear by Corner Pile



$$H_{eff} = 540.6 \text{ mm}$$

$$d_{top} = h - c - d_r / 2 = 837.5 \text{ mm}$$

$$d_v = \max(0.9d, 0.72h) = 653 \text{ mm}$$

$$e_d = 750 \text{ mm}$$

$$\beta = \frac{230}{(1000 + d_v)} \beta = 0.139$$

$$v_c = \beta \cdot \lambda \cdot \phi_c \cdot \sqrt{f_c'} \quad v_c = 0.46 \text{ N/mm}^2$$

width of critical section

$$b = (e_d + h_{eff} + \frac{d_{top}}{2}) + (d_{top} + h_{eff} + \frac{d_{top}}{2}) = 3419 \text{ mm}$$

$$V_r = V_c = v_c \cdot b \cdot d_v = 1020.3 \text{ kN}$$

$$V_f = R_2 = 420.7 \text{ kN}$$

$$\text{Stress Ratio, } V_f / V_r = 0.4$$

A23.3-04 Definition

A23.3-04 [11.3.6.3]

A23.3-04 [13.3.4]

15. Conclusion:

Pile FDN design of 10-CG-10001ABC are satisfactory.

15 Anchor Bolts Design

The anchor bolts design follows standard drawings of anchor bolts of the project and vendor drawings for diameter and material grade.

16 CONCLUSIONS:

The following is the summary of the analysis and design of the foundation:

1. Pile static capacity . Based on static analysis, maximum pile load is 421kN which well below the pile static capacity of 1271 kN as recommended by Thurber for 20 m embedment length. The anticipated foundation settlement is approximately 2.1 mm, therefore, acceptable.

2. Dynamic analysis results for machine performance . For the machine performance, amplitudes of vertical and horizontal vibration responses were checked at the compressor and motor. Analyses were carried out for two different soil shear wave velocities (high and low) to capture any possible worst case scenarios.

There are 3 pedestals side by side joined together by relatively thin pile cap. The results analyzing the the 3 foundations together as a rigid body may produce unconservative results. As an alternative of modeling the foundation with finite element software, another analysis considering only the mass and piles contributed to one pump are performed. The vibration responses were checked against the mechanical requirement, and there are satisfactory for both analysis model.

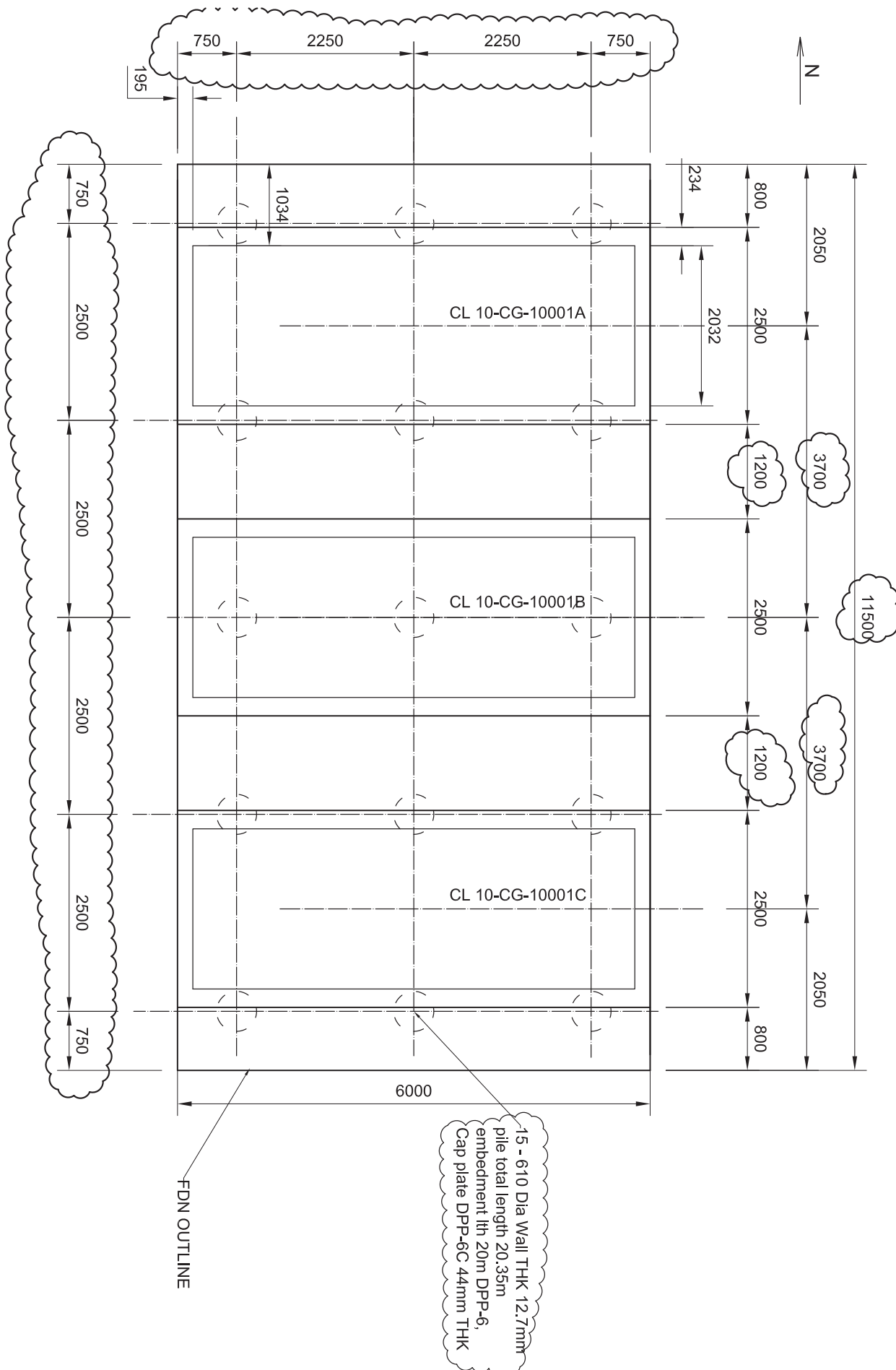
3. Resonance . From the dynamic analyses with the damping Safety Factor = 2.0, vibration responses do not show any resonance within the +/- 20% range of the operating frequencies. Therefore, considered acceptable.

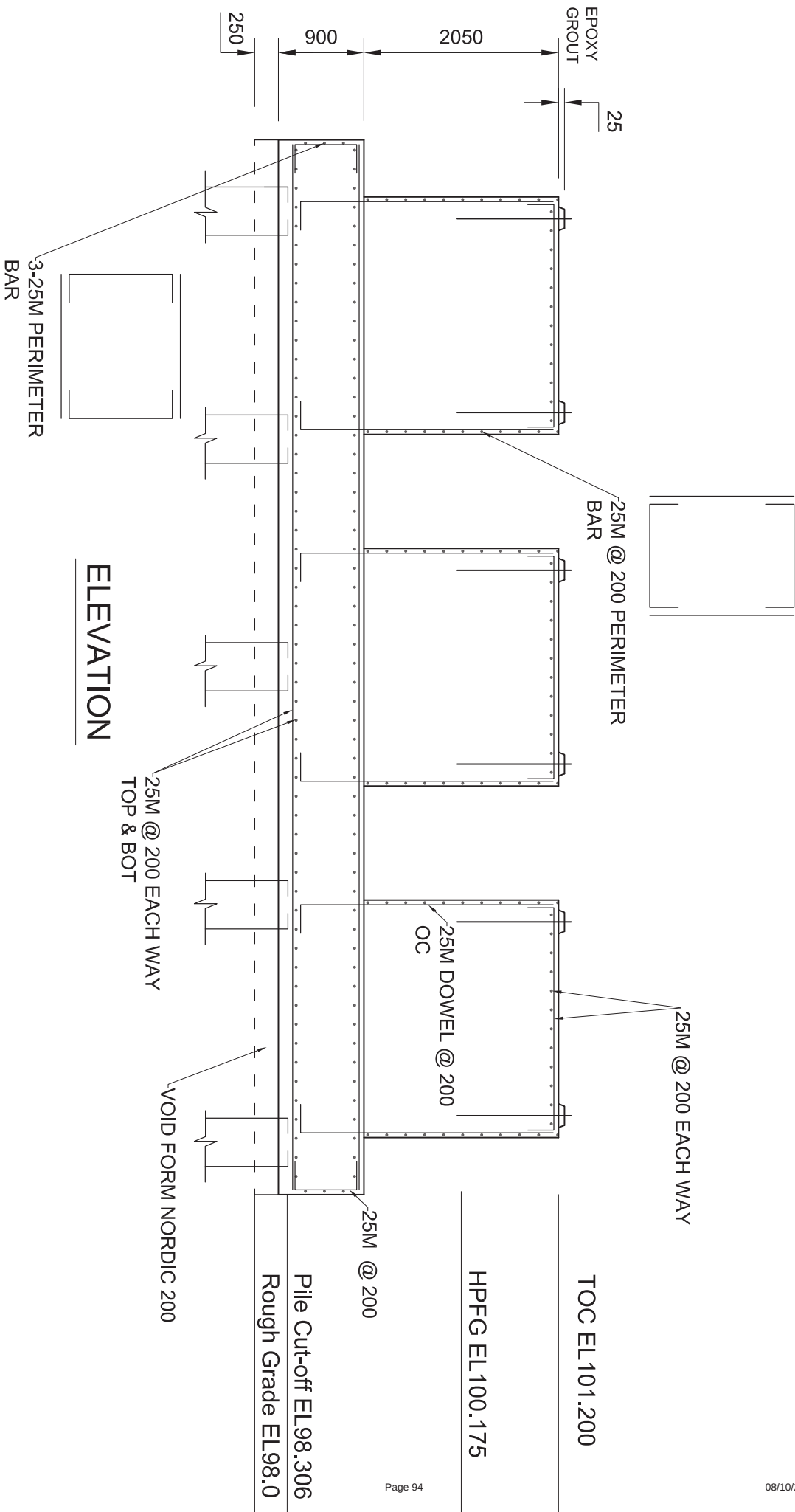
4. Foundation Vibration Velocity . From the dynamic analyses, the maximum foundation velocity up to the +20% of the operating frequency was 1.0mm/s for the models analyzed as "Isolated blocks" with one vibration source and 1.1m/s for the models analyzed as "one rigid block" with all 3 vibration sources accumulated. The allowable speed as per vendor drawing is 2.8 mm/s.

5. Dynamic analysis results for human perception . With respect to the human perception curves, the vibration responses generally within the "not noticeable to persons" range. Therefore, they are acceptable.

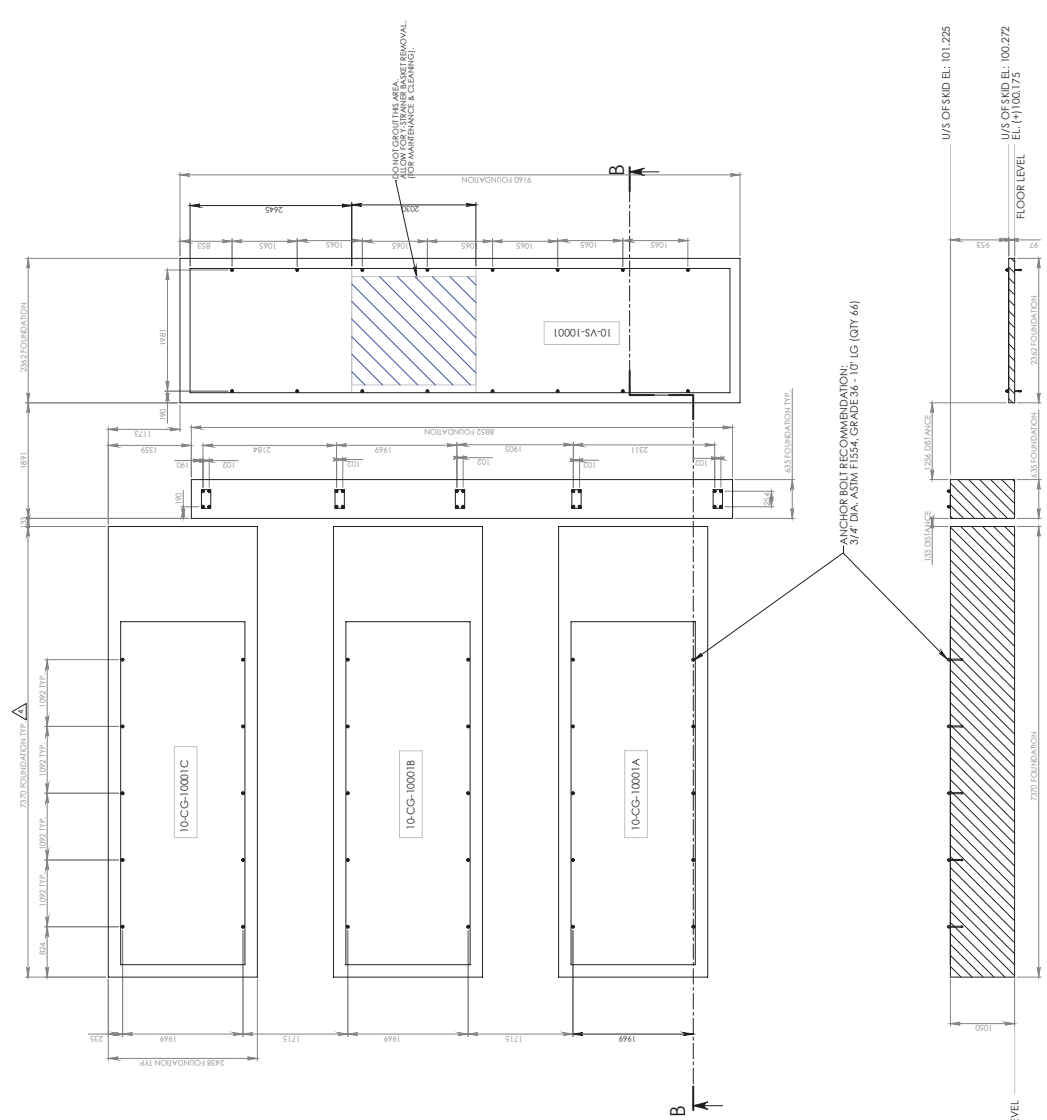
17.1 Appendix A: Foundation Sketches

10-CG-10001 ABC PILE LAYOUT





17.2 Appendix B: Vendor Documents and Drawings



ESTIMATED DRY WEIGHT (EACH SKID):
10-CG-10001 A/B/C

COMPRESSOR	3966 KG
MOTOR	5125 KG
BASE PLATE	2100 KG
PIPING & ACCESSORIES	1500 KG
TOTAL	12591 KG

ESTIMATED OPERATING WEIGHT (EACH SKID):
10-CG-10001 A/B/C

COMPRESSOR	3966 KG
MOTOR	5125 KG
BASE PLATE	2100 KG
PIPING & ACCESSORIES	1500 KG
TOTAL	12591 KG

ESTIMATED DRY WEIGHT:
10-VS-10001

SEPARATOR TANK	2750 KG
BASE PLATE	1010 KG
FOUNDATION	1500 KG
HEAT EXCHANGERS (2)	790 KG
PIPING & ACCESSORIES	1125 KG
TOTAL	7035 KG

ESTIMATED OPERATING WEIGHT:
10-VS-10001

SEPARATOR TANK	3070 KG
BASE PLATE	1010 KG
FOUNDATION	1500 KG
HEAT EXCHANGERS (2)	950 KG
PIPING & ACCESSORIES	1225 KG
TOTAL	7485 KG

NOTES:

- FOR CERTIFIED OVERALL GENERAL ARRANGEMENT DRAWING
SEE DOCUMENT A&V-4-0304-01-0001.4
- COMPRESSOR MODEL SB2-30-120H, 2ND STAGE-79 KG
MOTOR MODEL 1000 HP (820 KW) - 4 KV/3/60 HZ
VIBRATION ON 3.5 mm/s AT 1200 RPM
MASS MOMENT OF INERTIA = 12.5 KG-M² IMPELLERS WITH SHAFT
- MOTOR - 1000 HP (820 KW) - 4 KV/3/60 HZ
ROTOR WEIGHT = 953 KG
VIBRATION ON 3.5 mm/s AT 1200 RPM
MASS MOMENT OF INERTIA = 33.3 KG-M²
VIBRATION: 2.8 mm/s AT 1200 RPM
- FOR COG LOCATIONS OF MAJOR EQUIPMENT AND COMPLETE SKID,
REFER TO GA DRAWING DOCUMENT A&V-4-0304-01-0001.4.
- 25 mm EPOXY GROUT FOR COMPRESSOR SKID AND SEPARATOR TANK SKID
SHALL BE PROVIDED BY OTHERS UNLESS AS NOTED.

ALL DIMENSIONS ARE IN MILLIMETRES
UNLESS AS NOTED.

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Premier Fluid Systems

WILLIAMS CANADA PROPYLENE ULC

PROPANE DEHYDROGENERATION PLANT
TAN GAS COMPRESSOR

DESIGN BY	DATE	REV
PROJECT NO.	PROJECT DATE	PROJECT NAME
APPROVED BY	APPROVED DATE	APPROVED NAME
DATE ISSUED	21/7/21/5	A&V-4-0306-01-0001.0

REVISIONS	NO.	DATE	DESCRIPTION
1	1	21/7/21/5	ISSUED FOR PROJECT TENDERING BY
2	2	21/7/21/5	ISSUED AS PER CUSTOMER COMMENTS
3	3	21/7/21/5	ISSUED AS PER CUSTOMER COMMENTS
4	4	21/7/21/5	ISSUED AS PER CUSTOMER COMMENTS

FOUNDATION PLAN	FOUNDATION PLAN
FOUNDATION PLAN	FOUNDATION PLAN
FOUNDATION PLAN	FOUNDATION PLAN
FOUNDATION PLAN	FOUNDATION PLAN

Sheet 01 of 1



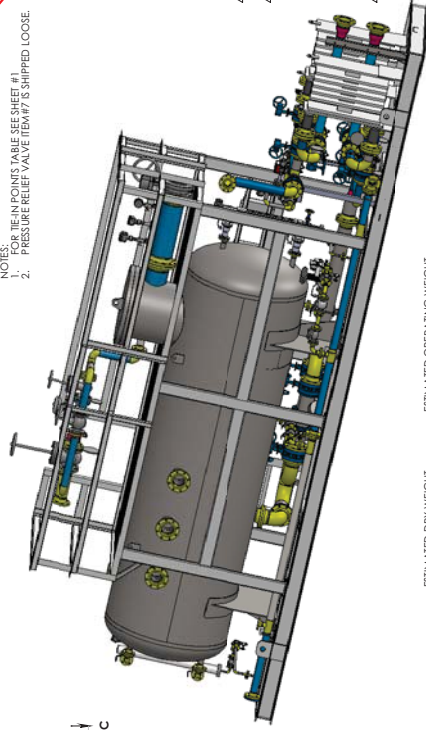
This LT need two transmitters. One for control and another for shut down.

Not found on P&ID. Supplier to check and correct tag number

INTER-CONNECTING TABLE OF CONNECTIONS

NO.	SIZE	TYPE	DESCRIPTION
N1	4"	AND 150# RF FLANGE	GAS INLET TO TANK
N2	8"	AND 150# RF FLANGE	GAS INLET (COMPRESSOR)
N3	6"	AND 300# RF FLANGE	GAS OUTLET (COMPRESSOR)
N4	2"	AND 150# RF FLANGE	SEALED OFF WATER IN
N5	1"	AND 150# RF FLANGE	DRAIN OUTLET
N6	4"	AND 150# RF FLANGE	SEALED OFF WATER RETURN
N7	6"	AND 150# RF FLANGE	GAS RECYCLATION TO COMPRESSORS

- NOTES:
1. TO BE RE-INSPECTED PER SHEET #1.
 2. PRESSURE RELIEF VALVE (PRV) IS SHIPPED LOOSE.



ESTIMATED DRY WEIGHT

SEPARATOR TANK	2250 KG
PLATFORM	1200 KG
HEAT EXCHANGERS	950 KG
PIPING & ACCESSORIES	1225 KG
TOTAL	7625 KG

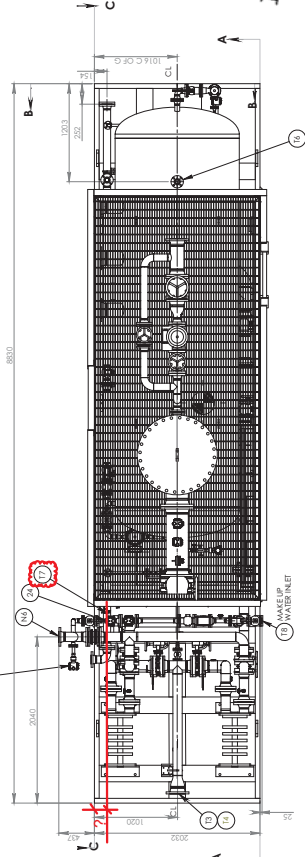
ESTIMATED OPERATING WEIGHT

SEPARATOR TANK	3070 KG
PLATFORM	1200 KG
HEAT EXCHANGERS	950 KG
PIPING & ACCESSORIES	1225 KG
TOTAL	7445 KG

ANVISED LONG 1.5 METERS HYDRAE MAX LOAD 12,000 KG

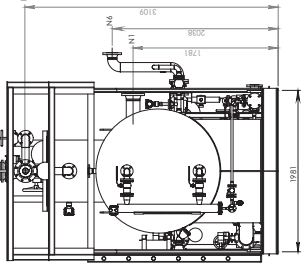
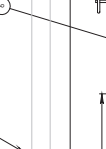
SEPARATOR TANK	2250 KG
PLATFORM	1200 KG
HEAT EXCHANGERS	950 KG
PIPING & ACCESSORIES	1225 KG
TOTAL	7625 KG

PLAN VIEW

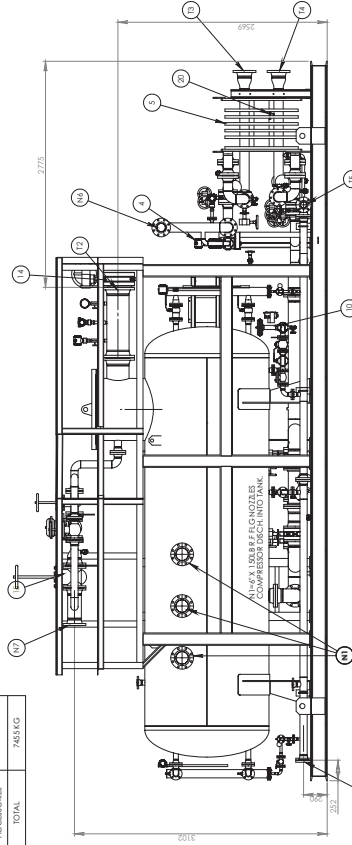


PLAN VIEW

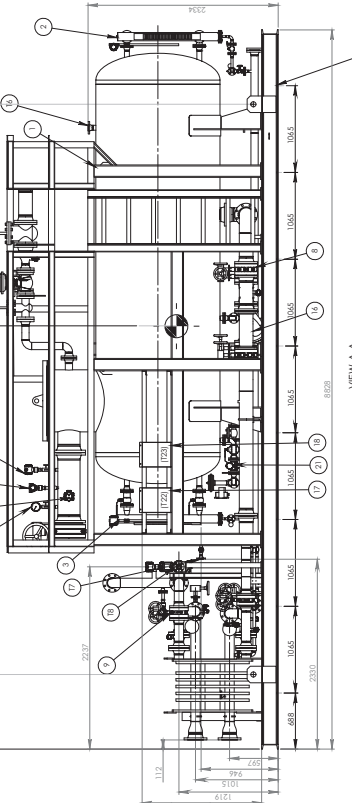
4X LIFTING SLINGS TO BE USED TO MEASURE OF PACKAGE



VIEW B-B



VIEW C-C



VIEW A-A



PRV: 150# 1/2" NPT. PRV: 150# 1/2" NPT. PRV: 150# 1/2" NPT. PRV: 150# 1/2" NPT.

DETAIL OF SEPARATOR TANK SKID - QTY: 1

NOTES:
1. ALL DIMENSIONS ARE IN MILLIMETERS UNLESS AS NOTED

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304-0043
Premier Fluid Systems

WILLIAMS CANADA PROPYLENE LLC
PROPANE DEHYDROGENATION PLANT
TAL GAS COMPRESSOR

REVISIONS

NO.	REV.	DESCRIPTION
1	1	ISSUED FOR PERMITS
2	2	REVISED AS PER CUSTOMER COMMENTS
3	3	REVISED AS PER CUSTOMER COMMENTS
4	4	REVISED AS PER CUSTOMER COMMENTS

PROJECT INFORMATION

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3	3	REVISED AS PER CUSTOMER COMMENTS
4	4	REVISED AS PER CUSTOMER COMMENTS

GENERAL ARRANGEMENT DWG

NO.	REV.	DESCRIPTION
1	1	ISSUED FOR PERMITS
2	2	REVISED AS PER CUSTOMER COMMENTS
3	3	REVISED AS PER CUSTOMER COMMENTS
4	4	REVISED AS PER CUSTOMER COMMENTS

DATE: 14/12/15

PROJECT NO: AGKY-4-0306-01-00014

PROJECT NAME: PROPANE DEHYDROGENATION PLANT

PROJECT LOCATION: TALEN, ALBERTA, CANADA

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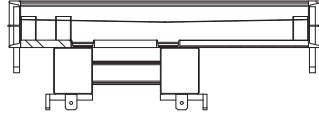
PROJECT NAME: PROPANE DEHYDROGENATION PLANT

PROJECT LOCATION: TALEN, ALBERTA, CANADA

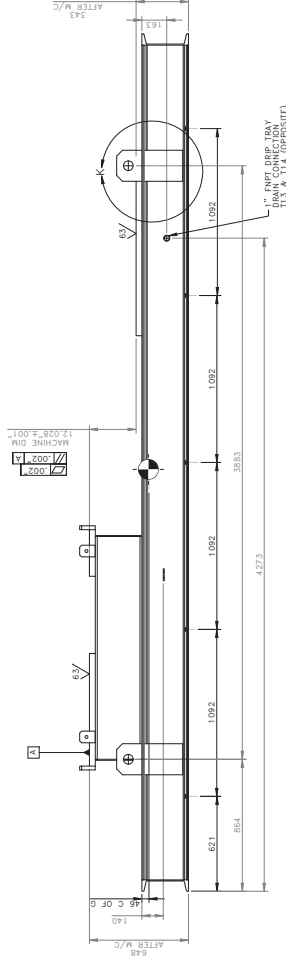
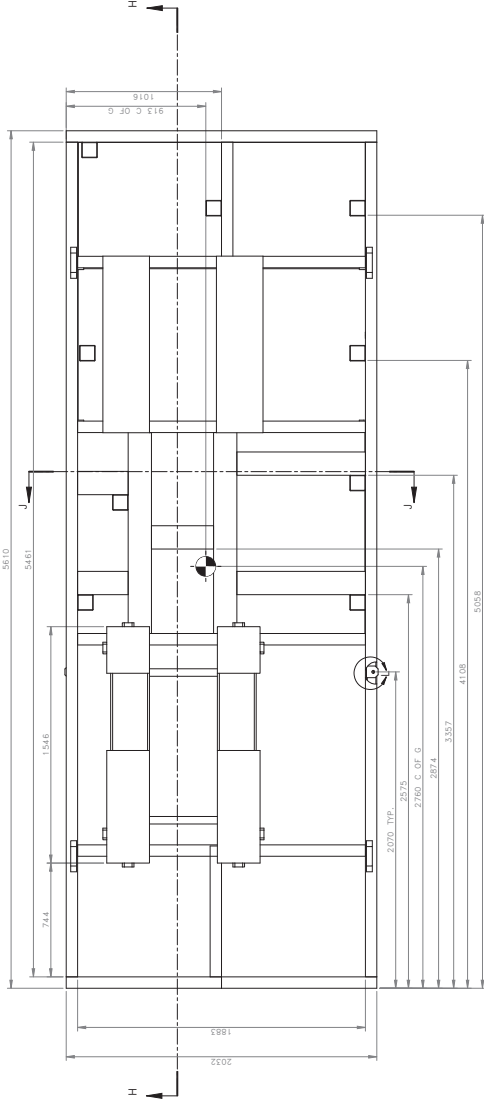
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PROJECT NAME: PROPANE DEHYDROGENATION PLANT

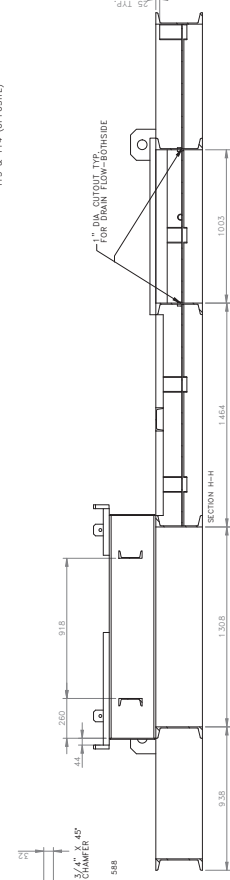
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2	2	C CHANNEL 12" X 20.2 X 80" LONG
3	4	C CHANNEL 12" X 20.2 X 74 1/8" LONG
4	2	C CHANNEL 12" X 20.2 X 51 1/2" LONG
5	1	C CHANNEL 12" X 20.2 X 33 11/16" LONG
6	1	C CHANNEL 12" X 20.2 X 32 1/16" LONG
7	2	C CHANNEL 12" X 20.2 X 38 1/4" LONG
8	2	RECTANGLE TUBE 12" X 10" X 1/2" THK X 57 3/8" LONG
9	2	C CHANNEL 6" X 8.2 X 51 11/16" LONG
10	2	C CHANNEL 6" X 8.2 X 38 1/4" LONG
11	2	C CHANNEL 6" X 8.2 X 13" LONG
12	1	C CHANNEL 6" X 8.2 X 16" LONG
13	2	C CHANNEL 6" X 8.2 X 30 7/8" LONG
14	2	MOTOR PAD 28" X 10" X 2" THK
14A	2	MOTOR PAD 45-1/2" X 12" X 2" THK
15	2	PUMP PAD 45-1/2" X 12" X 2" THK
16	4	LIFTING LUG 17" X 8" X 1 1/2" THK (CUT PER DWG)
17	9	GROUT SO TUBE 4" X 4" X 3/16" THK X 7" LG
18	1	DRIP TRAY COUPLING 6" W PLUG (RAIN)
19	2	GROUND LUG 3" X 1" X 1/4" (CUT PER DWG)
20	2	GROUND LUG 3" X 1" X 1/4" (CUT PER DWG)
21	8	ALIGNMENT LUG 3" X 4" X 1" THK (M/C TO SHARP)



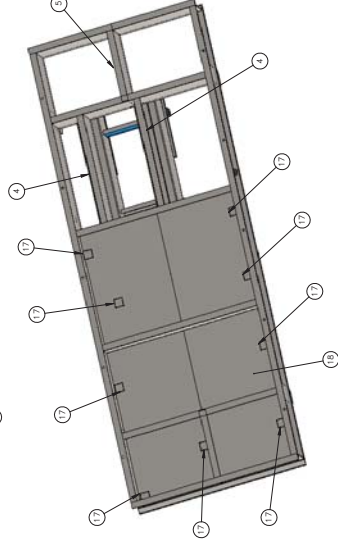
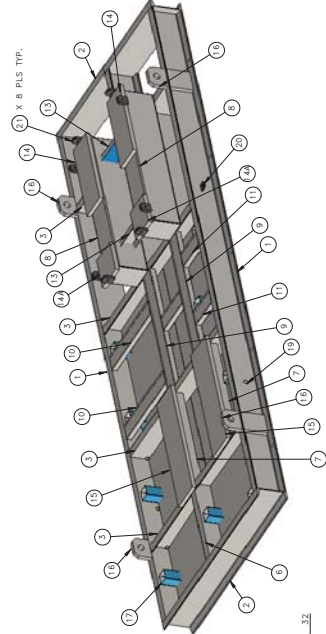
SECTION A-A
SCALE 1/4" = 1"



DETAIL X
MATERIAL: SKID BASE, SA 588 SUITABLE FOR -40°C MOUNT
= SAFE LIFTING LOAD 588KG/LUG



SECTION B-B
SCALE 1/4" = 1"



NOTES:
-MATERIAL: C.S. - SA 572 A/B or SA 588 FOR SKID BASE.

213-0232

REVISIONS

REV	DATE	DESCRIPTION
1	13/12/20	INITIAL DRAWING RELEASE

REFERENCE DRAWINGS

NO	REV	DATE	DESCRIPTION
1	1	13/12/20	INITIAL DRAWING RELEASE

PROJECT INFORMATION

NO	REV	DATE	DESCRIPTION
1	1	13/12/20	INITIAL DRAWING RELEASE

PROJECT: 213-0232

DATE: 13/12/20

PROJECT: 213-0232

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PROJECT: 213-0232

DATE: 13/12/20

PROJECT: 213-0232

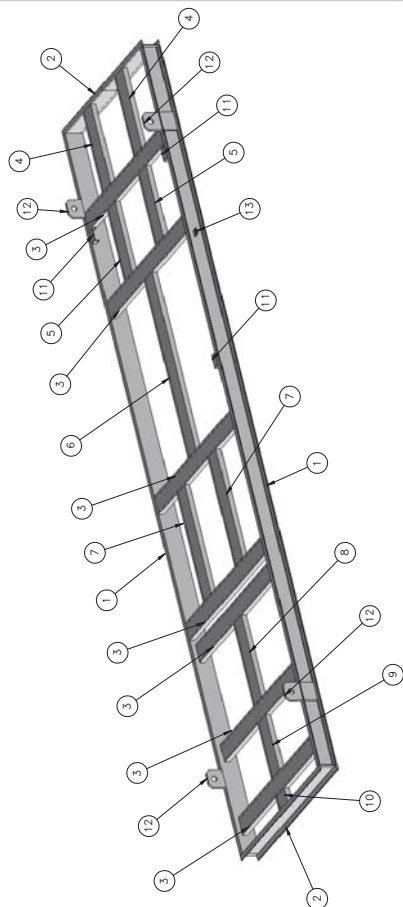
DATE: 13/12/20

PROJECT: 213-0232

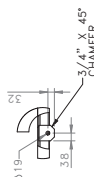
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PROJECT: 213-0232

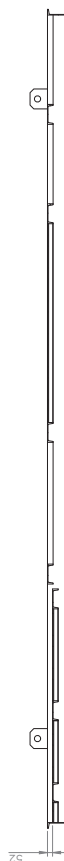
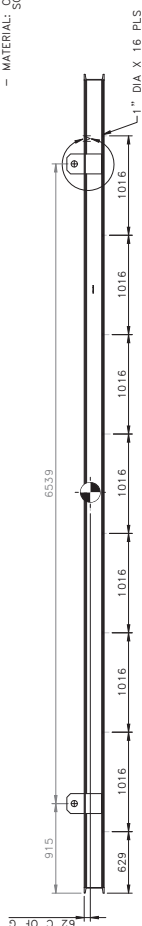
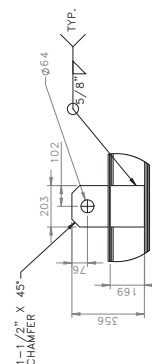
DATE: 13/12/20



MATERIAL: C.S. - SA 572 A/B
or SA 588



DETAIL N
GROUND LUG
- MATERIAL: C.S.-SA572 A/B or SA588
SCALE 1:12

SECTION L-L
SCALE 1 : 25

DETAIL M
LIFTING LUG
- MATERIAL: C.S.-SA572 A/B or SA588
- SUITABLE FOR -45°C MDMT
-SAFE LIFTING LOAD 3150 KG/LUG
SCALE 1 : 12

ITEM	QTY	DESCRIPTION
1	2	C CHANNEL B" X 11.5 X .315" LONG
2	2	C CHANNEL B" X 11.5 X .80" LONG
3	7	C CHANNEL B" X 11.5 X .75 1/2" LONG
4	2	C CHANNEL B" X 11.5 X .38" LONG
5	2	C CHANNEL B" X 11.5 X .32" LONG
6	1	C CHANNEL B" X 11.5 X .80" LONG
7	2	C CHANNEL B" X 11.5 X .48" LONG
8	1	C CHANNEL B" X 11.5 X .37" LONG
9	10	C CHANNEL B" X 11.5 X .25" LONG
10	1	C CHANNEL B" X 11.5 X .8" LONG
11	3	C CHANNEL B" X 8.5 X .3" LONG
12	4	JEFFING LOG 14" X 8" X 1" THK (CUT FIVE DOWS)



Williams Energy Canada ULC

Propane Dehydrogeneration Plant

Tal Gas Conversion

Strathcona, Alberta Canada

Scale

One

4 x 6

Title

Motor Pump Baseplate

Drawn By

044

Checked By

044

Project

044

Approved By

044

Date

044

 Premier Fluid Systems	CHARGER STAMP
POINT STAMP	

IT'S DRAWING NO.	213-0232
AH	AK
DG	DG

[illegible]

108-02.43	1	17/12/15	
PROG. SPS	REV.	DATA	

REFERENCE GRAMMAR	
1	REF CA LAVOUT PWS: ARIY-44-0304-21-00014 4/2/2016/2016

0.2
3 mm

NOTES:
ALL DIMENSIONS IN INCHES
UNLESS AS NOTED
TOLERANCE: ANGULAR
LINEAR: .015
PAINT: AS PER SPEC.
APPROX. WEIGHT: 100 LBS.

17.3 Appendix C: Communications and Emails



RE: 10-CG-10001 Tail Gas Compressor Anchor bolt 

A Sharif to: Arees Kham

08/09/2016 12:54 PM

"Charles.O.Christian@fluor.com", Dominic Gemmiti,
Cc: "Elzbieta.Szaro@Fluor.com", "Lebing.Liu@fluor.com",
"Mahendra.Z.Patel@Fluor.com", "Radoslav.Stefanovic@Fluor.com",

Hello Arees;

I will step in at this time. You are correct; anchor bolts are supplied by Fluor and we will make sure they are properly developed in the concrete foundations. However, as William's agent, when we see a flaw, or perhaps I should call it a slight deficiency in a vendor package, it is our obligation to point it out, even though the ultimate responsibility of the package is with the supplier. In this particular case, we felt that for a vibrating piece of equipment with a magnitude of horsepower such as the tail gas compressor, the anchor bolts are undersized and the short development length recommended is insufficient. In addition, we will not be specifying any type of post installed anchors (expansion nor grouted types) for vibratory equipment, regardless of your past performance. Thank you.

Regards,

Alexander Sharif, SE | **FLUOR Corp.** | Structural Engineering | a.sharif@fluor.com |

From: Arees Kham <Arees.Kham@pfspumps.com>
To: "Saugata.Ghosh@fluor.com" <Saugata.Ghosh@fluor.com>,
Cc: Dominic Gemmiti <dominic.gemmiti@pfspumps.com>, Vijay Rangisetty <vijay.r@pfspumps.com>,
"Mahendra.Z.Patel@Fluor.com" <Mahendra.Z.Patel@Fluor.com>, "Radoslav.Stefanovic@Fluor.com" <Radoslav.Stefanovic@Fluor.com>,
"Ritabrata.Pramanik@fluor.com" <Ritabrata.Pramanik@fluor.com>, "Lebing.Liu@fluor.com" <Lebing.Liu@fluor.com>,
"A.Sharif@fluor.com" <A.Sharif@fluor.com>, "Skyler.Chen@Fluor.com" <Skyler.Chen@Fluor.com>,
"Elzbieta.Szaro@Fluor.com" <Elzbieta.Szaro@Fluor.com>, "Charles.O.Christian@fluor.com" <Charles.O.Christian@fluor.com>
Date: 08/09/2016 12:37 PM
Subject: RE: 10-CG-10001 Tail Gas Compressor Anchor bolt

Hello Saugata,

Please see our response below, regarding point by point for the anchor bolts for the Tail Gas compressors. As discussed in our weekly telecom, PFS has given a recommendation for the anchor bolts. However, the anchor bolts are not in PFS scope of supply, so we cannot comment much on it.

Thanks and regards,

Arees S. Khambatta, P.Eng.
Operations Manager
Premier Fluid Systems Inc.
arees.kham@pfspumps.com
[905-6372611](tel:905-6372611)



From: Saugata.Ghosh@fluor.com [<mailto:Saugata.Ghosh@fluor.com>]

Sent: Wednesday, August 03, 2016 1:20 AM

To: Arees Kham

Cc: Dominic Gemmiti; Vijay Rangisetty; Mahendra.Z.Patel@Fluor.com; Radoslav.Stefanovic@Fluor.com; Ritabrata.Pramanik@fluor.com; Lebing.Liu@fluor.com; A.Sharif@fluor.com; Skyler.Chen@Fluor.com; Elzbieta.Szaro@Fluor.com; Charles.O.Christian@fluor.com

Subject: Fw: 10-CG-10001 Tail Gas Compressor Anchor bolt

Hello Arees

Kindly refer below communication from our Civil, Structural & Architectural Discipline (CSA) colleagues.

Our CSA colleagues would like to raise a few questions about the anchor bolts required on vendor drawing A6KV-4-0306-01-00010. On this drawing, the anchor bolts are described as "3/4" DIA, ASTM F1554, Grade 36-10" LG (QTY 66)". The anchor bolt sizes and length seems to be small compared with similar equipment from other vendors.

The questions they would like to raise are as follows:

1. Are the anchor bolts sufficient to resist the dynamic loads and piping thrusts etc produced by the equipment considering effects such as fatigue.

PFS had recommended the Hilti KWIK Bolt TZ Expansion Anchor 3.3.5, as we have excellent experience using this bolt for anchoring of our liquid ring compressor packages. This includes the minimum dynamic forces and minimum piping thrusts which may be experienced during operation. During the course of the design process, Fluor/Client advised that they would supply the anchor bolts, so the note above is only a recommendation as it is no longer in PFS scope of supply. We defer to Fluor's experience to install their preferred anchor bolts.

2. Do the anchor bolts provide sufficient clamping force to ensure the equipment skids and concrete block foundation vibrating together? If not, the equipment would vibrate by their own and the concrete foundations and piles will not help reducing the vibration of the equipment.

PFS - As noted in Q1 above, Fluor is providing the anchor bolts.

3. Do the anchor bolts require pretension?

PFS -As noted in Q1 above, Fluor is providing the anchor bolts.

4. if yes to the previous question, what is the magnitude of pretension forces of anchor bolts required?

PFS -As noted in Q1 above, Fluor is providing the anchor bolts.

5. The anchor bolts are only 10" long. Are the anchor bolts long enough to prevent pull out and break out

from concrete?

PFS -As noted in Q1 above, Fluor is providing the anchor bolts.

6. Do we need Nelson sleeves to allow adjustments of the locations of anchor bolts, and debonding of bolt shafts to provide free stretch lengths for pretension?

PFS -As noted in Q1 above, Fluor is providing the anchor bolts.

7. The vendor drawing is showing the anchor bolts lengths and it does not exclude the anchor bolt scope. Does it mean the vendor will supply the anchor bolts? If yes, please ask vendor to furnish further installation details such as thread lengths, projections, embedment lengths and anchorage rebars etc.

Response: Per Clause 13.0, PO Engineering Notes (included with PO Change Order 3), anchor bolts are excluded from Supplier's scope of supply.

Would request you to respond on Items 1 to 6 above

The relevant drawings are attached below

Regards

Saugata Ghosh | Fluor - New Delhi | Mechanical | saugata.ghosh@fluor.com | IODC 76.2434 | O
+91.124.4570700 | www.fluor.com

----- Forwarded by Saugata Ghosh/IN/FD/FluorCorp on 08/03/2016 10:37 AM -----

From: Lebing Liu/CA/FD/FluorCorp
To: Saugata Ghosh/IN/FD/FluorCorp@FluorCorp, Mahendra Z Patel/CA/FD/FluorCorp@FluorCorp,
Cc: Skyler Chen/CA/FD/FluorCorp@FluorCorp, A Sharif/CA/FD/FluorCorp@FluorCorp, Elzbieta
Szaro/CA/FD/FluorCorp@FluorCorp, Radoslav Stefanovic/CA/FD/FluorCorp@FluorCorp
Date: 08/02/2016 10:33 PM
Subject: 10-CG-10001 Tail Gas Compressor Anchor bolt

Hi Saugata,

Kindly please see the vendor drawings in the attachment for the tail gas compressor, 10-CG-10001A/B/C. These are drawings for 3 compressors side by side in building 02NH-102. Each compressor have 1100 horse power (820kW).

I would like to raise a few questions about the anchor bolts required on vendor drawing A6KV-4-0306-01-00010. On this drawing, the anchor bolts are described as "3/4" DIA, ASTM F1554, Grade 36-10" LG (QTY 66)". The anchor bolt sizes and length seems to be small compared with similar equipment from other vendors.

The questions I would like to raise is as follows:

1. Are the anchor bolts sufficient to resist the dynamic loads and piping thrusts etc produced by the equipment considering effects such as fatigue.

2. Do the anchor bolts provide sufficient clamping force to ensure the equipment skids and concrete block foundation vibrating together? If not, the equipment would vibrate by their own and the concrete foundations and piles will not help reducing the vibration of the equipment.
3. Do the anchor bolts require pretension?
4. if yes to the previous question, what is the magnitude of pretension forces of anchor bolts required?
5. The anchor bolts are only 10" long. Are the anchor bolts long enough to prevent pull out and break out from concrete?
6. Do we need Nelson sleeves to allow adjustments of the locations of anchor bolts, and debonding of bolt shafts to provide free stretch lengths for pretension?
7. The vendor drawing is showing the anchor bolts lengths and it does not exclude the anchor bolt scope. Does it mean the vendor will supply the anchor bolts? If yes, please ask vendor to furnish further installation details such as thread lengths, projections, embedment lengths and anchorage rebars etc.

[attachment "A6KV-4-0306-01-00010_FOUNDATION PLAN - 10-CG-10001ABC_R4.pdf" deleted by Saugata Ghosh/IN/FD/FluorCorp] [attachment "A6KV-4-0306-01-00014_GENERAL ARRANGEMENT DRAWING - 10-CG-10001ABC_R4.pdf" deleted by Saugata Ghosh/IN/FD/FluorCorp]

Best Regards

Lebing Liu | **Fluor Canada LTD** | Williams PDH Project | Structural Design Engineer | lebing.liu@fluor.com |
O:1-403-538-1646 |